

Outcome-free records and the minimax precision of conditional treatment effect estimation

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Abstract

We characterize the value of outcome-free covariate-treatment records for estimating a conditional average treatment effect at the center of the unit covariate cube. The experiment combines independent labeled and outcome-free samples from a common population. Under a known uniform covariate design, conditional exchangeability, a public overlap level $0 < \varepsilon < 1/2$ with propensity values in $[\varepsilon, 1 - \varepsilon]$, binary responses with arm means in $[0, 1]$, and public Hölder regularity, we establish the sharp minimax expected absolute-error rate uniformly over labeled counts of at least two and every nonnegative auxiliary count. The propensity and control-mean smoothness orders lie in $(0, 1]$, the contrast smoothness order lies in $[1, 3]$, and the common public Hölder radius satisfies $L > 1/2$. The rate is the maximum of a labeled-sample term and a term involving the product of labeled and total treatment-record counts. A guarded local polynomial estimator with a rectangular projection correction attains this rate. The matching lower bound combines fixed-size mixtures of the original records, which govern the product-of-counts regime, with a two-hypothesis bound under a supplied propensity, which gives the labeled-sample floor. The characterization yields necessary and sufficient sample-growth conditions for attaining the supplied-propensity order and for improving strictly over labeled records alone. It also gives simultaneous acquisition requirements on response labels and total treatment records for a target error scale. The rate exponents do not depend on ε , while the comparison constants may depend on it together with the other fixed public parameters.

1 Introduction

This paper establishes a sharp pointwise precision benchmark for conditional average treatment effect estimation when responses are observed on a smaller collection of records than covariates and treatment assignments. We consider $n \geq 2$, the labeled-record count, and $m \geq 0$, the independent outcome-free covariate-treatment record count. For the specified potential-outcome class in [Definition 3](#), with known uniform covariate design and the public parameter restrictions stated below, the minimax expected absolute error is

$$\underbrace{R(n, m)}_{\text{minimax pointwise absolute-error risk}} \asymp \max \left\{ n^{-\gamma/(2\gamma+d)}, [n(n+m)]^{-\gamma/(2\gamma+d+\gamma d/(\alpha+\beta))} \right\},$$

where d is the covariate dimension, and α, β, γ are the Hölder smoothness orders of the propensity, control response mean, and causal contrast, respectively. The target is the conditional

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causal contrast at the cube’s center. The comparison holds uniformly over both record counts, with constants depending on the fixed public class parameters. A fully specified decision based on the observed records attains this order, as established in [Theorem 2](#).

The model fixes the covariate distribution to be uniform on $[0, 1]^d$, with $d \geq 1$, and assumes conditional treatment exchangeability. Treatment and potential responses are binary. For a fixed public $\varepsilon \in (0, 1/2)$, the propensity lies in $[\varepsilon, 1 - \varepsilon]$, and both potential-response means lie in $[0, 1]$ throughout the cube. The public smoothness orders satisfy $\alpha, \beta \in (0, 1]$ and $\gamma \in [1, 3]$, with L , the common public Hölder radius, satisfying $L > 1/2$. These restrictions define the primitive law class in [Definition 3](#); the independent, common-population record blocks follow [Assumption 9](#). Admissible conditional-margin witnesses satisfy the restrictions over the whole cube. Their continuous contrast, together with full-support design and the identifying assumptions, determines the point target uniquely from the observed law.

The distinction between record types has a direct acquisition interpretation. Response ascertainment through chart review can yield labeled records containing covariates, treatment, and the true response, while routine records supply covariates and treatment. Surrogate-assisted treatment-effect estimation provides a related electronic-record setting [[Cheng et al., 2021](#), Sections 2.1, 2.5–2.6, and 4]. Within the present sampling model, adding a new independent labeled record changes (n, m) to $(n + 1, m)$. Labeling one already-counted outcome-free record changes the composition to $(n + 1, m - 1)$ at fixed total count; if the record index is fixed in advance, or chosen by a randomization independent of all records, the resulting dataset follows the stated sampling law at $(n + 1, m - 1)$. Adding a new outcome-free record changes (n, m) to $(n, m + 1)$. The sharp risk formula benchmarks these sample compositions under the stated independent-sampling assumptions.

The estimator combines a coarse local polynomial fit with a finer projection correction. It splits the original records into independent outcome and treatment roles, whose sizes are proportional to the labeled and total treatment-record counts. A rectangular average across these roles corrects the local estimating moments for the unknown nuisance functions. An empirical eigenvalue guard and clipping produce a finite Borel decision for every dataset. Under the stated model conditions, [Theorem 1](#) bounds its risk by effect approximation, nuisance approximation, localized response sampling, and rectangular sampling terms. Public tuning balances these terms to attain the sharp rate.

The matching lower bound retains the original fixed-size observation experiment. The construction in [Theorem 4](#) couples propensity and response perturbations while preserving identical auxiliary-only mixture laws across hypotheses. Its information bound supplies the nuisance converse in [Lemma 6](#) over the range where the product-of-counts term governs precision. The supplied-propensity comparison in [Theorem 3](#) establishes the labeled-sample order $n^{-\gamma/(2\gamma+d)}$ over the same primitive class and every auxiliary count. Together, these lower bounds and the attaining estimator give the uniform minimax characterization, including randomized decisions.

The characterization distinguishes strict improvement from attainment of the supplied-propensity order. Write $S = \alpha + \beta$, the combined nuisance smoothness, and let S_{crit} denote the smoothness threshold and q_* the total-record growth exponent:

$$S_{\text{crit}} = \frac{\gamma d}{2\gamma + d}, \quad q_* = \frac{\gamma d}{S(2\gamma + d)}.$$

At and above the threshold, labeled records alone attain the supplied-propensity order. Below it, attainment along an auxiliary-size sequence requires total treatment-record counts to remain

at least a positive constant times n^{q^*} . Below the smoothness threshold, strict asymptotic improvement over labeled records alone occurs exactly when the auxiliary-to-labeled ratio diverges. [Proposition 1](#) also translates a target absolute-error scale into simultaneous requirements on the labeled count and the product of labeled and total counts. These are risk requirements up to parameter-dependent constants, with exact equivalences for the displayed rate.

The closest supervised comparison is [Kennedy et al. \[2024, Section 3, Theorems 1–2 and equation \(14\)\]](#). At zero auxiliary records, our rate has the same numerical exponents and smoothness cutoff, as recorded in [Lemma 7](#). Their statistical guarantees concern their bounded-density model and stated attainment conditions; the lower bound for our known-uniform class combines the original-record construction in [Theorem 4](#) and [Lemma 6](#), which covers smoothness below S_{crit} with total count at most n^{q^*} , with the labeled-sample floor $n^{-\gamma/(2\gamma+d)}$ from the constant-nuisance subfamily in [Theorem 3](#). Semi-supervised analyses of average and quantile treatment effects characterize nuisance-dependent expansions and inference for marginal targets [[Chakraborty and Dai, 2022, Theorem 2.1, Corollary 2.1, Theorem 3.1, and Corollary 3.1](#)]. Our contribution characterizes worst-case pointwise absolute error and its acquisition requirements for the specified conditional target. The projection construction connects local residualized regression [[Kennedy, 2023, Theorem 3](#)] with finite-rank second-order correction [[Robins et al., 2009, Lemma 3](#)].

The exposition proceeds through related work, setup and assumptions, the estimator construction, sharp precision and sample requirements, and discussion and limitations. The upper-bound argument is organized in [Section A](#), the original-record converse in [Section B](#), and the acquisition algebra in [Section C](#).

2 Related work

The closest comparison is [Kennedy et al. \[2024, Section 3, Theorems 1–2 and equation \(14\)\]](#), who characterize supervised pointwise CATE estimation under Hölder smoothness. Write n for the labeled sample size, d for the covariate dimension, and α , β , and γ for the propensity, control-mean, and treatment-effect smoothness orders. With $s_{\text{pub}} = (\alpha + \beta)/2$, their absolute-error exponents are

$$\frac{1}{1 + d/(2\gamma) + d/(4s_{\text{pub}})} \quad \text{when} \quad s_{\text{pub}} < \frac{d/4}{1 + d/(2\gamma)}, \quad \frac{\gamma}{2\gamma + d} \quad \text{otherwise.}$$

Published attainment requires their matrix regularity, smooth nuisance-estimation errors, boundedness conditions, and control of the third-order remainder involving covariate-distribution estimation. Their localized second-order projection correction balances effect approximation, nuisance approximation, and sampling variation.

The algebraic comparison appears in [Lemma 7](#). It identifies the supervised exponents obtained by specializing the sample-size expression to zero auxiliary records. Its interpretation follows the published source model, which permits a bounded covariate density and uses a construction-specific design in the lower-bound argument. The present potential-outcome law class with known uniform design, specified in [Definition 3](#), fixes the covariate distribution throughout. Within that class, [Theorem 4](#) and [Lemma 6](#) supply the original-record testing argument. These results establish the lower bound in the experiment actually observed by the decision maker, while the published benchmark supplies the numerical supervised comparison.

This distinction becomes consequential when information arrives in unequal amounts. The sampling experiment combines n labeled records with m independent outcome-free covariate-treatment records; its sampling law is given in [Assumption 9](#). The pointwise absolute-error characterization in [Theorem 2](#) retains both record counts, and [Theorem 3](#) identifies the precision attainable when the propensity function is supplied. Together, these results connect the amount of treatment information to the accuracy of a conditional causal contrast under the paper’s fixed design, overlap, and smoothness conditions.

Related semi-supervised treatment-effect analyses study population functionals through asymptotic expansions. [Chakraborty and Dai \[2022, Theorem 2.1 and Corollary 2.1\]](#) separate labeled residual variation from averaging over the enlarged covariate sample for average treatment effect estimation. Their corollary addresses a vanishing labeled fraction with a correctly specified propensity model, retaining explicit nuisance remainders; asymptotic normality follows under the stated remainder conditions. The corresponding quantile treatment effect expansions additionally account for the density at the target quantile [[Chakraborty and Dai, 2022, Theorem 3.1 and Corollary 3.1](#)]. These expansions explain robustness and efficiency for marginal causal functionals. The present risk characterization instead evaluates a conditional contrast at a fixed interior profile and quantifies worst-case absolute error over the specified Hölder class.

The information carried by auxiliary records also distinguishes the surrogate-assisted setting of [Cheng et al. \[2021, Sections 2.1, 2.5–2.6, and 4\]](#). Their records include outcome-predictive surrogate variables, potentially measured after treatment, alongside baseline covariates and treatment. An imputation-based ATE estimator exploits these variables, with robustness to imputation-model misspecification and local efficiency under their ideal semi-supervised model; perturbation resampling supports inference. Their electronic health records application illustrates the role of surrogate information when chart-reviewed outcomes are scarce. Here, auxiliary observations contribute covariate-treatment information, and their value is characterized through the pointwise risk and sample requirements in [Theorem 2](#) and [Proposition 1](#).

A further comparison concerns the statistical criterion used to measure precision. [Kato \[2025, Theorems 4.2 and 4.4 and Corollary 4.5\]](#) derive two-sample ATE efficiency bounds for regular estimators and establish attainment through cross-fitted estimation under nuisance consistency and product-rate conditions. Their auxiliary sample contains covariates, and their analysis allows distinct labeled and auxiliary covariate distributions subject to support conditions. With coincident distributions and the stated evaluation weighting, the efficiency bound separates outcome residual variation from variation in the conditional effect across covariate profiles. Those conclusions characterize asymptotic variance at a fixed positive sampling fraction. The present minimax criterion, defined in [Definition 5](#), measures pointwise absolute error across the primitive class and accommodates every auxiliary-record count in the stated experiment.

Supplied-predictor conditional estimation offers another route for using auxiliary information. [Sui et al. \[2026, Sections 2.4 and 3 and Theorems 1–3\]](#) combine labeled observations, unlabeled covariates, and a supplied predictor to estimate conditional moment targets. Their Sobolev-equivalent reproducing kernel Hilbert space conditions support pointwise error bounds and confidence intervals, with prediction and residual components determining variance. Structure-adaptive CATE estimation uses different restrictions: [Kim \[2026, Sections 2.3 and 6, Corollary 4\]](#) give a pointwise squared-error result under bounded ambient-RKHS and overlap conditions when the contrast lies in the designated lower-complexity subspace; their broader analysis also treats integrated squared error, source conditions, known lower-dimensional rep-

representations, and model selection. Those results use supervised outcomes and RKHS/Sobolev structure. Our experiment instead fixes a Hölder class, measures pointwise absolute error, and studies the extra information supplied by outcome-free covariate-treatment records.

Methodologically, the projection correction belongs to the higher-order influence-function tradition. Finite-rank orthogonal projections provide a tractable representation of second-order corrections in nonlinear functional estimation [Robins et al., 2009, Lemma 3]. Orthogonal regression supplies the conditional-effect estimating structure: residualized treatment and outcome regression underlie the R-learner [Nie and Wager, 2021, Section 1], and local polynomial double-residual regression connects this structure to pointwise smoothness-based estimation [Kennedy, 2023, Theorem 3]. The rectangular projection estimator in Algorithm 2 and Definition 11 combines localization with independent treatment and outcome roles. Its upper bound in Theorem 1 makes the separate sampling contributions explicit, linking the projection architecture to the unequal information available from labeled and outcome-free records.

3 Setup and assumptions

Throughout, c and C denote finite positive constants whose values may change between appearances. At fixed public parameters, these constants are independent of the sample sizes. We write $a \lesssim b$ when $a \leq Cb$, and $a \asymp b$ when both $a \lesssim b$ and $b \lesssim a$. Vector norms are Euclidean unless otherwise indicated. We use $\mathbb{N} = \mathbb{N}_0 = \{0, 1, \dots\}$ and write $\mathbb{N}_{\geq 1}$ for the positive integers. Counts and polynomial multi-indices may be zero; dimensions, resolutions, and sample-index ranges carry their displayed positive lower bounds. The symbol $N = n + m$ always denotes the total treatment-record count.

Fix the public parameters

$$d \geq 1, \quad \alpha, \beta \in (0, 1], \quad \gamma \in [1, 3], \quad L > 1/2, \quad \varepsilon \in (0, 1/2).$$

Here d is the covariate dimension, α, β, γ are the propensity, control-mean, and causal-contrast smoothness orders, respectively, L is the common Hölder radius, and ε is the overlap level. These parameters remain fixed as the sample sizes vary.

3.1 Primitive variables and identifying restrictions

We use a coordinate-partial convention for Hölder regularity. For a function f and a regularity order s , the norm $\|f\|_{H^s}$ controls coordinate derivatives indexed by the nonnegative-integer multi-index κ , together with their highest-order Hölder variation. The following definition specifies the convention, including at integer smoothness orders.

Definition 1. [def:holder-norm] (Coordinate-partial Hölder norm) *The extended-real Hölder norm $\|f\|_{H^s}$, for $f : \mathbb{R}^d \rightarrow \mathbb{R}$, $d \in \{0, 1, \dots\}$, and $s \in \mathbb{R}$, is*

$$\|f\|_{H^s} = \begin{cases} \max \left\{ \sup_{\substack{\kappa \in \{0, 1, \dots\}^d, |\kappa| \leq \max\{\lceil s \rceil - 1, 0\} \\ x \in [0, 1]^d}} |\partial^\kappa f(x)|, \sup_{\substack{\kappa \in \{0, 1, \dots\}^d, |\kappa| = \max\{\lceil s \rceil - 1, 0\} \\ x, x' \in [0, 1]^d, x \neq x'}} \frac{|\partial^\kappa f(x) - \partial^\kappa f(x')|}{\|x - x'\|_2^{s - \max\{\lceil s \rceil - 1, 0\}}} \right\}, & f \in C \\ +\infty, & \text{otherwise} \end{cases}$$

Here $|\kappa| = \sum_{i=1}^d \kappa_i$, and $\partial^\kappa f(x)$ is the iterated derivative within $[0, 1]^d$, evaluated on coordinate directions with coordinate i repeated κ_i times. The differentiability condition means continuous

differentiability of the indicated order within the cube. Both suprema are taken in the extended nonnegative reals, with an empty supremum equal to zero.

For positive s , this convention uses derivatives through order $\lceil s \rceil - 1$ and a highest-order Hölder exponent of $s - \lceil s \rceil + 1$. Thus integer orders use a Lipschitz condition on derivatives of one lower order. Every regularity restriction below is evaluated within the unit cube.

A full record consists of a covariate vector X , a binary treatment indicator A , and binary potential responses $Y(0)$ and $Y(1)$ under control and treatment. Let $P = \mathcal{L}(X, A, Y(0), Y(1))$ denote their primitive Borel law. The observed response Y selects the potential response corresponding to the realized treatment.

Definition 2. [synth_5] (Observed response selection) *The observed response Y for a full record $(X, A, Y(0), Y(1))$, where $X \in \mathbb{R}^d$ is the covariate vector for a nonnegative integer d , $A \in \{0, 1\}$ is the treatment indicator, and $Y(0), Y(1) \in \{0, 1\}$ are the potential responses under control and treatment, is defined by*

$$Y = \begin{cases} Y(0), & A = 0, \\ Y(1), & A = 1. \end{cases}$$

The corresponding observed record is (X, A, Y) .

The evaluation profile is the fixed interior point $x_0 = (1/2, \dots, 1/2)$. We write e_P for an admissible propensity witness, $\mu_{0,P}$ and $\mu_{1,P}$ for admissible control and treated conditional-mean witnesses, and τ_P for the canonical continuous causal contrast. Their precise admissibility requirements are collected in Definition 3. The identifying restrictions begin with the covariate design and treatment assignment.

Assumption 1. [ass:uniform-design] (Uniform covariate design) *The covariate vector X satisfies*

$$\mathcal{L}_P(X) = \text{Unif}([0, 1]^d).$$

The known uniform design fixes the distribution used for localization and integration and gives full support on the cube. It is the maintained uniform-design restriction of the bounded-density framework considered by [Kennedy et al., 2024]. Treatment assignment is linked to the potential responses through conditional exchangeability.

Assumption 2. [ass:exchangeability] (Conditional treatment exchangeability) *The treatment indicator A and the potential responses $Y(0), Y(1)$ satisfy*

$$A \perp (Y(0), Y(1)) \mid X,$$

where X is the covariate vector.

Conditional exchangeability allows treatment-specific observed means to identify the corresponding potential-response means after conditioning on the covariates [Chakraborty and Dai, 2022]. Both treatment arms must also be represented throughout the evaluation domain.

Assumption 3. [ass:overlap] (Pointwise overlap on the cube) *The overlap condition for a primitive law P , at the overlap level $\varepsilon \in \mathbb{R}$, is $0 < \varepsilon$ together with the requirement that the raw Borel propensity witness carried by P takes values in $[\varepsilon, 1 - \varepsilon]$ at every $x \in [0, 1]^d$.*

Bounded overlap keeps both conditional treatment probabilities away from zero, as in the pointwise conditional-effect framework of [Kennedy et al., 2024]. Here the endpoints are the public values ε and $1 - \varepsilon$, with $0 < \varepsilon < 1/2$, and the restriction applies at every point of the cube.

The regularity conditions distinguish the two nuisance functions from the causal contrast. We first impose isotropic Hölder regularity on the propensity.

Assumption 4. [ass:propensity-holder] (Hölder smoothness of the propensity) *The designated Borel propensity witness e_P satisfies*

$$\|e_P\|_{H^\alpha} \leq L.$$

Isotropic Hölder propensity regularity controls how treatment probabilities vary across nearby covariate profiles [Kennedy et al., 2024]. The order α is common across coordinates, and the radius L bounds both the function and its Hölder variation. The control response mean has its own smoothness order.

Assumption 5. [ass:control-holder] (Hölder smoothness of the control mean) *The designated Borel control-mean witness $\mu_{0,P}$ satisfies*

$$\|\mu_{0,P}\|_{H^\beta} \leq L.$$

Isotropic Hölder control-mean regularity governs variation in the baseline response surface [Kennedy et al., 2024]. Allowing β to differ from α separates outcome regularity from treatment-assignment regularity. The target surface is controlled directly by the following restriction.

Assumption 6. [ass:effect-holder] (Hölder smoothness of the causal contrast) *The canonical continuous causal contrast τ_P satisfies*

$$\|\tau_P\|_{H^\gamma} \leq L.$$

Isotropic Hölder contrast regularity controls the local variation of the conditional treatment effect itself [Kennedy et al., 2024]. The separate order $\gamma \in [1, 3]$ permits a smoother causal contrast than either nuisance function and specifies the regularity used to estimate its value at x_0 .

The binary-response model also requires pointwise probability bounds on the designated arm means. We state these restrictions explicitly because admissible witnesses are functions defined throughout the cube.

Assumption 7. [ass:control-interior] (Control response bounds) *The designated Borel control-mean witness $\mu_{0,P}$ satisfies*

$$\mu_{0,P}(x) \in [0, 1] \quad \text{for every } x \in [0, 1]^d.$$

This pointwise control-response restriction is specific to the present analysis and keeps the designated control mean within the Bernoulli probability range, including its endpoints. The corresponding treated mean obeys the same numerical bounds.

Assumption 8. [ass:treated-interior] (Treated response bounds) *The designated Borel treated-mean witness $\mu_{1,P}$ satisfies*

$$\mu_{1,P}(x) \in [0, 1] \quad \text{for every } x \in [0, 1]^d.$$

The treated-response restriction is likewise specific to this analysis. Together, the two arm bounds ensure that the designated contrast takes values in $[-1, 1]$ everywhere on the cube.

Membership in the primitive class is existential: a law belongs when it admits conditional-mean witnesses satisfying the stated restrictions. Let $\mathcal{M}(d, \alpha, \beta, \gamma, L, \varepsilon)$ denote this class, abbreviated to $\mathcal{M}$ at the fixed public parameters.

Definition 3. `[def:primitive-class]` (Primitive law class $\mathcal{M}(d, \alpha, \beta, \gamma, L, \varepsilon)$) $\mathcal{M}(d, \alpha, \beta, \gamma, L, \varepsilon)$ is the class of primitive Borel laws $P = \mathcal{L}(X, A, Y(0), Y(1))$, where X is the covariate vector, A is the binary treatment indicator, and $Y(0)$ and $Y(1)$ are the binary potential responses under control and treatment. For public parameters $L > 1/2$ and $\varepsilon \in (0, 1/2)$, membership is defined by the following properties:

- **(Uniform design.)** The law satisfies [Assumption 1](#).
- **(Exchangeability.)** The law satisfies [Assumption 2](#).
- **(Conditional means.)** There exist Borel functions

$$g_e, g_0, g_1 : [0, 1]^d \rightarrow [0, 1]$$

such that

$$\begin{aligned} g_e(X) &= \mathbb{E}_P[A \mid X] && P\text{-almost surely,} \\ g_0(X) &= \mathbb{E}_P[Y(0) \mid X] && P\text{-almost surely,} \\ g_1(X) &= \mathbb{E}_P[Y(1) \mid X] && P\text{-almost surely.} \end{aligned}$$

- **(Overlap and smoothness.)** These functions satisfy [Assumptions 3 to 6](#), with g_e , g_0 , and $g_1 - g_0$ as the propensity, control mean, and causal contrast.
- **(Response bounds.)** The arm means satisfy [Assumptions 7 and 8](#).

For each admitted law, $e_P, \mu_{0,P}, \mu_{1,P}$ designate admissible functions g_e, g_0, g_1 , respectively. The function τ_P is the canonical continuous representative of the almost-sure conditional causal contrast, and the designated functions satisfy

$$\mu_{1,P}(x) - \mu_{0,P}(x) = \tau_P(x) \quad \text{for every } x \in [0, 1]^d.$$

The witness formulation separates almost-sure conditional-expectation identities from restrictions imposed at every covariate profile. Under the positive smoothness orders fixed above, e_P , $\mu_{0,P}$, and τ_P are continuous; the identity in [Definition 3](#) then gives continuity of $\mu_{1,P}$ as well.

Observed response selection, conditional exchangeability, and overlap yield

$$\mathbb{E}_P[Y \mid A = a, X] = \mu_{a,P}(X) \quad P\text{-almost surely,} \quad a \in \{0, 1\}.$$

Consequently, the observed-record law identifies the causal contrast almost everywhere under the covariate distribution. Uniform design assigns positive probability to every nonempty relatively open subset of the cube. Continuous representatives that agree almost everywhere therefore agree everywhere, which makes $\tau_P(x_0)$ a uniquely defined, identified point target. The continuous representative gives substantive meaning to evaluation at a fixed profile even though conditional expectations are initially defined through almost-sure identities.

3.2 Sampling experiment and risk

Write $P^o = \mathcal{L}_P(X, A, Y)$ for the observed-record law induced by [Definition 2](#), and $P_{XA} = \mathcal{L}_P(X, A)$ for the covariate-treatment marginal. For $n \geq 2$ and $m \geq 0$, the original dataset $\mathcal{D}_{n,m}$ consists of n independent labeled records with law P^o and m independent outcome-free records with law P_{XA} :

$$\mathcal{D}_{n,m} = \left(((X_i, A_i, Y_i))_{i=1}^n, ((\tilde{X}_j, \tilde{A}_j))_{j=1}^m \right).$$

The two blocks are independent. A uniform randomizer U on $[0, 1]$, independent of both blocks, is available to decisions. Thus the product sampling law $\mathbf{E}_{n,m}(P)$ in [Assumption 9](#) is

$$\mathbf{E}_{n,m}(P) = (P^o)^{\otimes n} \otimes P_{XA}^{\otimes m} \otimes \text{Unif}([0, 1]).$$

The following condition makes this sampling experiment explicit.

Assumption 9. `[ass:sample-law]` (Original experiment sampling law) *The original dataset $\mathcal{D}_{n,m}$ and independent uniform randomizer U have joint law*

$$\mathcal{L}_P(\mathcal{D}_{n,m}, U) = \mathbf{E}_{n,m}(P),$$

where $\mathbf{E}_{n,m}(P)$ is the product sampling law of the original dataset and randomizer.

Independent fixed-size labeled and outcome-free samples from a shared law provide the two-channel sampling structure used in semi-supervised causal estimation [[Chakraborty and Dai, 2022](#)]; the independent randomizer permits randomized decisions. In this experiment, each labeled record supplies covariates, treatment, and a response, while each outcome-free record supplies covariates and treatment. The common marginal P_{XA} makes both blocks informative about treatment assignment.

A decision T is a Borel real-valued function of the dataset and randomizer. Its absolute-error risk $\mathcal{R}_{n,m}(T, P)$ measures expected error in estimating the identified point contrast.

Definition 4. `[def:risk]` (Absolute-error risk $\mathcal{R}_{n,m}(T, P)$) *The absolute-error risk is*

$$\mathcal{R}_{n,m}(T, P) = \int |T(\mathcal{D}_{n,m}, U) - \tau_P(x_0)| d\mathbf{E}_{n,m}(P).$$

Here T is a Borel real-valued decision using the original dataset $\mathcal{D}_{n,m}$ and independent uniform randomizer U , x_0 is the fixed interior covariate profile, and $\mathbf{E}_{n,m}(P)$ is the product sampling law of the original dataset and randomizer. The target $\tau_P(x_0)$ is the canonical conditional causal contrast from [Definition 3](#).

Absolute loss expresses precision on the scale of the conditional response difference. To assess precision uniformly across admissible primitive laws, the minimax risk $R(n, m)$ optimizes the worst-case value of this loss over all Borel decisions.

Definition 5. `[def:minimax]` (Minimax risk $R(n, m)$)

$$R(n, m) = \inf_{T \text{ Borel}} \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P).$$

Here $\mathcal{M}$ is the primitive law class at the fixed public parameters, and $\mathcal{R}_{n,m}(T, P)$ is the expected absolute error for the point contrast under law P . The infimum ranges over Borel real-valued decisions T using the original dataset of n labeled records and m independent outcome-free treatment records, together with an independent uniform randomizer.

The sample sizes enter separately because response information is supplied by the labeled block and treatment-assignment information is supplied by both blocks. A complementary experiment supplies the entire admissible propensity function to the decision. Denote such a decision by T^e , and its minimax risk by $R^e(n, m)$.

Definition 6. `[def:oracle-risk]` (Supplied propensity minimax risk) *The supplied propensity minimax risk $R^e(n, m)$, for $d, n, m \in \mathbb{N}$ and $\alpha, \beta, \gamma, L, \varepsilon \in \mathbb{R}$, is the extended nonnegative quantity*

$$R^e(n, m) = \inf_{T^e} \sup_{P \in \mathcal{M}} \int |T^e(\mathcal{D}_{n,m}, U, e_P) - \tau_P(x_0)| d\mathbf{E}_{n,m}(P).$$

Here $\mathcal{M}$ is the primitive law class at these parameters, and e_P and τ_P are its designated admissible propensity function and canonical causal contrast, as defined in [Definition 3](#). The evaluation profile is $x_0 = (1/2, \dots, 1/2)$. The original dataset $\mathcal{D}_{n,m}$ comprises n labeled records and m independent outcome-free treatment records; U is the independent uniform randomizer on $[0, 1]$, and $\mathbf{E}_{n,m}(P)$ is their product sampling law from [Assumption 9](#). The infimum ranges over real-valued decisions T^e receiving the dataset, randomizer, and a supplied function from the covariate space to $\mathbb{R}$, with Borel measurability in $(\mathcal{D}_{n,m}, U)$ for every fixed supplied function.

Supplying e_P gives a reference experiment in which treatment probabilities are available throughout the cube. Measurability is required in the observed data and randomizer for each fixed supplied function. Comparing [Definitions 5](#) and [6](#) quantifies how the original sampling experiment approaches the precision available with supplied treatment probabilities.

3.3 Rate notation

We introduce the rate notation as a single family. Let $N = n + m$ be the total treatment-record count and $S = \alpha + \beta$ the sum of the two nuisance smoothness orders. Define the tuning denominator Δ , nuisance-smoothness cutoff S_{crit} , and total-record comparison exponent q_* by

$$\Delta = 2\gamma + d + \frac{\gamma d}{S}, \quad S_{\text{crit}} = \frac{\gamma d}{2\gamma + d}, \quad q_* = \frac{\gamma d}{S(2\gamma + d)} = \frac{S_{\text{crit}}}{S}.$$

The supplied-propensity reference order $r_o(n)$ and the unequal-channel upper-rate expression $r_{\text{up}}(n, m)$ are

$$r_o(n) = n^{-\gamma/(2\gamma+d)}, \quad r_{\text{up}}(n, m) = \max \left\{ r_o(n), (nN)^{-\gamma/\Delta} \right\}.$$

These expressions distinguish a term indexed by the response-bearing sample size from a term indexed by the product of the labeled and total treatment-record counts. Their statistical interpretation is supplied by [Theorems 1](#) to [3](#).

The notation also makes the sample-size comparison transparent. Algebraically,

$$(nN)^{-\gamma/\Delta} \leq r_o(n) \iff N \geq n^{q_*}.$$

When $S \geq S_{\text{crit}}$, the exponent satisfies $q_* \leq 1$, so $N \geq n$ meets this comparison. For $S < S_{\text{crit}}$, one has $q_* > 1$, and the comparison requires total treatment-record growth beyond the labeled count. This division organizes the precision and sample-requirement statements that follow.

4 Estimation with outcome-free records

The estimator allocates labeled and outcome-free records to two independent statistical roles. One role supplies response information; the other supplies treatment-assignment information. A local polynomial fit estimates the contrast at the target profile x_0 , while a finer projection corrects its empirical moments for the unknown propensity and control mean. Separate localization and projection scales allow the construction to use the smoothness of the effect and the combined smoothness of the nuisance functions.

We first specify the neighborhood associated with the localization scale h and its probability measure.

Definition 7. [synth_3] (Localization cube and uniform localization measure) *For an integer $d \geq 1$, let $x_0 = (1/2, \dots, 1/2) \in [0, 1]^d$. For $0 < h \leq 1/2$, define the localization cube of side length h , centered at x_0 , by*

$$C_h = x_0 + [-h/2, h/2]^d.$$

The uniform probability measure ν_h on C_h has density w_h with respect to d -dimensional Lebesgue measure dx on $[0, 1]^d$:

$$w_h(x) = h^{-d} \mathbf{1}\{x \in C_h\}, \quad d\nu_h(x) = w_h(x) dx.$$

Equivalently, for every Borel set $E \subseteq [0, 1]^d$,

$$\nu_h(E) = h^{-d} \int_{E \cap C_h} dx.$$

The localization cube C_h lies inside the covariate domain. Its uniform probability measure ν_h defines the inner product for both polynomial spaces below. Under [Assumption 1](#), weighting an observed covariate by the localization density w_h converts a population average into an integral under this local probability measure.

The coarse basis r consists of scaled tensor-product Legendre polynomials of total degree at most two. This degree accommodates local polynomial approximation throughout the stipulated effect-smoothness range.

Definition 8. [synth_1] (Scaled coarse polynomial basis) *The coarse polynomial basis vector r , for dimension $d \in \mathbb{N}$ and scale $h \in \mathbb{R}$, is defined at each $x \in \mathbb{R}^d$ by*

$$r(x) = \left(\prod_{i=1}^d \begin{cases} 1, & \kappa_i = 0, \\ \sqrt{12} \frac{x_i - \frac{1}{2}}{h}, & \kappa_i = 1, \\ \sqrt{5} \left(6 \left(\frac{x_i - \frac{1}{2}}{h} \right)^2 - \frac{1}{2} \right), & \kappa_i = 2 \end{cases} \right)_{\kappa \in \mathbb{N}_0^d: \sum_{i=1}^d \kappa_i \leq 2}.$$

Division by zero is interpreted as zero, so this definition also specifies r at $h = 0$.

Definition 9. [synth_7] (Localized cellwise polynomial basis) *For $d \geq 1$, $0 < h \leq 1/2$, and an integer $J \geq 1$, let $x_0 = (1/2, \dots, 1/2)$, $C_h = x_0 + [-h/2, h/2]^d$, and $\delta = h/J$. Partition each coordinate interval of C_h into J intervals of length δ , taking each interval half-open on the right*

except for the last interval, which includes the outer right endpoint. Their Cartesian products form the partition $\{C_{\mathbf{j}} : \mathbf{j} \in \{0, \dots, J-1\}^d\}$. The center of $C_{\mathbf{j}}$ is

$$c_{\mathbf{j}} = x_0 - \frac{h}{2}\mathbf{1} + \delta \left(\mathbf{j} + \frac{1}{2}\mathbf{1} \right).$$

Write

$$P_0(u) = 1, \quad P_1(u) = u, \quad P_2(u) = \frac{3u^2 - 1}{2}.$$

For each cell index $\mathbf{j}$ and multi-index $\kappa \in \mathbb{N}_0^d$ with $|\kappa| \leq 2$, define

$$b_{\mathbf{j},\kappa}(x) = J^{d/2} \mathbf{1}\{x \in C_{\mathbf{j}}\} \prod_{v=1}^d \sqrt{2\kappa_v + 1} P_{\kappa_v} \left(\frac{2(x_v - c_{\mathbf{j},v})}{\delta} \right).$$

The vector $\mathbf{b}_J(x)$ collects these functions, ordered lexicographically by cell index and then by multi-index. Its dimension is

$$k = J^d \binom{d+2}{2}.$$

These functions form the positive-leading-coefficient cellwise scaled Legendre orthonormal basis, under $d\nu_h(x) = h^{-d} \mathbf{1}\{x \in C_h\} dx$, of the space V_J of polynomials of total degree at most two on each partition cell. Set $\mathbf{b}_J(x) = 0$ for $x \notin C_h$. Thus the associated kernel is uniquely specified by

$$K_J(x, x') = \mathbf{b}_J(x)^\top \mathbf{b}_J(x').$$

The vector r is the coarse orthonormal basis and has $q = \binom{d+2}{2}$ coordinates. Write $r_0 = r(x_0)$ for its value at the target profile. A coefficient vector $v \in \mathbb{R}^q$ represents the coarse polynomial $p_v(x) = r(x)^\top v$; evaluating the fitted contrast therefore amounts to evaluating its coefficient vector against r_0 . The prescribed fine basis is $\mathbf{b}_J$ from the preceding definition. It has $k = qJ^d$ coordinates and spans the already specified cellwise space V_J ; the associated projection is defined next.

Definition 10. [def:projection] (Projection Π_J and kernel K_J)

$$\delta = \frac{h}{J}, \quad K_J(x, x') = \mathbf{b}_J(x)^\top \mathbf{b}_J(x'), \quad \Pi_J f(x) = \int K_J(x, x') f(x') d\nu_h(x').$$

Here δ is the fine-cell side length, h is the localization scale, J is the grid resolution, $\mathbf{b}_J$ is the cellwise orthonormal fine polynomial basis vector, and ν_h is the uniform probability measure on the localization cube C_h . Weighted basis expressions are zero outside C_h .

The projection kernel K_J represents the orthogonal projection Π_J onto V_J . The coarse fit operates across the entire localization cube, whereas the projection resolves nuisance variation on cells of side length δ . Refining the grid from J to a multiple J' of J subdivides every cell, so $V_J \subseteq V_{J'}$ and the correction space is enriched; at every resolution J , V_J contains the coarse polynomials, so the projection reproduces them.

We next assign the original observations to independent roles under the sampling law in [Assumption 9](#).

The concatenated treatment sequence $\mathcal{D}^T$ retains the covariate coordinate X_i^T and treatment coordinate A_i^T of each record. The outcome-role count ℓ is proportional to the labeled sample

size, while the treatment-role count t is proportional to the total treatment-record count. Their index sets, I_L and I_T , select disjoint records. Thus additional outcome-free observations enlarge the treatment role while the response information remains tied to the labeled sample. The split also makes products of empirical averages across roles suitable for estimating products of population moments.

To interpret the response correction, write

$$\eta_P(x) = \mathbb{E}_P[Y \mid X = x] = \mu_{0,P}(x) + e_P(x)\tau_P(x),$$

using the designated witnesses from [Definition 3](#). At the population level, residualizing treatment against its propensity gives

$$\mathbb{E}_P[(A - e_P(X))Y \mid X = x] = e_P(x)\{1 - e_P(x)\}\tau_P(x).$$

The empirical construction implements this local moment relation through a finite projection and independent roles. Its outcome projection uses the observed response, and its treatment role supplies the assignment factor in the projection correction.

Algorithm 1 (Record roles $\mathcal{D}^T, I_L, I_T$)

Given the original dataset $\mathcal{D}_{n,m}$ of n labeled records and m independent outcome-free treatment records, form the two roles as follows.

1. Concatenate the covariate-treatment pairs, with total treatment-record count N :

$$N = n + m, \quad \mathcal{D}^T = ((X_i^T, A_i^T))_{i=1}^N, \quad (X_i^T, A_i^T) = \begin{cases} (X_i, A_i), & 1 \leq i \leq n, \\ (\tilde{X}_{i-n}, \tilde{A}_{i-n}), & n < i \leq N. \end{cases}$$

Here X denotes the covariate vector and A the binary treatment indicator; tildes denote auxiliary records.

2. Define the outcome-role count ℓ , treatment-role count t , and their index sets:

$$\ell = \lfloor n/2 \rfloor, \quad t = N - \ell, \quad I_L = \{1, \dots, \ell\}, \quad I_T = \{\ell + 1, \dots, N\}.$$

3. Output the outcome-role labeled records and treatment-role covariate-treatment pairs:

$$((X_j, A_j, Y_j))_{j \in I_L}, \quad ((X_i^T, A_i^T))_{i \in I_T}.$$

Here Y_j is the observed response. Outcomes from treatment-role records are discarded.

The empirical outcome projection $\hat{\eta}_J^L$ estimates $\Pi_J \eta_P$. In the response vector $\hat{R}_{h,J}$, the first average estimates the treated-response moment and the rectangular average supplies its projected treatment-response correction. The raw matrix $\hat{Q}_{h,J}^{\text{raw}}$ applies the corresponding correction to the local polynomial design. Symmetrization produces the matrix $\hat{Q}_{h,J}$ used for inversion.

Define the population matrix $Q_{P,h,J}$ and population response vector $\bar{R}$ by

$$Q_{P,h,J} = \mathbb{E}_P[\hat{Q}_{h,J}], \quad \bar{R} = \mathbb{E}_P[\hat{R}_{h,J}],$$

where expectations are taken under the original sampling experiment. These are the population counterparts of the projected local estimating equations. Polynomial reproduction links those equations to approximation of the contrast, and overlap supplies the population positivity used to control inversion; the respective statements are [Lemmas 1](#) and [2](#).

For the sample decision, an eigenvalue guard supplies a criterion for inversion at the public overlap level. We use the clipping operator

$$\text{clip}_{[-1,1]}(z) = \max\{-1, \min\{1, z\}\},$$

which projects a real number onto the contrast range implied by the probability-valued arm means in [Definition 3](#). The guard level is

$$g(\varepsilon) = \frac{\varepsilon(1 - \varepsilon)}{2},$$

half the population Gram margin supplied by overlap. The complete decision combines the empirical moments, guard, and clipping.

Algorithm 2 (Empirical moments $\hat{R}_{h,J}, \hat{Q}_{h,J}$)

Given the original dataset $\mathcal{D}_{n,m}$, localization scale h , grid resolution J , localization cube C_h , localization density w_h , coarse polynomial basis vector r , and fine projection kernel K_J , compute the moments as follows.

1. Form the independent outcome and treatment roles, with counts and index sets

$$N = n + m, \quad \ell = \lfloor n/2 \rfloor, \quad t = N - \ell, \quad I_L = \{1, \dots, \ell\}, \quad I_T = \{\ell + 1, \dots, N\}.$$

The treatment sequence consists of the covariate-treatment pairs

$$(X_i^T, A_i^T) = \begin{cases} (X_i, A_i), & 1 \leq i \leq n, \\ (\tilde{X}_{i-n}, \tilde{A}_{i-n}), & n < i \leq N. \end{cases}$$

Use labeled records (X_j, A_j, Y_j) for $j \in I_L$ in the outcome role and pairs (X_i^T, A_i^T) for $i \in I_T$ in the treatment role. Treatment-role outcomes are discarded. Weighted basis expressions are zero outside C_h .

2. Define the outcome-role empirical projection:

$$\hat{\eta}_J^L(x) = \begin{cases} \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j) K_J(x, X_j) Y_j, & x \in C_h, \\ 0, & x \notin C_h. \end{cases}$$

3. Define the empirical response vector:

$$\begin{aligned} \hat{R}_{h,J} &= \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j) r(X_j) A_j Y_j - \frac{1}{t} \sum_{i \in I_T} w_h(X_i^T) r(X_i^T) A_i^T \hat{\eta}_J^L(X_i^T) \\ &= \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j) r(X_j) A_j Y_j - \frac{1}{t\ell} \sum_{i \in I_T} \sum_{j \in I_L} w_h(X_i^T) w_h(X_j) r(X_i^T) A_i^T K_J(X_i^T, X_j) Y_j. \end{aligned}$$

The response factor in the rectangular correction is Y_j , and the first summand uses $A_j Y_j$. The nested and rectangular expressions define the same vector.

4. Define the unsymmetrized empirical matrix:

$$\begin{aligned}\widehat{Q}_{h,J}^{\text{raw}} &= \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j) r(X_j) r(X_j)^\top A_j \\ &\quad - \frac{1}{t\ell} \sum_{i \in I_T} \sum_{j \in I_L} w_h(X_i^T) w_h(X_j) r(X_i^T) r(X_j)^\top A_i^T A_j K_J(X_i^T, X_j).\end{aligned}$$

5. Output the response vector and symmetrized matrix:

$$\left(\widehat{R}_{h,J}, \widehat{Q}_{h,J} \right), \quad \widehat{Q}_{h,J} = \frac{\widehat{Q}_{h,J}^{\text{raw}} + (\widehat{Q}_{h,J}^{\text{raw}})^\top}{2}.$$

All inputs are observed in $\mathcal{D}_{n,m}$, including when $m = 0$.

The guarded decision $\widehat{T}_{h,J}$ evaluates the fitted polynomial when the empirical matrix meets the eigenvalue cutoff and returns zero otherwise. Clipping bounds the decision on both branches. Together, these operations permit a risk guarantee over every admissible localization scale and grid resolution, including choices with sparse local observations.

The public choices of localization scale h_* and grid resolution J_* balance approximation and sampling error. They determine the tuned decision $\widehat{T}_{\text{up}}$ using the sample sizes and the public class parameters.

Definition 11. `[def:estimator]` (Guarded local polynomial decision $\widehat{T}_{h,J}$) *The local polynomial decision $\widehat{T}_{h,J}$ is defined by*

$$\widehat{T}_{h,J} = \begin{cases} \text{clip}_{[-1,1]}(r_0^\top \widehat{Q}_{h,J}^{-1} \widehat{R}_{h,J}), & \lambda_{\min}(\widehat{Q}_{h,J}) \geq g(\varepsilon), \\ 0, & \lambda_{\min}(\widehat{Q}_{h,J}) < g(\varepsilon). \end{cases}$$

Here r_0 is the coarse polynomial basis vector evaluated at the target profile, $\widehat{Q}_{h,J}$ is the symmetrized corrected empirical matrix, and $\widehat{R}_{h,J}$ is the corrected empirical response vector. The operator $\text{clip}_{[-1,1]}$ projects a real value onto $[-1, 1]$, and $\lambda_{\min}$ denotes the smallest eigenvalue. For the public overlap level ε , the inversion threshold is

$$g(\varepsilon) = \frac{\varepsilon(1 - \varepsilon)}{2}.$$

When $S \geq \gamma$, the grid choice is $J_* = 1$. When $S < \gamma$, the grid becomes finer as the localization scale shrinks, making δ^S comparable in order to h^γ . Thus nuisance approximation can use a finer spatial scale than the local effect fit. The localization choice then balances these approximation terms against the two sampling terms stated below.

For use in the decision-theoretic comparison, the same estimator is recorded as the original-record decision T_* .

Definition 12. `[def:tuning]` (Public tuning $h_*, J_*, \widehat{T}_{\text{up}}$) *The localization scale h_* , grid resolution J_* , and tuned decision $\widehat{T}_{\text{up}}$ are*

$$h_* = \frac{1}{2} \max \left\{ n^{-1/(2\gamma+d)}, (nN)^{-1/\Delta} \right\}, \quad J_* = \left\lceil h_*^{-\max\{0, \gamma/S-1\}} \right\rceil, \quad \widehat{T}_{\text{up}} = \widehat{T}_{h_*, J_*},$$

where N is the total treatment-record count, S is the sum of the propensity and control-mean Hölder orders, and Δ is the denominator governing the unequal-channel rate exponent.

Definition 13. [def:sharp-decision] (Sharp-rate decision T_*)

$$T_*(\mathcal{D}_{n,m}, U) = \widehat{T}_{h_*, J_*}(\mathcal{D}_{n,m}) = \widehat{T}_{\text{up}}(\mathcal{D}_{n,m}).$$

For every original dataset $\mathcal{D}_{n,m}$ and independent uniform randomizer U , this decision uses the prescribed empirical moments, empirical eigenvalue cutoff, clipping, role split, and public tuning parameters h_* and J_* . Its value is independent of U .

The following guarantee gives the construction's four statistical terms once: effect approximation h^γ , nuisance approximation $\delta^{\alpha+\beta}$, localized response sampling $(nh^d)^{-1/2}$, and rectangular sampling $(nNh^d\delta^d)^{-1/2}$. The public tuning balances them. The approximation, positivity, and sampling ingredients are isolated in [Lemmas 1 to 3](#); their proofs are organized in [Section A](#).

Theorem 1. [thm:rectangular-upper] (Rectangular estimation upper bounds) *Fix public parameters satisfying*

- **(Dimension.)** d is an integer with $d \geq 1$.
- **(Smoothness.)** $0 < \alpha \leq 1$, $0 < \beta \leq 1$, and $1 \leq \gamma \leq 3$.
- **(Radius.)** $L > 1/2$.
- **(Overlap.)** $0 < \varepsilon < 1/2$.

Let $\mathcal{M}$ be the primitive class at these fixed parameters from [Definition 3](#), and use the absolute-error risk from [Definition 4](#). There exists a finite constant $C > 0$, depending only on these public parameters, with both of the following guarantees.

For every pair of integers $n \geq 2$ and $m \geq 0$, every $h \in (0, 1/2]$, and every integer $J \geq 1$, the estimator $\widehat{T}_{h,J}$ from [Definition 11](#), viewed as a decision on the original dataset and randomizer that ignores the randomizer, is Borel measurable and satisfies, for every $P \in \mathcal{M}$,

$$\mathcal{R}_{n,m}(\widehat{T}_{h,J}, P) \leq C \min \left\{ 1, h^\gamma + (h/J)^{\alpha+\beta} + (nh^d)^{-1/2} + (n(n+m)h^d(h/J)^d)^{-1/2} \right\}.$$

Define the supplied-propensity point-estimation order and the denominator governing the unequal-channel rate exponent by

$$r_o(n) = n^{-\gamma/(2\gamma+d)}, \quad \Delta = 2\gamma + d + \frac{\gamma d}{\alpha + \beta},$$

and define the upper rate by

$$r_{\text{up}}(n, m) = \max \left\{ r_o(n), (n(n+m))^{-\gamma/\Delta} \right\}.$$

For every pair of integers $n \geq 2$ and $m \geq 0$, the publicly tuned decision $\widehat{T}_{\text{up}}$ from [Definition 12](#), likewise viewed as a decision that ignores the randomizer, is Borel measurable and satisfies, for every $P \in \mathcal{M}$,

$$\mathcal{R}_{n,m}(\widehat{T}_{\text{up}}, P) \leq C r_{\text{up}}(n, m).$$

This theorem completes the main-body estimator construction. The approximation, sampling, and inversion arguments are organized in [Section A](#).

5 Sharp precision and sample requirements

The sharp precision bound separates the information supplied by response labels from that supplied by all covariate-treatment records. The rate and attaining decision are defined once in [Definitions 13](#) and [14](#). The following result gives their uniform minimax comparison, explicit sample-size regimes, and extension to randomized decisions.

Definition 14. `[def:sharp-rate]` (Sharp rate r_*) *The sharp rate at the public class parameters is*

$$r_*(n, m; d, \alpha, \beta, \gamma) = \max \left\{ n^{-\gamma/(2\gamma+d)}, [n(n+m)]^{-\gamma/(2\gamma+d+\gamma d/(\alpha+\beta))} \right\} = r_{\text{up}}(n, m),$$

where $r_{\text{up}}(n, m)$ is the maximum of the supplied-propensity and unequal-channel rate orders.

Theorem 2. `[thm:sharp-annotation-frontier]` (Sharp annotation frontier) *Fix the covariate dimension $d \in \mathbb{N}$, the propensity and control-mean smoothness orders α, β , the effect smoothness order γ , the Hölder radius L , and the overlap level ε , satisfying:*

- *(Dimension.)* $d \geq 1$.
- *(Nuisance smoothness.)* $0 < \alpha \leq 1$ and $0 < \beta \leq 1$.
- *(Effect smoothness.)* $1 \leq \gamma \leq 3$.
- *(Radius.)* $L > 1/2$.
- *(Overlap.)* $0 < \varepsilon < 1/2$.

Let $\mathcal{M}$ denote the primitive class at these fixed parameters, and let r_* , T_* , and $R(n, m)$ be the sharp rate, total data-only decision, and minimax risk of [Definitions 5](#), [13](#) and [14](#), respectively. There exist real constants $0 < c \leq C < \infty$, depending only on the fixed parameters, such that, simultaneously for every $n, m \in \mathbb{N}$ with $n \geq 2$, T_* is Borel measurable and

$$c r_* \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P) \leq C r_*.$$

Moreover, every Borel real-valued decision T using the original dataset and the independent uniform randomizer satisfies

$$c r_* \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P).$$

Define the total treatment-record count N , the combined nuisance smoothness S , the exponent denominator Δ , the smoothness threshold S_{crit} , the total-record growth exponent q_* , and the supplied-propensity order $r_o(n)$ by

$$\begin{aligned} N &= n + m, & S &= \alpha + \beta, & \Delta &= 2\gamma + d + \frac{\gamma d}{\alpha + \beta}, \\ S_{\text{crit}} &= \frac{\gamma d}{2\gamma + d}, & q_* &= \frac{\gamma d}{(\alpha + \beta)(2\gamma + d)}, & r_o(n) &= n^{-\gamma/(2\gamma+d)}. \end{aligned}$$

For these sample sizes, the sharp rate has the branches

$$r_* = \begin{cases} r_o(n), & S \geq S_{\text{crit}}, \\ (nN)^{-\gamma/\Delta}, & S < S_{\text{crit}}, \quad N \leq n^{q_*}, \\ r_o(n), & S < S_{\text{crit}}, \quad N \geq n^{q_*}. \end{cases}$$

Whenever $N = n^{q_*}$, the two rate formulas coincide:

$$(nN)^{-\gamma/\Delta} = r_o(n).$$

The two terms express complementary precision requirements. The supplied-propensity term reflects the labeled-sample requirement for estimating the smooth contrast at a point; the product term reflects the joint requirement on labeled and treatment-record counts when the nuisance functions are unknown. If $S \geq S_{\text{crit}}$, the labeled count already supports the supplied-propensity order. Equality $S = S_{\text{crit}}$ belongs to this regime: then $q_* = 1$, and the necessary inequality $N \geq n$ places every admissible sample pair at or beyond the saturation threshold. If $S < S_{\text{crit}}$, the exponent $q_* > 1$ creates an intermediate regime $n \leq N \leq n^{q_*}$, in which precision depends on the product nN . At $N = n^{q_*}$, the two terms coincide; for larger total counts, the supplied-propensity term governs the rate.

The lower bound concerns the original records and includes every Borel decision using the independent uniform randomizer. Thus the comparison applies to both deterministic and randomized estimation procedures. The original-record construction in [Theorem 4](#) underlies the sparse-range nuisance converse in [Lemma 6](#). That converse supplies the product-of-counts lower bound in the range where it governs precision. Combined with the supplied-propensity lower bound below, it yields the sharp comparison across all sample imbalances. The matching upper bound is attained by the publicly tuned decision through [Theorem 1](#).

To interpret saturation, the next result evaluates the same primitive class when the designated propensity function is supplied to the decision. This benchmark identifies the labeled-sample precision against which the value of outcome-free records is measured. Its attaining known-design decision uses

$$h = n^{-1/(2\gamma+d)}, \quad T^e(\mathcal{D}_{n,m}, U; e_P) = \text{clip}_{[-1,1]} \left[\frac{1}{n} \sum_{i=1}^n w_h(X_i) r_0^\top r(X_i) \left\{ \frac{A_i Y_i}{e_P(X_i)} - \frac{(1 - A_i) Y_i}{1 - e_P(X_i)} \right\} \right].$$

Here $r_0 = r(x_0)$, and w_h and the degree-two basis r use the centered localization measure from [Definitions 7](#) and [8](#). These formulas are defined for every bandwidth $h > 0$; the supplied-propensity smoother below uses only $h_n = n^{-1/(2\gamma+d)}$, which lies in $(0, 1)$ for $n \geq 2$ and can exceed $1/2$ for small n . This is a known-design polynomial evaluation average, not an empirical local least-squares fit.

Theorem 3. [thm:supplied-propensity] (Supplied-propensity point estimation rate) *Fix public parameters satisfying*

- (**Dimension.**) $d \in \mathbb{N}$ and $d \geq 1$.
- (**Nuisance smoothness.**) $0 < \alpha \leq 1$ and $0 < \beta \leq 1$.
- (**Effect smoothness.**) $1 \leq \gamma \leq 3$.
- (**Radius.**) $L > 1/2$.
- (**Overlap.**) $0 < \varepsilon < 1/2$.

Let $\mathcal{M}$ be the primitive class at these parameters in [Definition 3](#), and let $R^e(n, m)$ be the supplied-propensity minimax absolute-error risk in [Definition 6](#). Define the supplied-propensity rate and the evaluation profile by

$$r_o(n) = n^{-\gamma/(2\gamma+d)}, \quad x_0 = (1/2, \dots, 1/2).$$

There exist real constants $0 < c \leq C < \infty$, depending only on the fixed public parameters, such that, for every $n, m \in \mathbb{N}$ with $n \geq 2$,

$$c r_o(n) \leq R^e(n, m) \leq C r_o(n).$$

The upper bound is attained by the explicit degree-two smoother under the known uniform design, applied to the inverse-propensity pseudo-outcomes

$$\frac{A_i Y_i}{e_P(X_i)} - \frac{(1 - A_i) Y_i}{1 - e_P(X_i)}, \quad i = 1, \dots, n,$$

at bandwidth

$$h = n^{-1/(2\gamma+d)},$$

with final clipping to $[-1, 1]$. Here the smoother uses its localization weight and coarse polynomial basis at this bandwidth for every $h > 0$, including $h > 1/2$. For every $P \in \mathcal{M}$, the resulting decision T^e is measurable as a function of the dataset and randomizer when supplied with the designated admissible propensity e_P , satisfies

$$|T^e(\mathcal{D}_{n,m}, U; e_P)| \leq 1$$

for every dataset and randomizer value, and obeys

$$\mathbb{E}_{\mathbf{E}_{n,m}(P)} [|T^e(\mathcal{D}_{n,m}, U; e_P) - \tau_P(x_0)|] \leq C r_o(n).$$

The sampling law $\mathbf{E}_{n,m}(P)$ is the product law of n independent labeled records, m independent outcome-free covariate-treatment records, and an independent uniform randomizer U , as specified in [Assumption 9](#).

The same lower bound holds for the supplied-propensity minimax risk over the same-class subfamily with constant propensity and control mean:

$$c r_o(n) \leq \inf_{T^e} \sup_{\substack{P \in \mathcal{M} \\ e_P = \mu_0, P = 1/2}} \mathbb{E}_{\mathbf{E}_{n,m}(P)} [|T^e(\mathcal{D}_{n,m}, U; e_P) - \tau_P(x_0)|],$$

where the constant-witness equalities hold on the unit cube and the infimum ranges over the supplied-propensity decisions of [Definition 6](#).

The upper-bound localization formula uses the same centered cube and uniform localization measure as [Definition 7](#), evaluated at the single bandwidth $h_n = n^{-1/(2\gamma+d)} \in (0, 1)$, which may exceed $1/2$. Because the cube has side length $h_n < 1$ and is centered at $x_0 = (1/2, \dots, 1/2)$, it lies inside $[0, 1]^d$. The rectangular estimator in [Theorem 1](#) retains its stated restriction $0 < h \leq 1/2$.

With the propensity supplied, inverse-propensity weighting gives a response whose conditional mean is the causal contrast. Local polynomial estimation can therefore target the contrast's smoothness directly, yielding the order $r_o(n)$. The same-class lower bound with constant propensity and control mean isolates the remaining response information requirement: even within this subfamily, the minimax risk is bounded below by a constant times $r_o(n)$ for every auxiliary count. Since a supplied-propensity decision can also implement any decision based on the original data, this lower bound provides the labeled-sample floor in [Theorem 2](#).

The sample requirements for attaining this benchmark can be expressed as a region of admissible counts. For a tolerance ζ , the annotation region $\mathcal{B}_*(\zeta)$ records the pairs whose sharp rate lies within that tolerance of the supplied-propensity order.

Definition 15. `[def:annotation-region]` (Annotation region $\mathcal{B}_*(\zeta)$) *The annotation region at tolerance ζ is*

$$\mathcal{B}_*(\zeta) = \{(n, m) \in \mathbb{N}_{\geq 2} \times \mathbb{N}_0 : r_*(n, m; d, \alpha, \beta, \gamma) \leq \zeta r_o(n)\},$$

where r_* is the sharp rate and $r_o(n)$ is the supplied-propensity point-estimation order.

For $\zeta \geq 1$, the explicit rate gives the equivalent count condition

$$(n, m) \in \mathcal{B}_*(\zeta) \iff n + m \geq \zeta^{-\Delta/\gamma} n^{q_*}, \quad n \geq 2, \quad m \geq 0.$$

In particular, $\zeta = 1$ gives the exact rate-level saturation region. Allowing a fixed finite tolerance gives attainment of the supplied-propensity order up to constants. The complete sequence equivalence, strict-improvement criterion, and target-error conditions are collected in [Proposition 1](#) and proved in [Section C](#).

Attainment of the supplied-propensity order and strict improvement over the labeled-only order impose distinct count requirements. Under $S < S_{\text{crit}}$, strict improvement occurs exactly when $m_n/n \rightarrow +\infty$, whereas supplied-propensity attainment requires the total count to remain at least a positive constant times n^{q_*} . Consequently, diverging auxiliary-to-labeled ratios can improve the minimax order while leaving precision in the intermediate product-of-counts regime. A bounded auxiliary-to-labeled ratio preserves the labeled-only order up to constants. At and above the smoothness threshold, every auxiliary-size sequence attains the supplied-propensity order.

For acquisition planning, [Proposition 1](#) gives two simultaneous requirements. For a target absolute-error scale $0 < \eta < 1$, the labeled count must support point-estimation precision at scale $\eta^{-(2+d/\gamma)}$, and the product of labeled and total counts must support nuisance adjustment at scale $\eta^{-(2+d/\gamma+d/S)}$. These requirements are necessary up to a constant depending on $(d, \alpha, \beta, \gamma, L, \varepsilon)$ and sufficient up to another such constant for uniform expected absolute error at most η . Their rate-level form is exact. Thus increasing either count can satisfy the product requirement, while the labeled-count requirement specifies the response information needed at the target precision.

A one-dimensional example makes the threshold concrete. Set $d = 1$, $\gamma = 1$, and $\alpha = \beta = 1/8$, with any fixed $L > 1/2$ and $\varepsilon \in (0, 1/2)$. Then

$$S = \frac{1}{4}, \quad S_{\text{crit}} = \frac{1}{3}, \quad \Delta = 7, \quad q_* = \frac{4}{3},$$

and

$$r_*(n, m) = \max \left\{ n^{-1/3}, [n(n+m)]^{-1/7} \right\}.$$

The labeled-only rate is $n^{-2/7}$, and the exact branch boundary is $N = n^{4/3}$. Total counts of order $n^{4/3}$ therefore attain the supplied-propensity order up to constants. For example, with $n = 1000$, the branch boundary is $N = 10000$, corresponding to $m = 9000$ outcome-free records. These counts describe equality of the rate formulas; the minimax risk retains constants that may depend on L and ε . In the same example, the exact rate-level requirements for target scale η are

$$n \geq \eta^{-3}, \quad nN \geq \eta^{-7}.$$

Taking the labeled count at order η^{-3} consequently requires a total count at order η^{-4} , illustrating how response labels and routine covariate-treatment records jointly determine attainable precision.

6 Discussion and limitations

The acquisition region in [Definition 15](#) provides a benchmark for allocating response labels and routine covariate-treatment records under the model in [Definition 3](#) and [Assumption 9](#). It identifies sample-size pairs for which the sharp rate is within a specified factor of the supplied-propensity order. Through [Theorem 2](#), this rate comparison translates into a minimax precision comparison up to constants depending on the public class parameters. For an absolute-error target, [Proposition 1](#) instead gives simultaneous requirements on the labeled count and the product of labeled and total treatment-record counts. These requirements offer a basis for comparing acquisition plans once their costs are specified.

A chart-review setting illustrates the distinction between the two record types. A labeled record contains covariates, treatment, and a true response ascertained through review; an outcome-free record contains the covariates and treatment available from routine records. Adding a new independent labeled record changes (n, m) to $(n + 1, m)$ and increases the total count. Reviewing a response for one record already present in the auxiliary block changes the composition to $(n + 1, m - 1)$ at fixed total count; if the record index is fixed in advance, or chosen by a randomization independent of all records, the resulting dataset follows the stated sampling law at $(n + 1, m - 1)$. Adding a new routine record changes (n, m) to $(n, m + 1)$. The sample requirements benchmark these alternative compositions under the specified independent, common-population experiment; acquisition costs determine which transition is relevant in an application.

The two channels address distinct sources of statistical difficulty. Response labels carry information about the probability-valued conditional response means and the causal contrast. The arm-mean conditions in [Assumptions 7](#) and [8](#) place both means in $[0, 1]$, and [Theorem 3](#) characterizes the labeled-sample precision when the propensity function is supplied. Outcome-free records carry information about treatment assignment conditional on covariates. Their contribution enters the second component of the sharp rate in [Definition 14](#), where the product of labeled and total counts governs the cost of nuisance adjustment. Increasing the routine-record count can reduce that component until the supplied-propensity order determines the overall rate.

Public nuisance regularity determines how much routine information is needed to reach this benchmark. The rate exponent depends on the sum of the propensity and control-mean Hölder orders, while the effect Hölder order governs the supplied-propensity precision. Greater nuisance regularity can reduce the total-record growth needed at a given labeled count. In the sufficiently smooth regime characterized by [Theorem 2](#), labeled records alone attain the supplied-propensity order; in the lower-regularity regime, additional outcome-free records can improve the minimax order until saturation. These comparisons concern rates, with constants depending on the fixed class parameters. Because the regularity parameters are public, they also determine the estimator’s localization and projection choices in [Definition 12](#). Acquisition planning and estimator tuning thus use the same description of the statistical model.

6.1 Limitations and future work

The analysis uses a known uniform covariate distribution on the unit cube, a fixed public overlap level $\varepsilon \in (0, 1/2)$, binary responses with arm means in $[0, 1]$, and independent record blocks from a common population. Extending the acquisition benchmark to general, potentially unknown covariate distributions is a direction for future work. Covariate-dependent or adaptive selection

of records for review would also change the labeled-record law and requires a sampling or missingness model beyond the experiment studied here. Covariate shift between reviewed and routine records would introduce an additional transport problem. These extensions would help connect the benchmark to populations with uneven support and heterogeneous record availability.

Unknown smoothness raises a separate question of adaptation. The present tuning and acquisition comparisons use public Hölder orders and a public radius. Procedures that select tuning parameters from the data would need risk guarantees over multiple regularity classes, together with an assessment of any adaptation cost. Uncertainty quantification also requires further analysis: the expected absolute-error bounds characterize point-estimation precision, whereas confidence intervals require coverage guarantees and control of approximation bias.

Alternative losses and causal targets may change the allocation requirements. Squared-error loss, integrated estimation of the treatment-effect function, average treatment effects, and policy-value targets each call for their own precision benchmark. Richer auxiliary information is another direction, including response proxies, repeated measurements, and additional variables predictive of the response. Such information would enlarge the observation model beyond covariate-treatment pairs and require an explicit account of how the auxiliary measurements relate to the true response.

Appendices

A Upper-bound proofs

The upper bounds in [Theorems 1](#) and [3](#) combine local polynomial approximation with control of the corresponding empirical moments. For the rectangular estimator, the two-scale construction in [Definition 10](#) and the public role split in [Algorithm 1](#) are statistical inputs: the coarse polynomial targets the smooth contrast, the fine projection controls nuisance approximation, and the independent roles separate response information from treatment-assignment information. The auxiliary results below follow this dependency order.

A.1 Local approximation and projection

Fix a primitive law P from [Definition 3](#) and admissible localization and resolution inputs h, J . Retain the notation $\bar{R}$ for the expected response vector and $Q_{P,h,J}$ for the expected symmetrized matrix:

$$\bar{R} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{R}_{h,J}, \quad Q_{P,h,J} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{Q}_{h,J}.$$

Both expectations concern the empirical moments in [Algorithm 2](#). Let p denote the Taylor polynomial of τ_P at x_0 , with degree $p_* = \lceil \gamma \rceil - 1$, and let ϑ denote its coefficient vector in the coarse basis. Since $1 \leq \gamma \leq 3$, this Taylor degree lies within the degree-two polynomial space used by the estimator. The following result collects the approximation and projection properties needed for the population estimating equations and their sampling bounds.

Lemma 1. [[lem:local-polynomial-projection-facts](#)] (Local polynomial projection bounds) *Fix public parameters $d \in \mathbb{N}$ and $\alpha, \beta, \gamma, L, \varepsilon \in \mathbb{R}$ satisfying:*

- (*Dimension.*) $d \geq 1$.
- (*Smoothness.*) $0 < \alpha \leq 1$, $0 < \beta \leq 1$, and $1 \leq \gamma \leq 3$.

- **(Radius.)** $L > 1/2$.
- **(Overlap.)** $0 < \varepsilon < 1/2$.

There exists a finite constant $C = C(d, \alpha, \beta, \gamma, L, \varepsilon) > 0$ such that, for every primitive law $P \in \mathcal{M}(d, \alpha, \beta, \gamma, L, \varepsilon)$ as defined in [Definition 3](#), every $h \in \mathbb{R}$ with $0 < h \leq 1/2$, and every integer $J \geq 1$, the following conclusions hold for the bases and projection of [Definition 10](#).

Write $x_0 = (1/2, \dots, 1/2)$ for the evaluation center, $C_h = [1/2 - h/2, 1/2 + h/2]^d$ for the localization cube, and ν_h for its uniform probability measure. Let p be the Taylor polynomial of the canonical contrast τ_P at x_0 , with Taylor degree $p_* = \lceil \gamma \rceil - 1$:

$$p(x) = \sum_{\substack{\kappa \in \mathbb{N}^d \\ \sum_{i=1}^d \kappa_i \leq p_*}} \frac{\partial^\kappa \tau_P(x_0)}{\prod_{i=1}^d \kappa_i!} \prod_{i=1}^d (x_i - 1/2)^{\kappa_i}.$$

There exists a coarse coefficient vector ϑ whose represented polynomial

$$p_\vartheta(x) = r(x)^\top \vartheta$$

satisfies

$$\sup_{x \in C_h} |\tau_P(x) - p_\vartheta(x)| \leq Ch^\gamma, \quad \|\vartheta\|_2 \leq C, \quad p_\vartheta(x_0) = \tau_P(x_0),$$

and

$$p_\vartheta(x) = p(x) \quad \text{for every } x \in C_h.$$

The coarse basis vector r and fine projection kernel K_J satisfy

$$\sup_{x \in C_h} \|r(x)\|_2 \leq C, \quad \sup_{x \in C_h} \int |K_J(x, x')| d\nu_h(x') \leq C,$$

and

$$\iint K_J(x, x')^2 d\nu_h(x) d\nu_h(x') = qJ^d,$$

where q is the number of coarse multi-indices of total degree at most two:

$$q = \# \left\{ \kappa \in \mathbb{N}^d : \sum_{i=1}^d \kappa_i \leq 2 \right\}.$$

Finally, with fine-cell side length $\delta = h/J$, every coarse basis coordinate r_u satisfies

$$\|e_P r_u - \Pi_J(e_P r_u)\|_{L_2(\nu_h)} \leq C\delta^\alpha,$$

where e_P is the designated admissible propensity witness, and the designated admissible control-mean witness $\mu_{0,P}$ satisfies

$$\|\mu_{0,P} - \Pi_J \mu_{0,P}\|_{L_2(\nu_h)} \leq C\delta^\beta.$$

Proof of [Lemma 1](#). All constants constructed below depend only on the fixed public parameters. Write

$$\begin{aligned} \mathcal{U}_{\text{coarse}} &= \left\{ \kappa \in \mathbb{N}^d : |\kappa| \leq 2 \right\}, \quad |\kappa| = \sum_{i=1}^d \kappa_i, \quad \kappa! = \prod_{i=1}^d \kappa_i!, \quad q = |\mathcal{U}_{\text{coarse}}|, \\ \mathsf{L}_0(v) &= 1, \quad \mathsf{L}_1(v) = \sqrt{12}v, \quad \mathsf{L}_2(v) = \sqrt{5}(6v^2 - 1/2) \quad (v \in \mathbb{R}). \end{aligned}$$

Thus the coarse coordinates are

$$r_\kappa(x) = \prod_{i=1}^d \mathbf{L}_{\kappa_i} \left(\frac{x_i - 1/2}{h} \right), \quad p_\vartheta(x) = \sum_{\kappa \in \mathcal{U}_{\text{coarse}}} r_\kappa(x) \vartheta_\kappa \quad (x \in \mathbb{R}^d).$$

We use the designated functions supplied by [Definition 3](#) and the localized measure and projection of [Definition 10](#).

1. *Taylor coefficients and approximation.* We first construct a positive constant C_T , uniform over admitted laws and localization scales.

If $\gamma = 1$, then $p_* = 0$. Let

$$\mathbf{0} = (0, \dots, 0), \quad \vartheta_\kappa = \begin{cases} \tau_P(x_0), & \kappa = \mathbf{0}, \\ 0, & \kappa \neq \mathbf{0}, \end{cases} \quad C_T = L(\sqrt{d} + 1).$$

Since $r_{\mathbf{0}} = 1$, this gives

$$p_\vartheta(x) = \tau_P(x_0) = p(x).$$

By [Definitions 1](#) and [3](#),

$$|\tau_P(x_0)| \leq L, \quad |\tau_P(x) - \tau_P(x')| \leq L\|x - x'\|_2 \quad (x, x' \in [0, 1]^d).$$

For equal arguments the second inequality follows from its zero left-hand side; for distinct arguments it is the order-zero Hölder bound. Moreover, for $x \in C_h$,

$$\|x - x_0\|_2^2 = \sum_{i=1}^d |x_i - 1/2|^2 \leq dh^2.$$

The inclusion $C_h \subset [0, 1]^d$ follows from $h \leq 1/2$. Consequently,

$$|\tau_P(x) - p_\vartheta(x)| \leq L\sqrt{d}h \leq C_T h, \quad \|\vartheta\|_2 = |\tau_P(x_0)| \leq L \leq C_T.$$

The representation also gives $p_\vartheta(x_0) = \tau_P(x_0)$.

Suppose now that $\gamma > 1$. Define

$$p_* = \lceil \gamma \rceil - 1, \quad s_T = \gamma - p_*, \quad D_{\text{dim}} = d + 1.$$

The public range of γ implies

$$p_* \in \{1, 2\}, \quad 0 < s_T \leq 1, \quad p_* + s_T = \gamma.$$

In particular, at $\gamma = 2$ and $\gamma = 3$, the pairs (p_*, s_T) are $(1, 1)$ and $(2, 1)$, respectively. The same two definitions supply continuous derivatives through order p_* and the bounds

$$\begin{aligned} |\partial^\kappa \tau_P(x)| &\leq L & (|\kappa| \leq p_*, \ x \in [0, 1]^d), \\ |\partial^\kappa \tau_P(x) - \partial^\kappa \tau_P(x')| &\leq L\|x - x'\|_2^{s_T} & (|\kappa| = p_*, \ x, x' \in [0, 1]^d). \end{aligned}$$

The second inequality includes equal arguments, where both sides vanish.

For $x \in C_h$, set

$$z_x = x - x_0, \quad F_x(v) = \tau_P(x_0 + vz_x) \quad (0 \leq v \leq 1).$$

The segment lies in the design cube. The chain rule, together with symmetry of the continuous derivatives, yields

$$F_x^{(p_*)}(v) = \sum_{(i_1, \dots, i_{p_*}) \in \{1, \dots, d\}^{p_*}} \left(\prod_{j=1}^{p_*} z_{x, i_j} \right) (\partial_{i_1} \cdots \partial_{i_{p_*}} \tau_P)(x_0 + vz_x).$$

Each coordinate of z_x has absolute value at most $\|z_x\|_2$. There are $d^{p_*} \leq D_{\dim}^{p_*}$ terms, so the top-derivative modulus gives

$$\left| F_x^{(p_*)}(v) - F_x^{(p_*)}(v') \right| \leq D_{\dim}^{p_*} L \|z_x\|_2^{p_* + s_T} |v - v'|^{s_T} \quad (0 \leq v, v' < 1).$$

Grouping the coordinate derivatives by their multiplicities gives

$$p(x) = \sum_{j=0}^{p_*} \frac{F_x^{(j)}(0)}{j!} = \sum_{|\kappa| \leq p_*} \frac{\partial^\kappa \tau_P(x_0)}{\kappa!} \prod_{i=1}^d (x_i - 1/2)^{\kappa_i}.$$

The Lagrange remainder for the Taylor polynomial of degree $p_* - 1$ provides $v_x \in (0, 1)$ such that, after subtracting the degree- p_* term,

$$\tau_P(x) - p(x) = \frac{F_x^{(p_*)}(v_x) - F_x^{(p_*)}(0)}{p_*!}.$$

Since $p_*! \geq 1$ and $v_x^{s_T} \leq 1$, this implies

$$|\tau_P(x) - p(x)| \leq D_{\dim}^{p_*} L \|z_x\|_2^\gamma.$$

To obtain a constant uniform in the fractional order, observe that

$$\|z_x\|_2 \leq \sqrt{d} \frac{h}{2} \leq D_{\dim} h, \quad D_{\dim} \geq 1, \quad \gamma = p_* + s_T \leq p_* + 1.$$

Hence, with

$$C_{\text{rem}} = D_{\dim}^{p_*} D_{\dim}^{p_*+1},$$

we have

$$|\tau_P(x) - p(x)| \leq C_{\text{rem}} L h^\gamma \quad (x \in C_h).$$

We next represent p in the coarse basis by an explicit change of coefficients. Define the matrix, whose rows and columns are indexed by $0, 1, 2$,

$$\mathbf{M}_h = \begin{pmatrix} 1 & 0 & 0 \\ 0 & h/\sqrt{12} & 0 \\ h^2/12 & 0 & h^2/(6\sqrt{5}) \end{pmatrix}.$$

Direct substitution gives

$$(hv)^j = \sum_{\ell=0}^2 (\mathbf{M}_h)_{j\ell} \mathbf{L}_\ell(v) \quad (j \in \{0, 1, 2\}, v \in \mathbb{R}).$$

Its entries satisfy

$$(\mathbf{M}_h)_{j\ell} = 0 \quad (\ell > j), \quad |(\mathbf{M}_h)_{j\ell}| \leq h^j.$$

Put

$$\begin{aligned} \mathcal{U}_T &= \{\kappa \in \mathbb{N}^d : |\kappa| \leq p_*\}, \quad N_T = |\mathcal{U}_T|, \\ a_{T,\kappa} &= \frac{\partial^\kappa \tau_P(x_0)}{\kappa!} \quad (\kappa \in \mathcal{U}_T), \\ \mathbf{M}_h[\kappa, \kappa'] &= \prod_{i=1}^d (\mathbf{M}_h)_{\kappa_i \kappa'_i} \quad (\kappa, \kappa' \in \mathcal{U}_T). \end{aligned}$$

Since $\kappa! \geq 1$,

$$|a_{T,\kappa}| \leq L, \quad |\mathbf{M}_h[\kappa, \kappa']| \leq h^{|\kappa|} \leq 1.$$

Expanding the product of the one-dimensional identities gives

$$\prod_{i=1}^d (x_i - 1/2)^{\kappa_i} = \sum_{\kappa' \in \mathcal{U}_T} \mathbf{M}_h[\kappa, \kappa'] r_{\kappa'}(x).$$

Indeed, a nonzero term has $\kappa'_i \leq \kappa_i$ in every coordinate, so its total degree remains at most p_* .

Define the coarse vector by zero extension:

$$\vartheta_{\kappa'} = \begin{cases} \sum_{\kappa \in \mathcal{U}_T} a_{T,\kappa} \mathbf{M}_h[\kappa, \kappa'], & \kappa' \in \mathcal{U}_T, \\ 0, & \kappa' \in \mathcal{U}_{\text{coarse}} \setminus \mathcal{U}_T. \end{cases}$$

Exchanging the finite sums proves

$$p_\vartheta(x) = p(x) \quad (x \in \mathbb{R}^d).$$

Furthermore,

$$|\vartheta_{\kappa'}| \leq N_T L, \quad \sum_{\kappa' \in \mathcal{U}_{\text{coarse}}} \vartheta_{\kappa'}^2 \leq N_T (N_T L)^2.$$

Thus, defining

$$C_{\text{coef}} = \sqrt{N_T} N_T + 1, \quad C_T = L \max\{C_{\text{rem}}, C_{\text{coef}}\},$$

we obtain

$$\|\vartheta\|_2 \leq C_{\text{coef}} L \leq C_T, \quad \sup_{x \in C_h} |\tau_P(x) - p_\vartheta(x)| \leq C_T h^\gamma.$$

At x_0 , every positive-degree centered monomial vanishes and the zero-degree coefficient is $\tau_P(x_0)$; therefore $p_\vartheta(x_0) = \tau_P(x_0)$. Both cases construct a finite positive C_T .

2. *Basis and kernel bounds.* For $|v| \leq 1/2$, the three reference polynomials satisfy

$$|\mathbf{L}_0(v)| = 1, \quad |\mathbf{L}_1(v)| \leq \frac{\sqrt{12}}{2} \leq 4, \quad |\mathbf{L}_2(v)| \leq \sqrt{5} \leq 4.$$

For the last inequality we used $0 \leq v^2 \leq 1/4$, which gives $|6v^2 - 1/2| \leq 1$. It follows that

$$|r_\kappa(x)| \leq 4^d, \quad \|r(x)\|_2^2 \leq q(4^d)^2 \quad (x \in C_h).$$

Here is the fine-cell apparatus used for the kernel calculations. Define

$$\delta = \frac{h}{J} > 0, \quad \mathcal{Z}_J = \{0, \dots, J-1\}^d, \quad c_{\mathbf{j},i} = 1/2 - h/2 + (j_i + 1/2)\delta \quad (\mathbf{j} \in \mathcal{Z}_J).$$

For each coordinate, let

$$I_{j_i} = \begin{cases} [1/2 - h/2 + j_i\delta, 1/2 - h/2 + (j_i + 1)\delta), & j_i < J-1, \\ [1/2 - h/2 + (J-1)\delta, 1/2 + h/2], & j_i = J-1, \end{cases} \quad D_{\mathbf{j}} = \prod_{i=1}^d I_{j_i}.$$

These cells partition C_h , with every terminal face assigned to the last interval. To see coverage explicitly, for $x \in C_h$ define

$$t_{x,i} = \frac{x_i - (1/2 - h/2)}{\delta} \in [0, J], \quad j_i(x) = \begin{cases} J-1, & t_{x,i} = J, \\ \lfloor t_{x,i} \rfloor, & t_{x,i} < J. \end{cases}$$

Then $x \in D_{\mathbf{j}(x)}$. Distinct indices give disjoint cells: in a coordinate where their indices differ, the smaller interval's right endpoint is at or before the larger interval's left endpoint, and the smaller interval excludes its right endpoint. These conventions also apply when $J = 1$.

The fine coordinates are

$$b_{\mathbf{j},\kappa}(x) = \begin{cases} \sqrt{J^d} \prod_{i=1}^d \mathsf{L}_{\kappa_i} \left(\frac{x_i - c_{\mathbf{j},i}}{\delta} \right), & x \in D_{\mathbf{j}}, \\ 0, & x \notin D_{\mathbf{j}}, \end{cases} \quad (\mathbf{j} \in \mathcal{Z}_J, \kappa \in \mathcal{U}_{\text{coarse}}).$$

In particular, all fine coordinates vanish outside C_h . The orthonormality specified in [Definition 10](#) reads

$$\int b_{\mathbf{j},\kappa}(x) b_{\mathbf{j}',\kappa'}(x) d\nu_h(x) = \begin{cases} 1, & (\mathbf{j}, \kappa) = (\mathbf{j}', \kappa'), \\ 0, & (\mathbf{j}, \kappa) \neq (\mathbf{j}', \kappa'). \end{cases}$$

Its normalization can also be seen directly: the reference integrals satisfy

$$\int_{-1/2}^{1/2} \mathsf{L}_j(v) \mathsf{L}_\ell(v) dv = \mathbf{1}_{\{j=\ell\}} \quad (j, \ell \in \{0, 1, 2\}),$$

by integrating the displayed polynomials. On a common cell, the change of variables $x_i = c_{\mathbf{j},i} + \delta v_i$ contributes

$$h^{-d} J^d \delta^d = 1.$$

Different cells have disjoint supports, and the assigned faces have zero Lebesgue measure.

The coordinate bound gives, for every $x \in \mathbb{R}^d$,

$$|b_{\mathbf{j},\kappa}(x)| \leq \sqrt{J^d} 4^d.$$

The constant coordinate on a cell is

$$b_{\mathbf{j},\mathbf{0}}(x) = \sqrt{J^d} \mathbf{1}_{D_{\mathbf{j}}}(x).$$

Consequently,

$$|b_{\mathbf{j},\kappa}(x)| \leq \frac{4^d}{\sqrt{J^d}} b_{\mathbf{j},\mathbf{0}}(x)^2, \quad \int |b_{\mathbf{j},\kappa}(x)| d\nu_h(x) \leq \frac{4^d}{\sqrt{J^d}},$$

where the second inequality uses $\int b_{\mathbf{j},\mathbf{0}}^2 d\nu_h = 1$.

If $x \in D_{\mathbf{j}}$, disjointness reduces the kernel to

$$K_J(x, x') = \sum_{\kappa \in \mathcal{U}_{\text{coarse}}} b_{\mathbf{j},\kappa}(x) b_{\mathbf{j},\kappa}(x').$$

The triangle inequality and the preceding bounds therefore give

$$\begin{aligned} \int |K_J(x, x')| d\nu_h(x') &\leq \sum_{\kappa \in \mathcal{U}_{\text{coarse}}} |b_{\mathbf{j},\kappa}(x)| \int |b_{\mathbf{j},\kappa}(x')| d\nu_h(x') \\ &\leq q(\sqrt{J^d} 4^d) \frac{4^d}{\sqrt{J^d}} = q(4^d)^2. \end{aligned}$$

For x outside all cells, $K_J(x, x') = 0$, so the same bound holds there.

There are $|\mathcal{Z}_J| = J^d$ cells and q coordinates per cell; hence the fine index set has qJ^d elements. Expanding the squared finite kernel and factoring the product integrals yields

$$\begin{aligned} \iint K_J(x, x')^2 d\nu_h(x) d\nu_h(x') &= \sum_{\mathbf{j},\kappa} \sum_{\mathbf{j}',\kappa'} \left(\int b_{\mathbf{j},\kappa}(x) b_{\mathbf{j}',\kappa'}(x) d\nu_h(x) \right)^2 \\ &= qJ^d = k. \end{aligned}$$

All terms are integrable by the square integrability of the basis coordinates. Define

$$C_B = \max\{\sqrt{q} 4^d + 1, q(4^d)^2 + 1\} > 0.$$

We have proved

$$\sup_{x \in C_h} \|r(x)\|_2 \leq C_B, \quad \sup_{x \in C_h} \int |K_J(x, x')| d\nu_h(x') \leq C_B,$$

together with the exact squared-kernel identity.

3. *Control-mean residual.* For square-integrable functions, write

$$\|f\|_h = \left(\int f(x)^2 d\nu_h(x) \right)^{1/2}.$$

The coarse constant coordinate equals one, so coarse orthonormality gives

$$\nu_h(C_h) = \int 1 d\nu_h = \int r_{\mathbf{0}}(x)^2 d\nu_h(x) = 1.$$

Thus ν_h is a probability measure supported on C_h .

The centers satisfy $c_{\mathbf{j}} \in D_{\mathbf{j}}$. For two points in the same cell, coordinatewise interval bounds give

$$|x_i - x'_i| \leq \delta, \quad \|x - x'\|_2^2 \leq d\delta^2 \quad (x, x' \in D_{\mathbf{j}}).$$

By [Definitions 1](#) and [3](#) and $0 < \beta \leq 1$,

$$|\mu_{0,P}(x)| \leq L, \quad |\mu_{0,P}(x) - \mu_{0,P}(x')| \leq L\|x - x'\|_2^\beta \quad (x, x' \in C_h).$$

Indeed, $\lceil \beta \rceil - 1 = 0$, so the Hölder norm bounds the function and its order-zero modulus; equal arguments satisfy the modulus inequality directly.

Define the fine coefficient vector and its expansion by

$$v_{\mathbf{j},\kappa}^{(0,J)} = \begin{cases} \mu_{0,P}(c_{\mathbf{j}})/\sqrt{J^d}, & \kappa = \mathbf{0}, \\ 0, & \kappa \neq \mathbf{0}, \end{cases} \quad f_{0,J}(x) = \sum_{\mathbf{j},\kappa} b_{\mathbf{j},\kappa}(x) v_{\mathbf{j},\kappa}^{(0,J)}.$$

The cell assignments imply

$$f_{0,J}(x) = \begin{cases} \mu_{0,P}(c_{\mathbf{j}}), & x \in D_{\mathbf{j}}, \\ 0, & x \notin C_h. \end{cases}$$

Set

$$E_{0,J} = L(\sqrt{d})^\beta \delta^\beta \geq 0.$$

For $x \in D_{\mathbf{j}}$, the modulus and distance bounds give

$$|\mu_{0,P}(x) - f_{0,J}(x)| \leq L\|x - c_{\mathbf{j}}\|_2^\beta \leq L(\sqrt{d}\delta)^\beta = E_{0,J}.$$

Because ν_h is a probability measure,

$$\|\mu_{0,P} - f_{0,J}\|_h^2 \leq \int E_{0,J}^2 d\nu_h = E_{0,J}^2.$$

We now derive the projection comparison used here and in the next step. For any $f \in L_2(\nu_h)$ and any real fine coefficient vector $v_J = (v_{J,\mathbf{j},\kappa})$, put

$$f_J(x) = \sum_{\mathbf{j},\kappa} b_{\mathbf{j},\kappa}(x) v_{J,\mathbf{j},\kappa}.$$

Expanding the kernel integral in [Definition 10](#) gives

$$\Pi_J f(x) = \sum_{\mathbf{j},\kappa} b_{\mathbf{j},\kappa}(x) \int b_{\mathbf{j},\kappa}(x') f(x') d\nu_h(x').$$

Orthonormality then implies

$$\int b_{\mathbf{j},\kappa}(x) f_J(x) d\nu_h(x) = v_{J,\mathbf{j},\kappa}, \quad \Pi_J(f - f_J) = \Pi_J f - f_J,$$

and

$$\int b_{\mathbf{j},\kappa}(x) (f(x) - \Pi_J f(x)) d\nu_h(x) = 0.$$

The last identity makes the cross term in the squared norm vanish, giving

$$\|f\|_h^2 = \|\Pi_J f\|_h^2 + \|f - \Pi_J f\|_h^2.$$

Apply this identity to $f - f_J$ and use the displayed projection identity:

$$\|f - f_J\|_h^2 = \|\Pi_J f - f_J\|_h^2 + \|f - \Pi_J f\|_h^2 \geq \|f - \Pi_J f\|_h^2.$$

The bounded control mean and its finite-basis approximant are square integrable. Taking $f = \mu_{0,P}$ and $f_J = f_{0,J}$, and then taking square roots, proves

$$\|\mu_{0,P} - \Pi_J \mu_{0,P}\|_h \leq E_{0,J}.$$

Therefore the positive constant

$$C_m = L(\sqrt{d})^\beta + 1$$

satisfies

$$\|\mu_{0,P} - \Pi_J \mu_{0,P}\|_h \leq C_m \delta^\beta.$$

4. *Propensity-weighted residuals.* Similarly, $0 < \alpha \leq 1$ and [Definitions 1](#) and [3](#) give

$$|e_P(x)| \leq L, \quad |e_P(x) - e_P(x')| \leq L\|x - x'\|_2^\alpha \quad (x, x' \in C_h).$$

Fix $u \in \mathcal{U}_{\text{coarse}}$, and define

$$f_{e,u}(x) = e_P(x)r_u(x) \quad (x \in \mathbb{R}^d).$$

On C_h , $|f_{e,u}(x)| \leq L4^d$, so it is square integrable under ν_h .

We construct a fine-basis expansion that equals $e_P(c_j)r_u$ on each cell. Define

$$\zeta_{j,i} = \frac{c_{j,i} - 1/2}{h}, \quad \sigma_J = \frac{1}{J}, \quad A_{j,i} = \begin{pmatrix} 1 & 0 & 0 \\ \sqrt{12}\zeta_{j,i} & \sigma_J & 0 \\ \sqrt{5}(6\zeta_{j,i}^2 + \sigma_J^2/2 - 1/2) & \sqrt{5}\sqrt{12}\zeta_{j,i}\sigma_J & \sigma_J^2 \end{pmatrix}.$$

Rows and columns are indexed by 0, 1, 2. Substitution into the three reference polynomials verifies

$$\mathbb{L}_j(\zeta_{j,i} + \sigma_J v_{\text{aff}}) = \sum_{\ell=0}^j (A_{j,i})_{j\ell} \mathbb{L}_\ell(v_{\text{aff}}) \quad (j \in \{0, 1, 2\}, v_{\text{aff}} \in \mathbb{R}).$$

For $x \in D_j$,

$$\frac{x_i - 1/2}{h} = \zeta_{j,i} + \sigma_J \frac{x_i - c_{j,i}}{\delta}.$$

Define the tensor coefficients

$$A_j[u, \kappa] = \prod_{i=1}^d (A_{j,i})_{u_i \kappa_i} \quad (\kappa \in \mathcal{U}_{\text{coarse}}).$$

Expanding the product of the affine identities gives

$$r_u(x) = \sum_{\kappa \in \mathcal{U}_{\text{coarse}}} A_j[u, \kappa] \prod_{i=1}^d \mathbb{L}_{\kappa_i} \left(\frac{x_i - c_{j,i}}{\delta} \right).$$

Only indices with $\kappa_i \leq u_i$ contribute, and hence every contributing index has total degree at most two.

Now set

$$v_{\mathbf{j},\kappa}^{(e,u,J)} = \frac{e_P(c_{\mathbf{j}})A_{\mathbf{j}}[u,\kappa]}{\sqrt{J^d}}, \quad f_{e,u,J}(x) = \sum_{\mathbf{j},\kappa} b_{\mathbf{j},\kappa}(x)v_{\mathbf{j},\kappa}^{(e,u,J)}.$$

Disjointness of the cells and the tensor identity imply the everywhere defined formula

$$f_{e,u,J}(x) = \begin{cases} e_P(c_{\mathbf{j}})r_u(x), & x \in D_{\mathbf{j}}, \\ 0, & x \notin C_h. \end{cases}$$

Define

$$E_{e,J} = L(\sqrt{d})^\alpha 4^d \delta^\alpha \geq 0.$$

For $x \in D_{\mathbf{j}}$,

$$\begin{aligned} |f_{e,u}(x) - f_{e,u,J}(x)| &= |e_P(x) - e_P(c_{\mathbf{j}})| |r_u(x)| \\ &\leq L\|x - c_{\mathbf{j}}\|_2^\alpha 4^d \\ &\leq L(\sqrt{d}\delta)^\alpha 4^d = E_{e,J}. \end{aligned}$$

The probability normalization therefore gives

$$\|f_{e,u} - f_{e,u,J}\|_h^2 \leq E_{e,J}^2.$$

Applying the projection comparison from the preceding step to this fine-basis expansion yields

$$\|e_P r_u - \Pi_J(e_P r_u)\|_h \leq \|f_{e,u} - f_{e,u,J}\|_h \leq E_{e,J}.$$

Thus the positive constant

$$C_e = L(\sqrt{d})^\alpha 4^d + 1$$

satisfies, uniformly in u ,

$$\|e_P r_u - \Pi_J(e_P r_u)\|_h \leq C_e \delta^\alpha.$$

5. *A common residual constant.* Define

$$C_R = \max\{C_e, C_m\} > 0.$$

Since $\delta^\alpha, \delta^\beta \geq 0$, the preceding estimates imply

$$\|e_P r_u - \Pi_J(e_P r_u)\|_h \leq C_R \delta^\alpha \quad (u \in \mathcal{U}_{\text{coarse}}), \quad \|\mu_{0,P} - \Pi_J \mu_{0,P}\|_h \leq C_R \delta^\beta.$$

6. *Combination of the bounds.* Finally, choose

$$C = \max\{C_T, \max\{C_B, C_R\}\}.$$

This is finite, positive, and depends only on the public parameters. The inequalities

$$C_T \leq C, \quad C_B \leq C, \quad C_R \leq C, \quad h^\gamma \geq 0, \quad \delta^\alpha \geq 0, \quad \delta^\beta \geq 0$$

allow every bound above to be enlarged to the claimed constant C . The same Taylor coefficient vector retains its exact center value and its equality with p on C_h , while the squared-kernel integral retains the exact value $qJ^d = k$. These conclusions hold for every admitted P , every $0 < h \leq 1/2$, and every integer $J \geq 1$.

□

The Taylor approximation controls variation of the contrast across the localization cube while reproducing its value exactly at the evaluation profile. The bounded coefficient vector and coarse basis provide uniform control of polynomial evaluation. For the fine projection, the integral bound controls its action on bounded approximation errors, whereas the squared-kernel identity records the sampling cost of its rank. The two nuisance residual bounds concern different functions: the propensity multiplied by a coarse basis coordinate, and the control mean. Their respective orders account for the combined nuisance approximation term in [Theorem 1](#).

A.2 Population positivity

The corrected empirical matrix has a population counterpart whose stability is determined by overlap and orthogonal projection geometry. The next identity separates its quadratic form into an overlap-weighted polynomial norm and a squared projection residual.

Lemma 2. [lem:population-positivity] (Population quadratic form positivity) *Suppose the following conditions hold:*

- **(Public parameters.)** *The dimension d is an integer with $d \geq 1$, the smoothness parameters satisfy $0 < \alpha, \beta \leq 1$ and $1 \leq \gamma \leq 3$, the radius satisfies $L > 1/2$, and the overlap parameter satisfies $0 < \varepsilon < 1/2$.*
- **(Primitive law.)** *The law P belongs to the primitive class $\mathcal{M}$ at these public parameters, as specified in [Definition 3](#). Let e_P be its designated admissible propensity witness.*
- **(Sample sizes.)** *The sample counts n, m are nonnegative integers with $n \geq 2$.*
- **(Localization and projection.)** *The localization scale satisfies $0 < h \leq 1/2$, and the grid resolution J is an integer with $J \geq 1$. Use the projection Π_J from [Definition 10](#).*

Let ν_h be the uniform localization measure on the localization cube C_h , with

$$d\nu_h(x) = h^{-d} \mathbf{1}_{C_h}(x) dx.$$

For every coefficient vector $v \in \mathbb{R}^q$, indexed by the multi-indices of total degree at most two, define its coarse-basis polynomial and the population matrix by

$$p_v(x) = \sum_u r_u(x) v_u, \quad Q_{P,h,J} = \mathbb{E}_{\mathbf{E}_{n,m}(P)}[\hat{Q}_{h,J}],$$

where r_u are the scaled orthonormal coarse polynomial basis functions and $\hat{Q}_{h,J}$ is the symmetrized corrected empirical matrix. Then

$$\begin{aligned} v^\top Q_{P,h,J} v &= \int e_P(x) (1 - e_P(x)) p_v(x)^2 d\nu_h(x) \\ &\quad + \int [e_P(x) p_v(x) - \Pi_J(e_P p_v)(x)]^2 d\nu_h(x) \\ &\geq \varepsilon(1 - \varepsilon) \sum_u v_u^2. \end{aligned}$$

Proof of Lemma 2. Fix $v \in \mathbb{R}^q$. We compute the population entries, contract them with v , and then obtain the energy decomposition and lower bound.

1. The localization density and fine-basis coordinates are

$$w_h(x) = h^{-d} \mathbf{1}_{C_h}(x), \quad k = qJ^d, \quad b_{J,z_f}(x) = [\mathbf{b}_J(x)]_{z_f}, \quad z_f \in \{1, \dots, k\}.$$

Here $x \in \mathbb{R}^d$, and the density is as in Definition 7. The role counts from Algorithm 1 satisfy

$$\ell = \lfloor n/2 \rfloor \geq 1, \quad t = n + m - \ell \geq n - \ell \geq 1.$$

For fixed h, J , the localized basis products are bounded and measurable, so all expectations and integrals below are finite.

For coarse indices $u, w \in \{1, \dots, q\}$, expand the kernel in Definition 10 inside the raw matrix of Algorithm 2. This gives

$$\begin{aligned} \hat{Q}_{h,J;uw}^{\text{raw}} &= \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j) r_u(X_j) r_w(X_j) A_j \\ &\quad - \sum_{z_f=1}^k \frac{1}{t\ell} \sum_{i \in I_T} \sum_{j \in I_L} [w_h(X_i^T) r_u(X_i^T) b_{J,z_f}(X_i^T) A_i^T] [w_h(X_j) r_w(X_j) b_{J,z_f}(X_j) A_j]. \end{aligned}$$

Under the product sampling law of Assumption 9, each covariate-treatment pair has marginal law P_{XA} . Each treatment-role record is independent of each outcome-role record: for a labeled treatment-role record their indices are distinct, and for an auxiliary record independence follows from the product sampling law. Taking expectations term by term therefore yields

$$\begin{aligned} \mathbb{E}_{\mathbf{E}_{n,m}(P)}[\hat{Q}_{h,J;uw}^{\text{raw}}] &= \frac{\ell}{\ell} \mathbb{E}_P[w_h(X) r_u(X) r_w(X) A] \\ &\quad - \sum_{z_f=1}^k \frac{t\ell}{t\ell} \mathbb{E}_P[w_h(X) r_u(X) b_{J,z_f}(X) A] \mathbb{E}_P[w_h(X) r_w(X) b_{J,z_f}(X) A]. \end{aligned}$$

For any of the functions

$$f_{\text{loc}} \in \{r_u r_w, r_u b_{J,z_f}, r_w b_{J,z_f}\}, \quad f_{\text{loc}} : \mathbb{R}^d \longrightarrow \mathbb{R},$$

the designated conditional treatment mean and uniform design in Definition 3 give

$$\begin{aligned} \mathbb{E}_P[w_h(X) f_{\text{loc}}(X) A] &= \mathbb{E}_P[w_h(X) f_{\text{loc}}(X) e_P(X)] \\ &= h^{-d} \int_{C_h} f_{\text{loc}}(x) e_P(x) dx \\ &= \int f_{\text{loc}}(x) e_P(x) d\nu_h(x). \end{aligned}$$

The localization cube lies in $[0, 1]^d$, so these identities use the designated witness on its specified domain. Substitution gives a formula symmetric in u, w . Consequently, the symmetrization in Algorithm 2 preserves its expectation, and

$$\begin{aligned} (Q_{P,h,J})_{uw} &= \int r_u(x) e_P(x) r_w(x) d\nu_h(x) \\ &\quad - \sum_{z_f=1}^k \left(\int b_{J,z_f}(x) e_P(x) r_u(x) d\nu_h(x) \right) \left(\int b_{J,z_f}(x) e_P(x) r_w(x) d\nu_h(x) \right). \end{aligned}$$

2. Define the weighted coarse-basis coefficients by

$$c_{z_f, u} = \int b_{J, z_f}(x) e_P(x) r_u(x) d\nu_h(x), \quad 1 \leq z_f \leq k, \quad 1 \leq u \leq q.$$

Multiplying the entry formula by $v_u v_w$, summing, and using $p_v(x) = \sum_{u=1}^q r_u(x) v_u$, we obtain

$$\begin{aligned} v^\top Q_{P, h, J} v &= \int e_P(x) \left(\sum_{u=1}^q r_u(x) v_u \right)^2 d\nu_h(x) - \sum_{z_f=1}^k \left(\sum_{u=1}^q c_{z_f, u} v_u \right)^2 \\ &= \int e_P(x) p_v(x)^2 d\nu_h(x) - \sum_{z_f=1}^k \left(\sum_{u=1}^q c_{z_f, u} v_u \right)^2. \end{aligned}$$

Indeed, the integral term follows by expanding the finite double sum, and the correction follows from

$$\sum_{u, w=1}^q v_u v_w \sum_{z_f=1}^k c_{z_f, u} c_{z_f, w} = \sum_{z_f=1}^k \left(\sum_{u=1}^q c_{z_f, u} v_u \right)^2.$$

The finite kernel expansion in [Definition 10](#) and linearity of integration give

$$\begin{aligned} \Pi_J(e_P p_v)(x) &= \sum_{z_f=1}^k b_{J, z_f}(x) \int b_{J, z_f}(y) e_P(y) p_v(y) d\nu_h(y) \\ &= \sum_{z_f=1}^k b_{J, z_f}(x) \sum_{u=1}^q c_{z_f, u} v_u. \end{aligned}$$

The fine basis in that definition is orthonormal. Expanding the square and integrating therefore gives

$$\begin{aligned} \int [\Pi_J(e_P p_v)(x)]^2 d\nu_h(x) &= \sum_{z_f, z'_f=1}^k \left(\sum_{u=1}^q c_{z_f, u} v_u \right) \left(\sum_{w=1}^q c_{z'_f, w} v_w \right) \int b_{J, z_f} b_{J, z'_f} d\nu_h \\ &= \sum_{z_f=1}^k \left(\sum_{u=1}^q c_{z_f, u} v_u \right)^2. \end{aligned}$$

Thus

$$v^\top Q_{P, h, J} v = \int e_P p_v^2 d\nu_h - \int [\Pi_J(e_P p_v)]^2 d\nu_h.$$

3. The finite coarse expansion makes p_v square-integrable, and bounded overlap makes $e_P p_v$ square-integrable. The projection expansion just established also yields

$$\begin{aligned} \int e_P p_v \Pi_J(e_P p_v) d\nu_h &= \sum_{z_f=1}^k \left(\int b_{J, z_f} e_P p_v d\nu_h \right)^2 \\ &= \int [\Pi_J(e_P p_v)]^2 d\nu_h. \end{aligned}$$

Expanding the residual square and using this equality gives the Pythagorean identity

$$\int (e_P p_v)^2 d\nu_h = \int [\Pi_J(e_P p_v)]^2 d\nu_h + \int [e_P p_v - \Pi_J(e_P p_v)]^2 d\nu_h.$$

Meanwhile, the pointwise identity $e_P(1 - e_P)p_v^2 = e_P p_v^2 - (e_P p_v)^2$ gives

$$\int e_P(1 - e_P)p_v^2 d\nu_h = \int e_P p_v^2 d\nu_h - \int (e_P p_v)^2 d\nu_h.$$

Combining these two identities with the quadratic-form formula proves

$$v^\top Q_{P,h,J} v = \int e_P(1 - e_P)p_v^2 d\nu_h + \int [e_P p_v - \Pi_J(e_P p_v)]^2 d\nu_h.$$

4. The scaled coarse polynomials of [Definition 8](#) satisfy

$$\int r_u(x)r_w(x) d\nu_h(x) = \begin{cases} 1, & u = w, \\ 0, & u \neq w. \end{cases}$$

To see this, integrate their tensor products over the localization cube and rescale each coordinate to $[-1/2, 1/2]$. Direct integration of the three normalized one-dimensional polynomials in that definition gives unit squared norms and zero pairwise inner products. When $u = w$, every coordinate factor is one; when $u \neq w$, some multi-index coordinate differs and its factor is zero. Hence finite expansion gives

$$\int p_v(x)^2 d\nu_h(x) = \sum_{u,w=1}^q v_u v_w \int r_u r_w d\nu_h = \sum_{u=1}^q v_u^2.$$

Class membership supplies [Assumption 3](#). Since ν_h is supported on $C_h \subseteq [0, 1]^d$, for ν_h -almost every x ,

$$e_P(x)(1 - e_P(x)) - \varepsilon(1 - \varepsilon) = (e_P(x) - \varepsilon)(1 - \varepsilon - e_P(x)) \geq 0.$$

Multiplying by $p_v(x)^2$ and integrating therefore yields

$$\int e_P(1 - e_P)p_v^2 d\nu_h \geq \varepsilon(1 - \varepsilon) \int p_v^2 d\nu_h = \varepsilon(1 - \varepsilon) \sum_{u=1}^q v_u^2.$$

Finally,

$$\int [e_P p_v - \Pi_J(e_P p_v)]^2 d\nu_h \geq 0,$$

so the decomposition from the preceding step implies

$$v^\top Q_{P,h,J} v \geq \varepsilon(1 - \varepsilon) \sum_{u=1}^q v_u^2.$$

□

The overlap restriction in [Assumption 3](#) supplies the lower bound through the first integral. The second integral is nonnegative and makes the contribution of finite projection explicit. Thus the population matrix retains a fixed positivity margin at every admissible resolution. This margin supplies the population reference for the empirical eigenvalue guard in [Definition 11](#).

A.3 Rectangular sampling

The public split assigns disjoint observations to the outcome and treatment roles under [Assumption 9](#). Their counts in [Algorithm 1](#) are proportional to n and $N = n + m$, respectively, including when $m = 0$. Consequently, the sampling analysis distinguishes ordinary localized averages from the rectangular averages formed across the two roles. The next result controls both the response vector and the matrix used for inversion.

Lemma 3. [lem:rectangular-sampling-bound] (Rectangular sampling bound) *Fix the covariate dimension $d \in \mathbb{N}$, the smoothness parameters $\alpha, \beta, \gamma \in \mathbb{R}$, the Hölder radius $L \in \mathbb{R}$, and the overlap parameter $\varepsilon \in \mathbb{R}$, satisfying*

- *(Dimension.)* $d \geq 1$.
- *(Nuisance smoothness.)* $0 < \alpha \leq 1$ and $0 < \beta \leq 1$.
- *(Effect smoothness.)* $1 \leq \gamma \leq 3$.
- *(Radius.)* $L > 1/2$.
- *(Overlap parameter.)* $0 < \varepsilon < 1/2$.

There exists a finite constant $C = C(d, \alpha, \beta, \gamma, L, \varepsilon) > 0$ such that the following bound holds for every primitive law $P \in \mathcal{M}(d, \alpha, \beta, \gamma, L, \varepsilon)$ from [Definition 3](#), and every $n, m, J \in \mathbb{N}$ and $h \in \mathbb{R}$ satisfying

- *(Sample sizes.)* $n \geq 2$ and $m \geq 0$.
- *(Localization.)* $0 < h \leq 1/2$.
- *(Resolution.)* $J \geq 1$.

Take expectations under $\mathbb{E}_{n,m}(P)$, the product sampling law specified in [Assumption 9](#). Write N for the total treatment-record count, q for the number of polynomial multi-indices of total degree at most two, and k for the fine projection rank:

$$N = n + m, \quad q = \binom{d+2}{2}, \quad k = qJ^d.$$

For the empirical response vector $\hat{R}_{h,J}$ and symmetrized corrected empirical matrix $\hat{Q}_{h,J}$ from [Algorithm 2](#), define their population expectations by

$$\bar{R} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{R}_{h,J}, \quad Q_{P,h,J} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{Q}_{h,J}.$$

Then

$$\mathbb{E}_{\mathbb{E}_{n,m}(P)} \left\| \hat{R}_{h,J} - \bar{R} \right\|_2^2 + \mathbb{E}_{\mathbb{E}_{n,m}(P)} \left\| \hat{Q}_{h,J} - Q_{P,h,J} \right\|_F^2 \leq C \left\{ \frac{1}{nh^d} + \frac{k}{nNh^{2d}} \right\}.$$

Proof of [Lemma 3](#). 1. Define the dimension-dependent constants

$$\begin{aligned} M_d &= 4^{2d}, & D_d &= q4^{2d}, & C_{\text{ker},d} &= M_d^4(D_d^2 + 1), \\ C_{\text{coord},d} &= 6M_d^2 + 12C_{\text{ker},d}, & C &= (q + q^2 + 1)C_{\text{coord},d}. \end{aligned}$$

These constants are finite and positive. Fix an admitted law P and admissible n, m, h, J , and set

$$V_{n,m,h,J} = \frac{1}{nh^d} + \frac{k}{nNh^{2d}} \geq 0.$$

We will bound each response coordinate and each raw-matrix coordinate by $C_{\text{coord},d} V_{n,m,h,J}$, then sum the bounds after symmetrization.

2. We first establish the explicit basis bounds used in the coordinate estimates. Write

$$\mathcal{U}_d = \left\{ u \in \mathbb{N}^d : \sum_{i=1}^d u_i \leq 2 \right\}, \quad |\mathcal{U}_d| = q,$$

and define the reference polynomial factors, for $\xi_{\text{ref}} \in \mathbb{R}$, by

$$\begin{aligned} p_{\text{ref},0}(\xi_{\text{ref}}) &= 1, \\ p_{\text{ref},1}(\xi_{\text{ref}}) &= \sqrt{12} \xi_{\text{ref}}, \\ p_{\text{ref},2}(\xi_{\text{ref}}) &= \sqrt{5} \left(6\xi_{\text{ref}}^2 - \frac{1}{2} \right). \end{aligned}$$

For $|\xi_{\text{ref}}| \leq 1/2$,

$$0 \leq \xi_{\text{ref}}^2 \leq \frac{1}{4}, \quad \left| 6\xi_{\text{ref}}^2 - \frac{1}{2} \right| \leq 1,$$

and consequently

$$|p_{\text{ref},0}(\xi_{\text{ref}})| = 1 \leq 4, \quad |p_{\text{ref},1}(\xi_{\text{ref}})| \leq \frac{\sqrt{12}}{2} \leq 4, \quad |p_{\text{ref},2}(\xi_{\text{ref}})| \leq \sqrt{5} \leq 3 \leq 4.$$

The product formula in [Definition 8](#) therefore gives

$$|r_u(x)| \leq 4^d \quad (x \in C_h, u \in \mathcal{U}_d).$$

For completeness, the same factors give an explicit row-integral bound for the fine kernel. Set

$$\delta = \frac{h}{J}, \quad \mathcal{Z}_J = \{0, \dots, J-1\}^d, \quad x_{\text{cell},\mathbf{z},i} = \frac{1}{2} - \frac{h}{2} + \left(z_i + \frac{1}{2} \right) \delta \quad (\mathbf{z} \in \mathcal{Z}_J).$$

Let $C_{\text{cell},\mathbf{z}}$ be the product of the intervals

$$\left[\frac{1}{2} - \frac{h}{2} + z_i \delta, \frac{1}{2} - \frac{h}{2} + (z_i + 1) \delta \right),$$

with the right endpoint included when $z_i = J-1$. Thus these cells partition C_h . The fine basis coordinates of [Definition 10](#) are

$$b_{\mathbf{z},u}(x) = \sqrt{J^d} \mathbf{1}_{C_{\text{cell},\mathbf{z}}}(x) \prod_{i=1}^d p_{\text{ref},u_i} \left(\frac{x_i - x_{\text{cell},\mathbf{z},i}}{\delta} \right), \quad x \in \mathbb{R}^d, \quad (\mathbf{z}, u) \in \mathcal{Z}_J \times \mathcal{U}_d.$$

In particular, their values are zero outside their supporting cells, and

$$|b_{\mathbf{z},u}(x)| \leq \sqrt{J^d} 4^d.$$

With $\mathbf{0} = (0, \dots, 0)$, the constant coordinate satisfies

$$b_{\mathbf{z},\mathbf{0}}(x) = \sqrt{J^d} \mathbf{1}_{C_{\text{cell},\mathbf{z}}}(x), \quad \int b_{\mathbf{z},\mathbf{0}}(x)^2 d\nu_h(x) = 1,$$

where the last identity is its orthonormal normalization. Pointwise,

$$|b_{\mathbf{z},u}(x)| \leq \frac{4^d}{\sqrt{J^d}} b_{\mathbf{z},\mathbf{0}}(x)^2,$$

so integration yields

$$\int |b_{\mathbf{z},u}(x)| d\nu_h(x) \leq \frac{4^d}{\sqrt{J^d}}.$$

If $x \in C_{\text{cell},\mathbf{z}}$, only that cell contributes to the kernel, and hence

$$K_J(x, x') = \sum_{u \in \mathcal{U}_d} b_{\mathbf{z},u}(x) b_{\mathbf{z},u}(x').$$

It follows that

$$\begin{aligned} \int |K_J(x, x')| d\nu_h(x') &\leq \sum_{u \in \mathcal{U}_d} |b_{\mathbf{z},u}(x)| \int |b_{\mathbf{z},u}(x')| d\nu_h(x') \\ &\leq q(\sqrt{J^d} 4^d) \frac{4^d}{\sqrt{J^d}} = D_d. \end{aligned}$$

If x belongs to no fine cell, every fine basis coordinate at x is zero, and the same bound holds. Thus

$$\int |K_J(x, x')| d\nu_h(x') \leq D_d \quad (x \in \mathbb{R}^d).$$

The basis-product formula also gives

$$K_J(x, x') = K_J(x', x).$$

3. We now express the two coordinate types through a common marked kernel. Define the role counts

$$\ell = \lfloor n/2 \rfloor, \quad t = N - \ell.$$

Both are positive because $n \geq 2$ and $N \geq n$. Let W and Z be independent generic treatment-role and outcome-role records, respectively:

$$W = (X^T, A^T) \in [0, 1]^d \times \{0, 1\}, \quad Z = (X^L, A^L, Y^L) \in [0, 1]^d \times \{0, 1\}^2,$$

$$\mathcal{L}(W) = P_{XA}, \quad \mathcal{L}(Z) = \mathcal{L}_P(X, A, Y).$$

Their covariate marginals are uniform by [Definition 3](#). The product sampling law and the disjoint roles in [Algorithm 1](#) give independent records

$$W_i = (X_i^T, A_i^T) \quad (i \in I_T), \quad Z_j = (X_j, A_j, Y_j) \quad (j \in I_L),$$

independent within and between the two roles.

Use the coordinate tags

$$\diamond \in (\{\mathbf{R}\} \times \mathcal{U}_d) \sqcup (\{\mathbf{Q}\} \times \mathcal{U}_d \times \mathcal{U}_d).$$

For each tag, define the three mark functions by

$\diamond$	$f_{\diamond,0}(Z)$	$f_{\diamond,T}(W)$	$f_{\diamond,L}(Z)$
$(\mathbf{R}, u)$	$r_u(X^L)A^LY^L$	$r_u(X^T)A^T$	Y^L
$(\mathbf{Q}, u, v)$	$r_u(X^L)r_v(X^L)A^L$	$r_u(X^T)A^T$	$r_v(X^L)A^L$

on the full record domains displayed above. Binary marks have absolute value at most one. The coarse-coordinate bound from the preceding step therefore gives

$$\begin{aligned} |f_{\diamond,0}(Z)| &\leq M_d \quad (X^L \in C_h), \\ |f_{\diamond,T}(W)| &\leq M_d \quad (X^T \in C_h), \\ |f_{\diamond,L}(Z)| &\leq M_d \quad (X^L \in C_h). \end{aligned}$$

Define

$$\begin{aligned} H_{\diamond}(W, Z) &= w_h(X^T)w_h(X^L)K_J(X^T, X^L)f_{\diamond,T}(W)f_{\diamond,L}(Z), \\ \widehat{S}_{\diamond} &= \frac{1}{\ell} \sum_{j \in I_L} w_h(X_j)f_{\diamond,0}(Z_j), \quad \widehat{H}_{\diamond} = \frac{1}{t\ell} \sum_{i \in I_T} \sum_{j \in I_L} H_{\diamond}(W_i, Z_j), \quad F_{\diamond} = \widehat{S}_{\diamond} - \widehat{H}_{\diamond}. \end{aligned}$$

These expressions vanish termwise whenever a localization weight is zero. The rectangular formulas in [Algorithm 2](#) give exactly

$$F_{(\mathbf{R},u)} = (\widehat{R}_{h,J})_u, \quad F_{(\mathbf{Q},u,v)} = (\widehat{Q}_{h,J}^{\text{raw}})_{uv}.$$

In the response identity, expanding the empirical projection gives the rectangular formula on C_h ; outside C_h , its defining zero value and the zero localization weight give the same identity.

4. The localized mark and kernel energies admit explicit bounds. Since

$$w_h(x) = h^{-d} \mathbf{1}_{C_h}(x), \quad \mathbb{P}(X^L \in C_h) = \mathbb{P}(X^T \in C_h) = h^d,$$

uniform design gives

$$\mathbb{E} w_h(X^L)^2 = \mathbb{E} w_h(X^T)^2 = h^{-d}.$$

Thus

$$\mathbb{E} \left[(w_h(X^L)f_{\diamond,0}(Z))^2 \right] \leq \frac{M_d^2}{h^d}.$$

Similarly, bounding both marks and integrating over the two independent covariates gives

$$\begin{aligned} \mathbb{E} H_{\diamond}(W, Z)^2 &\leq M_d^4 h^{-4d} \iint_{C_h \times C_h} K_J(x, x')^2 dx dx' \\ &= \frac{M_d^4}{h^{2d}} \iint K_J(x, x')^2 d\nu_h(x) d\nu_h(x') \\ &= \frac{M_d^4 k}{h^{2d}}, \end{aligned}$$

using the kernel-square identity in [Lemma 1](#).

Define the integrated slices on the full record domains by

$$H_{\diamond,T}(W) = \mathbb{E}_Z H_{\diamond}(W, Z), \quad H_{\diamond,L}(Z) = \mathbb{E}_W H_{\diamond}(W, Z),$$

where each expectation integrates an independent record with the law specified above. The row-integral estimate gives

$$\begin{aligned} |\mathbb{E}_Z [w_h(X^L) K_J(x, X^L) f_{\diamond,L}(Z)]| &\leq M_d \mathbb{E}_Z [w_h(X^L) |K_J(x, X^L)|] \\ &= M_d \int |K_J(x, x')| d\nu_h(x') \\ &\leq M_d D_d. \end{aligned}$$

Consequently,

$$|H_{\diamond,T}(W)| \leq M_d^2 D_d w_h(X^T).$$

Kernel symmetry gives the corresponding bound with the roles reversed:

$$|H_{\diamond,L}(Z)| \leq M_d^2 D_d w_h(X^L).$$

Squaring and integrating yields

$$\mathbb{E} H_{\diamond,T}(W)^2 \leq \frac{M_d^4 D_d^2}{h^d}, \quad \mathbb{E} H_{\diamond,L}(Z)^2 \leq \frac{M_d^4 D_d^2}{h^d}.$$

All the displayed features and kernels are measurable and square-integrable. In particular, their means, centered squares, and finite-sum integrals are well defined.

5. We derive the rectangular variance formula used with these energies. Define

$$\begin{aligned} \mu_{\diamond} &= \mathbb{E} H_{\diamond}(W, Z), \\ H_{\diamond,0}(W, Z) &= H_{\diamond}(W, Z) - \mu_{\diamond}, \\ H_{\diamond,1}(W) &= H_{\diamond,T}(W) - \mu_{\diamond}, \\ H_{\diamond,2}(Z) &= H_{\diamond,L}(Z) - \mu_{\diamond}, \\ H_{\diamond,12}(W, Z) &= H_{\diamond,0}(W, Z) - H_{\diamond,1}(W) - H_{\diamond,2}(Z). \end{aligned}$$

Fubini's theorem gives

$$\mathbb{E} H_{\diamond,0} = \mathbb{E} H_{\diamond,1} = \mathbb{E} H_{\diamond,2} = 0,$$

and

$$\begin{aligned} \mathbb{E}[H_{\diamond,0} H_{\diamond,1}(W)] &= \mathbb{E} H_{\diamond,1}(W)^2, \\ \mathbb{E}[H_{\diamond,0} H_{\diamond,2}(Z)] &= \mathbb{E} H_{\diamond,2}(Z)^2, \\ \mathbb{E}[H_{\diamond,1}(W) H_{\diamond,2}(Z)] &= 0. \end{aligned}$$

Indeed, the first two identities follow by integrating the other record first, and the third follows from independence and centering. Expanding the residual square therefore gives

$$\mathbb{E} H_{\diamond,12}^2 = \mathbb{E} H_{\diamond,0}^2 - \mathbb{E} H_{\diamond,1}^2 - \mathbb{E} H_{\diamond,2}^2.$$

Also,

$$\mathbb{E} H_{\diamond,0}^2 = \mathbb{E} H_{\diamond}^2 - \mu_{\diamond}^2 \leq \mathbb{E} H_{\diamond}^2.$$

Hence

$$\mathbb{E}H_{\diamond,12}^2 \leq \mathbb{E}H_{\diamond}^2,$$

and centering each slice gives

$$\mathbb{E}H_{\diamond,1}^2 \leq \mathbb{E}H_{\diamond,T}^2, \quad \mathbb{E}H_{\diamond,2}^2 \leq \mathbb{E}H_{\diamond,L}^2.$$

To compute the rectangular second moment, recall the independent-sample identity. For $n_{\text{av}} \geq 1$ independent identically distributed square-integrable real variables $Z_{\text{av},j}$, define

$$\bar{Z}_{\text{av}} = \frac{1}{n_{\text{av}}} \sum_{j=1}^{n_{\text{av}}} Z_{\text{av},j}.$$

Expanding the square and factoring the mixed expectations gives

$$\mathbb{E}\bar{Z}_{\text{av}}^2 = \frac{1}{n_{\text{av}}} \mathbb{E}Z_{\text{av},1}^2 + \left(1 - \frac{1}{n_{\text{av}}}\right) (\mathbb{E}Z_{\text{av},1})^2.$$

Apply this identity first to the outcome-record average, with the treatment records fixed, and then to the treatment-record averages. Finite-sum integration commutes with taking the slice means. Since $H_{\diamond,0}$ has mean zero, the resulting identity is

$$\begin{aligned} \mathbb{E}(\hat{H}_{\diamond} - \mu_{\diamond})^2 &= \frac{\mathbb{E}H_{\diamond,0}^2}{t\ell} \\ &\quad + \frac{1}{t} \left(1 - \frac{1}{\ell}\right) \mathbb{E}H_{\diamond,1}^2 \\ &\quad + \left(1 - \frac{1}{t}\right) \frac{1}{\ell} \mathbb{E}H_{\diamond,2}^2. \end{aligned}$$

Substituting the residual-energy identity above reduces this to

$$\mathbb{E}(\hat{H}_{\diamond} - \mu_{\diamond})^2 = \frac{\mathbb{E}H_{\diamond,1}^2}{t} + \frac{\mathbb{E}H_{\diamond,2}^2}{\ell} + \frac{\mathbb{E}H_{\diamond,12}^2}{t\ell}.$$

The three preceding energy comparisons imply

$$\mathbb{E}(\hat{H}_{\diamond} - \mu_{\diamond})^2 \leq \frac{\mathbb{E}H_{\diamond,T}^2}{t} + \frac{\mathbb{E}H_{\diamond,L}^2}{\ell} + \frac{\mathbb{E}H_{\diamond}^2}{t\ell}.$$

6. We convert this inequality to the sampling envelope and then include the first-order term. The floor split satisfies

$$\ell \geq 1, \quad n \leq 2\ell + 1 \leq 3\ell, \quad 2\ell \leq n \leq N.$$

Therefore

$$\ell \geq \frac{n}{3}, \quad t = N - \ell \geq \frac{N}{2},$$

and

$$\frac{1}{\ell} \leq \frac{3}{n}, \quad \frac{1}{t} \leq \frac{2}{N} \leq \frac{2}{n}, \quad \frac{1}{t\ell} \leq \frac{6}{nN}.$$

These inequalities apply for every $m \geq 0$, including $m = 0$.

By the definition of $C_{\text{ker},d}$,

$$M_d^4 D_d^2 \leq C_{\text{ker},d}, \quad M_d^4 \leq C_{\text{ker},d}.$$

The slice and kernel bounds thus give

$$\begin{aligned} \mathbb{E}(\hat{H}_\diamond - \mu_\diamond)^2 &\leq \frac{5C_{\text{ker},d}}{nh^d} + \frac{6C_{\text{ker},d}k}{nNh^{2d}} \\ &\leq 6C_{\text{ker},d}V_{n,m,h,J}. \end{aligned}$$

The independent-sample mean identity also gives

$$\mathbb{E}\hat{S}_\diamond = \mathbb{E}[w_h(X^L)f_{\diamond,0}(Z)], \quad \mathbb{E}\hat{H}_\diamond = \mu_\diamond,$$

and

$$\begin{aligned} \mathbb{E}(\hat{S}_\diamond - \mathbb{E}\hat{S}_\diamond)^2 &= \frac{1}{\ell} \left\{ \mathbb{E} \left[(w_h(X^L)f_{\diamond,0}(Z))^2 \right] - (\mathbb{E}[w_h(X^L)f_{\diamond,0}(Z)])^2 \right\} \\ &\leq \frac{3M_d^2}{nh^d}. \end{aligned}$$

Linearity of expectation gives the centered-difference identity

$$F_\diamond - \mathbb{E}F_\diamond = (\hat{S}_\diamond - \mathbb{E}\hat{S}_\diamond) - (\hat{H}_\diamond - \mu_\diamond).$$

The pointwise inequality

$$(\xi_{\text{diff}} - \zeta_{\text{diff}})^2 \leq 2\xi_{\text{diff}}^2 + 2\zeta_{\text{diff}}^2 \quad (\xi_{\text{diff}}, \zeta_{\text{diff}} \in \mathbb{R})$$

follows from $(\xi_{\text{diff}} + \zeta_{\text{diff}})^2 \geq 0$. Applying it to the centered difference yields

$$\begin{aligned} \mathbb{E}(F_\diamond - \mathbb{E}F_\diamond)^2 &\leq \frac{6M_d^2}{nh^d} + 12C_{\text{ker},d}V_{n,m,h,J} \\ &\leq C_{\text{coord},d}V_{n,m,h,J}, \end{aligned}$$

because $1/(nh^d) \leq V_{n,m,h,J}$. This argument accommodates the shared outcome records in the two terms.

7. Applying the preceding estimate to the response tags gives

$$\mathbb{E} \left[((\hat{R}_{h,J})_u - \bar{R}_u)^2 \right] \leq C_{\text{coord},d}V_{n,m,h,J} \quad (u \in \mathcal{U}_d).$$

Applying it to the raw-matrix tags gives

$$\mathbb{E} \left[((\hat{Q}_{h,J}^{\text{raw}})_{uv} - \mathbb{E}(\hat{Q}_{h,J}^{\text{raw}})_{uv})^2 \right] \leq C_{\text{coord},d}V_{n,m,h,J} \quad (u, v \in \mathcal{U}_d).$$

The square-integrability established above applies to every coordinate in these two displays.

8. Define the centered raw matrix by

$$B_{\text{raw},uv} = (\hat{Q}_{h,J}^{\text{raw}})_{uv} - \mathbb{E}(\hat{Q}_{h,J}^{\text{raw}})_{uv}, \quad u, v \in \mathcal{U}_d.$$

The symmetrization formula in [Algorithm 2](#) and linearity of expectation give

$$(Q_{P,h,J})_{uv} = \frac{\mathbb{E}(\widehat{Q}_{h,J}^{\text{raw}})_{uv} + \mathbb{E}(\widehat{Q}_{h,J}^{\text{raw}})_{vu}}{2},$$

and therefore

$$(\widehat{Q}_{h,J} - Q_{P,h,J})_{uv} = \frac{B_{\text{raw},uv} + B_{\text{raw},vu}}{2}.$$

Every symmetrized coordinate is square-integrable as an average of two square-integrable raw coordinates.

For each sample,

$$\left(\frac{B_{\text{raw},uv} + B_{\text{raw},vu}}{2} \right)^2 \leq \frac{B_{\text{raw},uv}^2 + B_{\text{raw},vu}^2}{2},$$

since the difference between the right and left sides is $(B_{\text{raw},uv} - B_{\text{raw},vu})^2/4$. Summing and exchanging the two indices gives the pointwise contraction

$$\begin{aligned} \|\widehat{Q}_{h,J} - Q_{P,h,J}\|_F^2 &\leq \frac{1}{2} \sum_{u,v \in \mathcal{U}_d} (B_{\text{raw},uv}^2 + B_{\text{raw},vu}^2) \\ &= \sum_{u,v \in \mathcal{U}_d} B_{\text{raw},uv}^2. \end{aligned}$$

9. Integrability permits exchanging expectation and each finite sum. Summing the response-coordinate bounds gives

$$\begin{aligned} \mathbb{E}\|\widehat{R}_{h,J} - \bar{R}\|_2^2 &= \sum_{u \in \mathcal{U}_d} \mathbb{E} \left[((\widehat{R}_{h,J})_u - \bar{R}_u)^2 \right] \\ &\leq q C_{\text{coord},d} V_{n,m,h,J}. \end{aligned}$$

Integrating the pointwise matrix contraction and summing the raw-coordinate bounds gives

$$\begin{aligned} \mathbb{E}\|\widehat{Q}_{h,J} - Q_{P,h,J}\|_F^2 &\leq \sum_{u,v \in \mathcal{U}_d} \mathbb{E} B_{\text{raw},uv}^2 \\ &\leq q^2 C_{\text{coord},d} V_{n,m,h,J}. \end{aligned}$$

Adding these inequalities, and using $C_{\text{coord},d} V_{n,m,h,J} \geq 0$, yields

$$\begin{aligned} \mathbb{E}\|\widehat{R}_{h,J} - \bar{R}\|_2^2 + \mathbb{E}\|\widehat{Q}_{h,J} - Q_{P,h,J}\|_F^2 &\leq (q + q^2) C_{\text{coord},d} V_{n,m,h,J} \\ &\leq (q + q^2 + 1) C_{\text{coord},d} V_{n,m,h,J} \\ &= C \left\{ \frac{1}{nh^d} + \frac{k}{nNh^{2d}} \right\}. \end{aligned}$$

The chosen constant depends on d , and hence has the permitted dependence on the fixed public parameters. All estimates hold uniformly over the admitted law and the admissible sample sizes and tuning inputs.

□

The first term reflects labeled information within the localization cube. The second couples the two record counts and increases with the fine projection rank. Because $\delta = h/J$, its square-root order is the rectangular sampling term in [Theorem 1](#). The same bound covers fluctuations of the response moments and of the empirical matrix, so it supplies the sampling control needed for the guarded local fit.

A.4 Guarded inversion and public tuning

The decision in [Definition 11](#) uses the empirical cutoff $g(\varepsilon) = \varepsilon(1 - \varepsilon)/2$ against the population positivity margin $\varepsilon(1 - \varepsilon)$ in [Lemma 2](#). This half-margin gap connects matrix fluctuations to the stability of the inversion branch. The complementary branch returns zero, and clipping confines the decision to the contrast range $[-1, 1]$ supplied by [Assumptions 7](#) and [8](#). These operations give the total, finite, Borel decision covered by [Theorem 1](#); the resulting constants may depend on the fixed overlap level.

The roles of the auxiliary results are distinct. [Lemma 1](#) supplies the effect approximation order h^γ , the nuisance projection orders, and bounded polynomial coefficients. [Lemma 2](#) supplies population stability, while [Lemma 3](#) controls empirical departures from the population moments. Together, these ingredients underlie the four contributions in the uniform risk bound of [Theorem 1](#): effect approximation, combined nuisance approximation, labeled sampling, and rectangular sampling. Clipping supplies the constant bound represented by the minimum with one.

The public tuning in [Definition 12](#) assigns the fine resolution according to $S = \alpha + \beta$, the sum of nuisance smoothness orders. When $S < \gamma$, the resolution increases as the localization scale decreases, placing nuisance approximation at the effect approximation order. When $S \geq \gamma$, the choice $J_* = 1$ provides the required approximation control. The ceiling preserves an integer resolution while changing its order by a bounded factor.

The localization scale balances the labeled sampling requirement and the rectangular sampling requirement. Its two candidate orders correspond to the two terms in [Definition 14](#), with denominator $\Delta = 2\gamma + d + \gamma d/S$ for the unequal-count term. Taking their maximum yields the publicly tuned risk guarantee in [Theorem 1](#). The resulting precision has a direct statistical interpretation: response labels govern the ordinary local sampling term, while the total treatment-record count reduces the sampling cost of the fine projection correction.

A.5 The supplied-propensity upper bound

The decision displayed in [Theorem 3](#) uses the designated propensity from [Definition 3](#) to form an inverse-propensity-weighted response. Under exchangeability and overlap, this response has conditional mean equal to the canonical contrast and bounded magnitude. The known uniform covariate distribution then supplies the design measure for the degree-two local polynomial average.

Polynomial reproduction makes the contrast smoothness the relevant approximation order for this decision. Its bandwidth $n^{-1/(2\gamma+d)}$, specified in [Theorem 3](#), balances local approximation with labeled sampling. Clipping preserves the bounded contrast range. These are the statistical ingredients of the upper-bound component of that result, which attains the order $r_o(n) = n^{-\gamma/(2\gamma+d)}$ uniformly over the primitive class and every auxiliary-record count. This benchmark identifies the labeled-sample precision attained by the rectangular estimator when its combined approximation and sampling costs are controlled at the same order.

B Original-record lower bounds

The lower bounds use a marked-component lower-bound experiment within the primitive class of [Definition 3](#). Under each hypothesis, a finite prior on primitive laws induces a mixture of the original sampling experiment: each primitive law generates exactly n labeled records and m outcome-free treatment records before mixing. Write M_θ for this fixed-size mixture law under hypothesis $\theta \in \{-1, +1\}$. The construction separates the point contrasts while controlling the statistical distance between the two complete-data mixtures. We use the following normalization of that distance.

Definition 16. [\[synth_4\]](#) (Treatment-record marginal) *For a primitive Borel probability law $P = \mathcal{L}(X, A, Y(0), Y(1))$ on $[0, 1]^d \times \{0, 1\}^3$, define its treatment-record marginal by*

$$P_{XA} = \mathcal{L}_P(X, A).$$

Thus P_{XA} is the probability law on $[0, 1]^d \times \{0, 1\}$ satisfying

$$P_{XA}(E) = P((X, A) \in E)$$

for every Borel set $E \subseteq [0, 1]^d \times \{0, 1\}$.

Definition 17. [\[synth_6\]](#) (Fixed-size hypothesis mixtures) *For $n \in \mathbb{N}$ with $n \geq 2$, $m \in \mathbb{N}_0$, and the finite priors ϖ_θ produced by [Algorithm 3](#) on its probability-valued domain, define*

$$M_\theta := \int (P_o^{\otimes n} \otimes P_{XA}^{\otimes m}) d\varpi_\theta(P), \quad \theta \in \{-1, +1\},$$

where $P_o = \mathcal{L}_P(X, A, Y)$, $Y = (1 - A)Y(0) + AY(1)$, and $P_{XA} = \mathcal{L}_P(X, A)$. Thus M_{+1} and M_{-1} are the positive- and negative-hypothesis probability laws of the original-record dataset $\mathcal{D}_{n,m}$, with n labeled records and m outcome-free treatment records. Under each mixture, one primitive law P is drawn from the corresponding prior and all $n + m$ records are drawn conditionally independently from that law through their respective observation channels.

The unhalved squared Hellinger distance $H^2(M_{+1}, M_{-1})$ measures the distinguishability of the two hypotheses using both record types. The finite-prior construction $\mathfrak{C}$ of [Algorithm 3](#) below couples propensity and response perturbations through correlated sign pairs. A smooth localization bump controls the contrast near the evaluation profile, while a finer smooth frame controls the spatial dependence of the perturbations.

Definition 18. [\[def:hellinger\]](#) (Squared Hellinger distance H^2) *The squared Hellinger distance between the positive- and negative-hypothesis mixture laws M_{+1} and M_{-1} is*

$$H^2(M_{+1}, M_{-1}) = \int \left(\sqrt{\frac{dM_{+1}}{d(M_{+1} + M_{-1})}} - \sqrt{\frac{dM_{-1}}{d(M_{+1} + M_{-1})}} \right)^2 d(M_{+1} + M_{-1}).$$

Definition 19. [\[synth_2\]](#) (Constructed arm-mean functions) *The real-valued arm-mean functions $\mu_{0,\theta,\eta}$ and $\mu_{1,\theta,\eta}$ are defined for every dimension $d \in \mathbb{N}$, real scales h, δ , real propensity perturbation amplitude a , real response perturbation amplitude b , hypothesis sign $\theta \in \{-1, +1\}$, and response sign array $\eta \in \{-1, +1\}^{\mathcal{I}}$ by*

$$\begin{aligned} \mu_{0,\theta,\eta}(x) &= \frac{1}{2} + b \sum_{\mathbf{z} \in \mathcal{I}} \eta_{\mathbf{z}} \phi_{\mathbf{z}}(x) + \theta ab B_h(x)^2, \\ \mu_{1,\theta,\eta}(x) &= \frac{1}{2} + b \sum_{\mathbf{z} \in \mathcal{I}} \eta_{\mathbf{z}} \phi_{\mathbf{z}}(x) - \theta ab B_h(x)^2, \quad x \in \mathbb{R}^d. \end{aligned}$$

Here $\mathcal{I}$, the finite frame index set, B_h , the localization bump, and $\phi_{\mathbf{z}}$, the localized frame entries, use the defining formulas in [Algorithm 3](#), evaluated at dimension d and the stated real parameter values. The hypothesis sign and response signs encode Boolean values by assigning $+1$ to true and -1 to false. Each $\eta_{\mathbf{z}}$ is the encoded second component of the Boolean sign pair at $\mathbf{z}$; the first component is arbitrary. Control and treatment correspond to false and true arm values, respectively.

In [Algorithm 3](#), a is the propensity perturbation amplitude and b is the scalar response-mean perturbation amplitude. The arm means are centered at the shared function $1/2 + bG_{\eta}(x)$, with offsets $-\tau_{\theta}(x)/2$ under control and $+\tau_{\theta}(x)/2$ under treatment. Consequently, the shared perturbation cancels from their difference, leaving exactly $\tau_{\theta}(x)$. Here the scalar b is distinct from the estimator's fine polynomial basis vector $\mathbf{b}_J(x)$ in [Definition 10](#).

In [Algorithm 3](#), the product sign-pair weights $w_{\theta}(\lambda, \eta)$ encode the hypothesis through the correlation between the two perturbations. The construction itself depends only on d and (h, δ, a, b) ; the parameters $\alpha, \beta, \gamma, L, \varepsilon$ enter only through the admissibility, occupancy, and information constants of the construction certificate. The deterministic contrast τ_{θ} is shared by every primitive law within a hypothesis. Thus mixing over nuisance functions preserves an exact target separation. The compensating terms in the arm means link this separation to the singleton cancellation stated in the construction certificate below.

The information argument uses conditional factorization over spatial components. The calculus in [Lemma 4](#), stated after the construction, records how squared Hellinger distance behaves under products, conditioning on a common marginal, and the addition of a common independent coordinate.

Algorithm 3 (Finite-prior construction $\mathfrak{C}$)

The construction $\mathfrak{C}$ takes inputs (h, δ, a, b) satisfying $0 < \delta \leq h \leq 1/2$ and $a, b > 0$, at the public parameters $(d, \alpha, \beta, \gamma, L)$ and fixed interior covariate profile x_0 .

1. Define the real functions

$$\chi(u) = \begin{cases} \exp(-1/u), & u > 0, \\ 0, & u \leq 0, \end{cases} \quad \vartheta(u) = \frac{\pi}{2} \frac{\chi(u)}{\chi(u) + \chi(1-u)}, \quad u \in \mathbb{R},$$

and, for each $z \in \mathbb{Z}$,

$$\omega_z(u) = \begin{cases} \sin \vartheta(u - z + 1), & z - 1 \leq u \leq z, \\ \cos \vartheta(u - z), & z \leq u \leq z + 1, \\ 0, & u \notin [z - 1, z + 1]. \end{cases}$$

The two expressions at $u = z$ agree.

2. For $w \in \mathbb{R}^d$ and $x \in [0, 1]^d$, define the localization bump and fine-scale frame by

$$B(w) = \prod_{v=1}^d \frac{\chi(1 - 4w_v^2)}{\chi(1)}, \quad B_h(x) = B((x - x_0)/h),$$

$$\psi_{\mathbf{z}}(x) = \prod_{v=1}^d \omega_{z_v}((x_v - x_{0,v})/\delta),$$

$$\mathcal{I} = \left\{ \mathbf{z} \in \mathbb{Z}^d : |z_v| \leq 1 + \frac{h}{2\delta} \text{ for every } v \in \{1, \dots, d\} \right\}, \quad \phi_{\mathbf{z}}(x) = B_h(x)\psi_{\mathbf{z}}(x), \quad \mathbf{z} \in \mathcal{I}.$$

3. For sign arrays $\lambda, \eta \in \{-1, +1\}^{\mathcal{I}}$, define

$$F_{\lambda}(x) = \sum_{\mathbf{z} \in \mathcal{I}} \lambda_{\mathbf{z}} \phi_{\mathbf{z}}(x), \quad G_{\eta}(x) = \sum_{\mathbf{z} \in \mathcal{I}} \eta_{\mathbf{z}} \phi_{\mathbf{z}}(x).$$

For each hypothesis index $\theta \in \{-1, +1\}$, define the sign-pair correlation and product weights by

$$\rho_{\theta} = \frac{\theta}{2}, \quad w_{\theta}(\lambda, \eta) = \prod_{\mathbf{z} \in \mathcal{I}} \frac{1 + \rho_{\theta} \lambda_{\mathbf{z}} \eta_{\mathbf{z}}}{4}.$$

4. Define the propensity, contrast, and potential-response means by

$$e_{\lambda}(x) = \frac{1}{2} + aF_{\lambda}(x), \quad \tau_{\theta}(x) = -4ab\rho_{\theta}B_h(x)^2,$$

$$\mu_{0,\theta,\eta}(x) = \frac{1}{2} + bG_{\eta}(x) - \frac{1}{2}\tau_{\theta}(x), \quad \mu_{1,\theta,\eta}(x) = \frac{1}{2} + bG_{\eta}(x) + \frac{1}{2}\tau_{\theta}(x).$$

The probability-valued domain consists of inputs satisfying

$$e_{\lambda}(x), \mu_{0,\theta,\eta}(x), \mu_{1,\theta,\eta}(x) \in [0, 1] \quad \text{for every } \theta \in \{-1, +1\}, \lambda, \eta \in \{-1, +1\}^{\mathcal{I}}, x \in [0, 1]^d.$$

On this domain, define the primitive Borel law $P_{\theta,\lambda,\eta}$ by

$$X \sim \text{Unif}([0, 1]^d), \quad \mathcal{L}(A, Y(0), Y(1) \mid X = x) = \text{Bernoulli}(e_{\lambda}(x)) \otimes \text{Bernoulli}(\mu_{0,\theta,\eta}(x)) \otimes \text{Bernoulli}(\mu_{1,\theta,\eta}(x)).$$

Here X is the covariate vector, A is the binary treatment indicator, and $Y(0), Y(1)$ are the binary potential responses under control and treatment. The prescribed functions $e_{\lambda}, \mu_{0,\theta,\eta}, \mu_{1,\theta,\eta}$ are the designated conditional-margin functions of this law.

5. Output the contrasts and finite pushforward measures

$$\mathfrak{C}(h, \delta, a, b) = ((\tau_{\theta}, \varpi_{\theta}))_{\theta \in \{-1, +1\}}, \quad \varpi_{\theta} = \sum_{\lambda, \eta \in \{-1, +1\}^{\mathcal{I}}} w_{\theta}(\lambda, \eta) \text{Dirac}(P_{\theta,\lambda,\eta}),$$

where $\text{Dirac}(P)$ is the unit point mass at the primitive law P .

The product inequality in [Lemma 4](#) aggregates component distances. Its conditional identity integrates these distances over the common covariate distribution, subject to the stated measurable-density condition. The final identity accommodates the independent randomizer available to decisions in [Definition 4](#).

To translate mixture separation into expected absolute error, the analysis uses the fuzzy testing bound of [Kennedy et al. \[2024, Section 3, Lemma 1\]](#). [Lemma 5](#), stated after the Hellinger calculus, fixes both its loss and its Hellinger normalization.

Lemma 4. [lem:hellinger-block-calculus] (Hellinger calculus for blocks) *For probability laws $\mathbb{B}, \mathbb{C}$ on a common measurable space, use the normalization of [Definition 18](#):*

$$h^2(\mathbb{B}, \mathbb{C}) = \int \left(\sqrt{\frac{d\mathbb{B}}{d\xi}} - \sqrt{\frac{d\mathbb{C}}{d\xi}} \right)^2 d\xi, \quad \xi = \mathbb{B} + \mathbb{C}.$$

The following statements hold on arbitrary measurable spaces.

For any finite family of pairs of probability laws $\mathbb{B}_{u_}, \mathbb{C}_{u_*}$, each pair on its respective common measurable space,*

$$h^2 \left(\bigotimes_{u_*} \mathbb{B}_{u_*}, \bigotimes_{u_*} \mathbb{C}_{u_*} \right) \leq \sum_{u_*} h^2(\mathbb{B}_{u_*}, \mathbb{C}_{u_*}).$$

Let ν be a probability law on the conditioning space, and let $\mathbb{B}_z, \mathbb{C}_z$ be probability kernels into a common measurable space. Suppose there exists a common s -finite reference kernel such that both kernels are represented, at every z , by nonnegative extended-real densities with respect to that reference kernel, with both densities jointly measurable in the conditioning and outcome coordinates. Then

$$h^2(\nu(dz)\mathbb{B}_z(dw), \nu(dz)\mathbb{C}_z(dw)) = \int h^2(\mathbb{B}_z, \mathbb{C}_z) d\nu(z).$$

For any probability laws $\mathbb{B}, \mathbb{C}$ on a common measurable space and any probability law ν on another measurable space,

$$h^2(\mathbb{B} \otimes \nu, \mathbb{C} \otimes \nu) = h^2(\mathbb{B}, \mathbb{C}).$$

Proof of [Lemma 4](#). We first establish the finite-reference identity used below. Whenever a real Radon–Nikodym density is used, its value is taken to be zero where the extended-real derivative is infinite; for finite measures, this changes the derivative on a null set.

Let ρ_{ref} be a finite measure on the common space Ω , and suppose measurable nonnegative real densities satisfy

$$f_{\mathbb{B}, \text{ref}}, f_{\mathbb{C}, \text{ref}} : \Omega \longrightarrow [0, \infty), \quad \mathbb{B}(dw) = f_{\mathbb{B}, \text{ref}}(w) \rho_{\text{ref}}(dw), \quad \mathbb{C}(dw) = f_{\mathbb{C}, \text{ref}}(w) \rho_{\text{ref}}(dw).$$

Define

$$\xi_\Sigma := \mathbb{B} + \mathbb{C}, \quad f_{\mathbb{B}, \Sigma} := \frac{d\mathbb{B}}{d\xi_\Sigma}, \quad f_{\mathbb{C}, \Sigma} := \frac{d\mathbb{C}}{d\xi_\Sigma}, \quad r_{\Sigma, \text{ref}} := \frac{d\xi_\Sigma}{d\rho_{\text{ref}}},$$

using the real-density convention above. Radon–Nikodym reconstruction and the chain rule give

$$\mathbb{B}(dw) = f_{\mathbb{B}, \Sigma}(w) \xi_\Sigma(dw), \quad \mathbb{C}(dw) = f_{\mathbb{C}, \Sigma}(w) \xi_\Sigma(dw),$$

and, ρ_{ref} -almost everywhere,

$$f_{\mathbb{B}, \text{ref}} = f_{\mathbb{B}, \Sigma} r_{\Sigma, \text{ref}}, \quad f_{\mathbb{C}, \text{ref}} = f_{\mathbb{C}, \Sigma} r_{\Sigma, \text{ref}}.$$

For these nonnegative densities,

$$r_{\Sigma, \text{ref}} (\sqrt{f_{\mathbb{B}, \Sigma}} - \sqrt{f_{\mathbb{C}, \Sigma}})^2 = (\sqrt{f_{\mathbb{B}, \Sigma} r_{\Sigma, \text{ref}}} - \sqrt{f_{\mathbb{C}, \Sigma} r_{\Sigma, \text{ref}}})^2.$$

Thus changing measure in the normalization of [Definition 18](#) yields

$$\begin{aligned} h^2(\mathbb{B}, \mathbb{C}) &= \int r_{\Sigma, \text{ref}} (\sqrt{f_{\mathbb{B}, \Sigma}} - \sqrt{f_{\mathbb{C}, \Sigma}})^2 d\rho_{\text{ref}} \\ &= \int (\sqrt{f_{\mathbb{B}, \text{ref}}} - \sqrt{f_{\mathbb{C}, \text{ref}}})^2 d\rho_{\text{ref}}. \end{aligned}$$

1. *Finite products.* Write $\mathcal{U}_*$ for the finite indexing set and Ω_{u_*} for the measurable space of the pair indexed by u_* . Define

$$\xi_{u_*} := \mathbb{B}_{u_*} + \mathbb{C}_{u_*}, \quad f_{\mathbb{B}, u_*} := \frac{d\mathbb{B}_{u_*}}{d\xi_{u_*}}, \quad f_{\mathbb{C}, u_*} := \frac{d\mathbb{C}_{u_*}}{d\xi_{u_*}}, \quad u_* \in \mathcal{U}_*,$$

again as real densities. They are measurable, nonnegative, and integrable, and reconstruction gives

$$\mathbb{B}_{u_*}(dw) = f_{\mathbb{B}, u_*}(w)\xi_{u_*}(dw), \quad \mathbb{C}_{u_*}(dw) = f_{\mathbb{C}, u_*}(w)\xi_{u_*}(dw),$$

with

$$\int f_{\mathbb{B}, u_*} d\xi_{u_*} = \int f_{\mathbb{C}, u_*} d\xi_{u_*} = 1.$$

Define the coordinate affinity by

$$\mathbf{a}_{\text{aff}, u_*} := \int \sqrt{f_{\mathbb{B}, u_*}(w)f_{\mathbb{C}, u_*}(w)} d\xi_{u_*}(w).$$

The square roots are square-integrable, so their product is integrable. Expanding the square in the finite-reference identity gives

$$\begin{aligned} \mathbf{h}^2(\mathbb{B}_{u_*}, \mathbb{C}_{u_*}) &= \int \left(f_{\mathbb{B}, u_*} + f_{\mathbb{C}, u_*} - 2\sqrt{f_{\mathbb{B}, u_*}f_{\mathbb{C}, u_*}} \right) d\xi_{u_*} \\ &= 2(1 - \mathbf{a}_{\text{aff}, u_*}). \end{aligned}$$

On the product space, define

$$\xi_{\otimes} := \bigotimes_{u_* \in \mathcal{U}_*} \xi_{u_*}, \quad f_{\mathbb{B}, \otimes}(w) := \prod_{u_* \in \mathcal{U}_*} f_{\mathbb{B}, u_*}(w_{u_*}), \quad f_{\mathbb{C}, \otimes}(w) := \prod_{u_* \in \mathcal{U}_*} f_{\mathbb{C}, u_*}(w_{u_*}), \quad w \in \prod_{u_* \in \mathcal{U}_*} \Omega_{u_*}.$$

To verify the product-density representation, take measurable sets

$$W_{u_*} \subseteq \Omega_{u_*}, \quad u_* \in \mathcal{U}_*.$$

Product integration gives

$$\int_{\prod_{u_*} W_{u_*}} f_{\mathbb{B}, \otimes} d\xi_{\otimes} = \prod_{u_*} \int_{W_{u_*}} f_{\mathbb{B}, u_*} d\xi_{u_*} = \prod_{u_*} \mathbb{B}_{u_*}(W_{u_*}).$$

The same calculation applies to $\mathbb{C}$. Uniqueness of finite product measures therefore yields

$$\left(\bigotimes_{u_*} \mathbb{B}_{u_*} \right) (dw) = f_{\mathbb{B}, \otimes}(w)\xi_{\otimes}(dw), \quad \left(\bigotimes_{u_*} \mathbb{C}_{u_*} \right) (dw) = f_{\mathbb{C}, \otimes}(w)\xi_{\otimes}(dw).$$

In particular,

$$\int f_{\mathbb{B}, \otimes} d\xi_{\otimes} = \int f_{\mathbb{C}, \otimes} d\xi_{\otimes} = 1.$$

Applying the finite-reference identity and expanding the square as above gives

$$\mathbf{h}^2 \left(\bigotimes_{u_*} \mathbb{B}_{u_*}, \bigotimes_{u_*} \mathbb{C}_{u_*} \right) = 2 \left(1 - \int \sqrt{f_{\mathbb{B}, \otimes} f_{\mathbb{C}, \otimes}} d\xi_{\otimes} \right).$$

Nonnegativity of the coordinate densities implies the pointwise identity

$$\sqrt{f_{\mathbb{B},\otimes}(w)f_{\mathbb{C},\otimes}(w)} = \prod_{u_*} \sqrt{f_{\mathbb{B},u_*}(w_{u_*})f_{\mathbb{C},u_*}(w_{u_*})}.$$

Product integration consequently gives

$$\int \sqrt{f_{\mathbb{B},\otimes}f_{\mathbb{C},\otimes}} d\xi_{\otimes} = \prod_{u_*} \mathbf{a}_{\text{aff},u_*}.$$

The affinities satisfy

$$\mathbf{a}_{\text{aff},u_*} \geq 0, \quad 0 \leq \mathbf{h}^2(\mathbb{B}_{u_*}, \mathbb{C}_{u_*}) = 2(1 - \mathbf{a}_{\text{aff},u_*}),$$

and hence

$$0 \leq \mathbf{a}_{\text{aff},u_*} \leq 1.$$

For every subset $\mathcal{V}_* \subseteq \mathcal{U}_*$, induction on its cardinality gives

$$0 \leq \prod_{u_* \in \mathcal{V}_*} \mathbf{a}_{\text{aff},u_*} \leq 1.$$

A second induction gives the required affinity-defect bound. The empty product is one and the empty sum is zero. For $u_* \in \mathcal{U}_* \setminus \mathcal{V}_*$, the induction step is

$$\begin{aligned} 1 - \prod_{u'_* \in \mathcal{V}_* \cup \{u_*\}} \mathbf{a}_{\text{aff},u'_*} &= \left(1 - \prod_{u'_* \in \mathcal{V}_*} \mathbf{a}_{\text{aff},u'_*}\right) + (1 - \mathbf{a}_{\text{aff},u_*}) \prod_{u'_* \in \mathcal{V}_*} \mathbf{a}_{\text{aff},u'_*} \\ &\leq \sum_{u'_* \in \mathcal{V}_*} (1 - \mathbf{a}_{\text{aff},u'_*}) + (1 - \mathbf{a}_{\text{aff},u_*}). \end{aligned}$$

Taking $\mathcal{V}_* = \mathcal{U}_*$ and multiplying by two proves

$$\begin{aligned} \mathbf{h}^2\left(\bigotimes_{u_*} \mathbb{B}_{u_*}, \bigotimes_{u_*} \mathbb{C}_{u_*}\right) &= 2 \left(1 - \prod_{u_*} \mathbf{a}_{\text{aff},u_*}\right) \\ &\leq 2 \sum_{u_*} (1 - \mathbf{a}_{\text{aff},u_*}) = \sum_{u_*} \mathbf{h}^2(\mathbb{B}_{u_*}, \mathbb{C}_{u_*}). \end{aligned}$$

The empty family is included by the stated empty-product and empty-sum conventions.

2. *A common conditioning marginal.* Let Z and Ω be the conditioning and outcome spaces. Choose the s-finite reference kernel and jointly measurable densities supplied by the hypothesis, writing them as

$$\Xi_{\text{ref}} : Z \longrightarrow \{\text{s-finite measures on } \Omega\}, \quad f_{\mathbb{B},\text{ref}}, f_{\mathbb{C},\text{ref}} : Z \times \Omega \longrightarrow [0, \infty],$$

so that, for every $z \in Z$,

$$\mathbb{B}_z(dw) = f_{\mathbb{B},\text{ref}}(z, w) \Xi_{\text{ref},z}(dw), \quad \mathbb{C}_z(dw) = f_{\mathbb{C},\text{ref}}(z, w) \Xi_{\text{ref},z}(dw).$$

Define the sum kernel and normalized densities by

$$\mathbb{K}_{\Sigma,z} := \mathbb{B}_z + \mathbb{C}_z$$

and

$$(r_{\mathbb{B}}(z, w), r_{\mathbb{C}}(z, w)) := \begin{cases} \left(\frac{f_{\mathbb{B},\text{ref}}(z, w)}{f_{\mathbb{B},\text{ref}}(z, w) + f_{\mathbb{C},\text{ref}}(z, w)}, \frac{f_{\mathbb{C},\text{ref}}(z, w)}{f_{\mathbb{B},\text{ref}}(z, w) + f_{\mathbb{C},\text{ref}}(z, w)} \right), & 0 < f_{\mathbb{B},\text{ref}}(z, w) + f_{\mathbb{C},\text{ref}}(z, w) \\ (0, 0), & \text{otherwise.} \end{cases}$$

These are jointly measurable real functions on all of $Z \times \Omega$, and

$$0 \leq r_{\mathbb{B}}(z, w) \leq 1, \quad 0 \leq r_{\mathbb{C}}(z, w) \leq 1,$$

because each numerator lies between zero and its positive finite denominator; the other cases have value zero.

For every z ,

$$\mathbb{K}_{\Sigma,z}(dw) = (f_{\mathbb{B},\text{ref}}(z, w) + f_{\mathbb{C},\text{ref}}(z, w)) \Xi_{\text{ref},z}(dw),$$

and the nonnegative extended integral satisfies

$$\int (f_{\mathbb{B},\text{ref}}(z, w) + f_{\mathbb{C},\text{ref}}(z, w)) d\Xi_{\text{ref},z}(w) = 2.$$

Thus the sum of the densities is finite $\Xi_{\text{ref},z}$ -almost everywhere. Where that sum is zero, both densities are zero. Cancellation therefore gives, $\Xi_{\text{ref},z}$ -almost everywhere,

$$\begin{aligned} (f_{\mathbb{B},\text{ref}} + f_{\mathbb{C},\text{ref}})(z, w) r_{\mathbb{B}}(z, w) &= f_{\mathbb{B},\text{ref}}(z, w), \\ (f_{\mathbb{B},\text{ref}} + f_{\mathbb{C},\text{ref}})(z, w) r_{\mathbb{C}}(z, w) &= f_{\mathbb{C},\text{ref}}(z, w). \end{aligned}$$

Consequently, for every z ,

$$\mathbb{B}_z(dw) = r_{\mathbb{B}}(z, w) \mathbb{K}_{\Sigma,z}(dw), \quad \mathbb{C}_z(dw) = r_{\mathbb{C}}(z, w) \mathbb{K}_{\Sigma,z}(dw).$$

Define the finite joint reference measure by

$$\Xi_{\Sigma}(dz, dw) := \nu(dz) \mathbb{K}_{\Sigma,z}(dw).$$

The kernel density representations and joint measurability imply

$$\nu(dz) \mathbb{B}_z(dw) = r_{\mathbb{B}}(z, w) \Xi_{\Sigma}(dz, dw), \quad \nu(dz) \mathbb{C}_z(dw) = r_{\mathbb{C}}(z, w) \Xi_{\Sigma}(dz, dw).$$

Moreover,

$$\Xi_{\Sigma}(Z \times \Omega) = \int \mathbb{K}_{\Sigma,z}(\Omega) d\nu(z) = 2.$$

The normalized-density bounds give

$$0 \leq \sqrt{r_{\mathbb{B}}(z, w)} \leq 1, \quad 0 \leq \sqrt{r_{\mathbb{C}}(z, w)} \leq 1,$$

and hence

$$0 \leq (\sqrt{r_{\mathbb{B}}(z, w)} - \sqrt{r_{\mathbb{C}}(z, w)})^2 \leq 4.$$

The measurable square-root difference is therefore integrable against Ξ_Σ . Applying the finite-reference identity and then integration over the kernel gives

$$\begin{aligned} & \mathbf{h}^2(\nu(dz)\mathbb{B}_z(dw), \nu(dz)\mathbb{C}_z(dw)) \\ &= \int (\sqrt{r_\mathbb{B}(z, w)} - \sqrt{r_\mathbb{C}(z, w)})^2 d\Xi_\Sigma(z, w) \\ &= \int_Z \left[\int_\Omega (\sqrt{r_\mathbb{B}(z, w)} - \sqrt{r_\mathbb{C}(z, w)})^2 d\mathbb{K}_{\Sigma, z}(w) \right] d\nu(z). \end{aligned}$$

For each z , the same finite-reference identity and the conditional density representations yield

$$\int_\Omega (\sqrt{r_\mathbb{B}(z, w)} - \sqrt{r_\mathbb{C}(z, w)})^2 d\mathbb{K}_{\Sigma, z}(w) = \mathbf{h}^2(\mathbb{B}_z, \mathbb{C}_z).$$

Substitution proves

$$\mathbf{h}^2(\nu(dz)\mathbb{B}_z(dw), \nu(dz)\mathbb{C}_z(dw)) = \int \mathbf{h}^2(\mathbb{B}_z, \mathbb{C}_z) d\nu(z).$$

3. *A common independent probability factor.* For the given $\mathbb{B}, \mathbb{C}$, use

$$\xi_\Sigma := \mathbb{B} + \mathbb{C}, \quad f_{\mathbb{B}, \Sigma} := \frac{d\mathbb{B}}{d\xi_\Sigma}, \quad f_{\mathbb{C}, \Sigma} := \frac{d\mathbb{C}}{d\xi_\Sigma},$$

with the real-density convention above. The sum of the product laws is

$$(\mathbb{B} \otimes \nu) + (\mathbb{C} \otimes \nu) = \xi_\Sigma \otimes \nu.$$

The product Radon–Nikodym identities give, $(\xi_\Sigma \otimes \nu)$ -almost everywhere,

$$\frac{d(\mathbb{B} \otimes \nu)}{d(\xi_\Sigma \otimes \nu)}(w, z) = f_{\mathbb{B}, \Sigma}(w), \quad \frac{d(\mathbb{C} \otimes \nu)}{d(\xi_\Sigma \otimes \nu)}(w, z) = f_{\mathbb{C}, \Sigma}(w).$$

Substituting into the defining square-root formula and integrating a function of the first coordinate gives

$$\begin{aligned} \mathbf{h}^2(\mathbb{B} \otimes \nu, \mathbb{C} \otimes \nu) &= \int_{\Omega \times Z} (\sqrt{f_{\mathbb{B}, \Sigma}(w)} - \sqrt{f_{\mathbb{C}, \Sigma}(w)})^2 d(\xi_\Sigma \otimes \nu)(w, z) \\ &= \nu(Z) \int_\Omega (\sqrt{f_{\mathbb{B}, \Sigma}(w)} - \sqrt{f_{\mathbb{C}, \Sigma}(w)})^2 d\xi_\Sigma(w) \\ &= \mathbf{h}^2(\mathbb{B}, \mathbb{C}), \end{aligned}$$

where the last equality uses $\nu(Z) = 1$.

□

For the original-record lower bound, the testing comparison concerns the law of the entire fixed-size dataset. At this level, the observation space contains both record blocks, and the testing statement can be used with product count one. This formulation preserves the unequal record counts in the experiment. The certificate also records an exact equality of the auxiliary mixture laws; the relevant marginal is the treatment-record marginal of [Definition 16](#).

Lemma 5. [lem:published-fuzzy-testing] (Absolute-loss fuzzy testing bound) *Let $(\mathbb{B}_z)$ and $(\mathbb{C}_z)$ be probability kernels from a common measurable index space to a common measurable observation space, and let ω be a probability prior on the index space. Let $v_0 \geq 1$ be an integer, let $\mathcal{M}$ be a model of measures containing $\mathbb{B}_z$ and $\mathbb{C}_z$ for every z , and let Ψ be a real-valued functional on measures. Use the squared Hellinger distance of [Definition 18](#), with normalization*

$$h^2(\mathbb{B}, \mathbb{C}) = \int \left(\sqrt{\frac{d\mathbb{B}}{d\xi}} - \sqrt{\frac{d\mathbb{C}}{d\xi}} \right)^2 d\xi, \quad \xi = \mathbb{B} + \mathbb{C}.$$

Suppose the following conditions hold:

- **(Constants.)** The separation bound s_0 and distance bound u_0 satisfy $s_0 > 0$ and $0 \leq u_0 < 2$.
- **(Separation.)** For every pair of indices z, z' ,

$$\Psi(\mathbb{B}_z) - \Psi(\mathbb{C}_{z'}) \geq s_0.$$

- **(Mixture distance.)** The product-law mixtures satisfy

$$h^2 \left(\int \mathbb{B}_z^{\otimes v_0} d\omega(z), \int \mathbb{C}_z^{\otimes v_0} d\omega(z) \right) \leq u_0.$$

Then every measurable real-valued decision $\widehat{\Psi}$ on the v_0 -observation sample space satisfies

$$\sup_{P \in \mathcal{M}} \int \left| \widehat{\Psi}(w) - \Psi(P) \right| dP^{\otimes v_0}(w) \geq \frac{s_0}{4} \left\{ 1 - \sqrt{u_0(1 - u_0/4)} \right\},$$

where the nonnegative loss integral is interpreted in the extended sense. This is the absolute-loss specialization of [[Kennedy et al., 2024](#), Section 3, Lemma 1].

Proof of Lemma 5. Fix a measurable decision $\widehat{\Psi}$. All loss integrals and their suprema below take values in $[0, \infty]$.

1. Define the testing submodel, its component risks, and its worst-case risk by

$$\begin{aligned} \mathcal{M}_{\text{test}} &:= \{\mathbb{B}_z : z \text{ is an index}\} \cup \{\mathbb{C}_z : z \text{ is an index}\}, \\ \mathcal{R}_{\text{test}}(P) &:= \int |\widehat{\Psi}(w) - \Psi(P)| dP^{\otimes v_0}(w), \quad P \text{ a measure on the observation space,} \\ \mathcal{S}_{\text{test}} &:= \sup_{P \in \mathcal{M}_{\text{test}}} \mathcal{R}_{\text{test}}(P). \end{aligned}$$

Every member of $\mathcal{M}_{\text{test}}$ is a probability measure. The finite-product maps $z \mapsto \mathbb{B}_z^{\otimes v_0}$ and $z \mapsto \mathbb{C}_z^{\otimes v_0}$ are probability kernels. Define their mixture laws by

$$\mathbb{B}_{\text{mix}} := \int \mathbb{B}_z^{\otimes v_0} d\omega(z), \quad \mathbb{C}_{\text{mix}} := \int \mathbb{C}_z^{\otimes v_0} d\omega(z).$$

Both mixtures are probability measures, since their component laws and the prior are probability measures. The distance hypothesis therefore gives

$$h^2(\mathbb{B}_{\text{mix}}, \mathbb{C}_{\text{mix}}) \leq u_0.$$

2. The probability prior ensures that the index space is nonempty. Choose an index z_0 . Separation gives

$$\Psi(\mathbb{C}_z) \leq \Psi(\mathbb{B}_{z_0}) - s_0 \quad \text{for every } z.$$

Thus the nonempty set of values $\Psi(\mathbb{C}_z)$ is bounded above. Define the finite real numbers

$$\psi_{\mathbb{C}}^{\sup} := \sup_z \Psi(\mathbb{C}_z), \quad a_{\text{test}} := \psi_{\mathbb{C}}^{\sup} + \frac{s_0}{2}.$$

For each index z , separation with every z' gives

$$\Psi(\mathbb{C}_{z'}) \leq \Psi(\mathbb{B}_z) - s_0, \quad \psi_{\mathbb{C}}^{\sup} \leq \Psi(\mathbb{B}_z) - s_0.$$

Consequently,

$$a_{\text{test}} + \frac{s_0}{2} \leq \Psi(\mathbb{B}_z), \quad \Psi(\mathbb{C}_z) \leq a_{\text{test}} - \frac{s_0}{2}.$$

Define the measurable threshold event on the sample space by

$$\mathcal{A}_{\text{test}} := \{w : \hat{\Psi}(w) < a_{\text{test}}\}.$$

3. We derive the Hellinger testing comparison for these mixtures. Define

$$\begin{aligned} \xi_{\text{test}} &:= \mathbb{B}_{\text{mix}} + \mathbb{C}_{\text{mix}}, \\ f_{\mathbb{B}}(w) &:= \frac{d\mathbb{B}_{\text{mix}}}{d\xi_{\text{test}}}(w), \quad f_{\mathbb{C}}(w) := \frac{d\mathbb{C}_{\text{mix}}}{d\xi_{\text{test}}}(w), \\ h_{\text{test}}^2 &:= h^2(\mathbb{B}_{\text{mix}}, \mathbb{C}_{\text{mix}}). \end{aligned}$$

Choose nonnegative versions of the densities. They are integrable, and

$$\int f_{\mathbb{B}} d\xi_{\text{test}} = \int f_{\mathbb{C}} d\xi_{\text{test}} = 1, \quad \int (f_{\mathbb{B}} - f_{\mathbb{C}}) d\xi_{\text{test}} = 0.$$

For every measurable sample event $\mathcal{A}$,

$$\begin{aligned} \mathbb{B}_{\text{mix}}(\mathcal{A}) - \mathbb{C}_{\text{mix}}(\mathcal{A}) &= \int_{\mathcal{A}} (f_{\mathbb{B}} - f_{\mathbb{C}}) d\xi_{\text{test}}, \\ \int_{\mathcal{A}^c} (f_{\mathbb{B}} - f_{\mathbb{C}}) d\xi_{\text{test}} &= - \int_{\mathcal{A}} (f_{\mathbb{B}} - f_{\mathbb{C}}) d\xi_{\text{test}}. \end{aligned}$$

Applying the integral triangle inequality on each of $\mathcal{A}$ and $\mathcal{A}^c$, then adding, yields

$$2 \left| \int_{\mathcal{A}} (f_{\mathbb{B}} - f_{\mathbb{C}}) d\xi_{\text{test}} \right| \leq \int |f_{\mathbb{B}} - f_{\mathbb{C}}| d\xi_{\text{test}}.$$

The normalization in [Definition 18](#) gives

$$h_{\text{test}}^2 = \int (\sqrt{f_{\mathbb{B}}} - \sqrt{f_{\mathbb{C}}})^2 d\xi_{\text{test}}.$$

The square roots of the densities are square integrable. The pointwise identities

$$\begin{aligned} (\sqrt{f_{\mathbb{B}}} + \sqrt{f_{\mathbb{C}}})^2 + (\sqrt{f_{\mathbb{B}}} - \sqrt{f_{\mathbb{C}}})^2 &= 2f_{\mathbb{B}} + 2f_{\mathbb{C}}, \\ |f_{\mathbb{B}} - f_{\mathbb{C}}| &= |\sqrt{f_{\mathbb{B}}} - \sqrt{f_{\mathbb{C}}}| |\sqrt{f_{\mathbb{B}}} + \sqrt{f_{\mathbb{C}}}| \end{aligned}$$

therefore imply

$$\int (\sqrt{f_{\mathbb{B}}} + \sqrt{f_{\mathbb{C}}})^2 d\xi_{\text{test}} = 4 - h_{\text{test}}^2.$$

Also, by nonnegativity and the pointwise bound $(\sqrt{f_{\mathbb{B}}} - \sqrt{f_{\mathbb{C}}})^2 \leq f_{\mathbb{B}} + f_{\mathbb{C}}$,

$$0 \leq h_{\text{test}}^2 \leq \int (f_{\mathbb{B}} + f_{\mathbb{C}}) d\xi_{\text{test}} = 2.$$

Cauchy–Schwarz now gives

$$\begin{aligned} \int |f_{\mathbb{B}} - f_{\mathbb{C}}| d\xi_{\text{test}} &\leq \sqrt{h_{\text{test}}^2} \sqrt{4 - h_{\text{test}}^2} \\ &= \sqrt{h_{\text{test}}^2 (4 - h_{\text{test}}^2)}. \end{aligned}$$

Combining the preceding bounds proves, for every measurable $\mathcal{A}$,

$$\begin{aligned} |\mathbb{B}_{\text{mix}}(\mathcal{A}) - \mathbb{C}_{\text{mix}}(\mathcal{A})| &\leq \frac{1}{2} \sqrt{h_{\text{test}}^2 (4 - h_{\text{test}}^2)} \\ &= \sqrt{h_{\text{test}}^2 (1 - h_{\text{test}}^2/4)}. \end{aligned}$$

4. Define the total variation distance of the mixtures by

$$V_{\text{test}} := \sup_{\mathcal{A} \text{ measurable}} |\mathbb{B}_{\text{mix}}(\mathcal{A}) - \mathbb{C}_{\text{mix}}(\mathcal{A})|.$$

Taking the supremum in the preceding event bound gives

$$V_{\text{test}} \leq \sqrt{h_{\text{test}}^2 (1 - h_{\text{test}}^2/4)}.$$

The complementary-event identity implies

$$\begin{aligned} \mathbb{B}_{\text{mix}}(\mathcal{A}_{\text{test}}) + \mathbb{C}_{\text{mix}}(\mathcal{A}_{\text{test}}^c) &= 1 + \mathbb{B}_{\text{mix}}(\mathcal{A}_{\text{test}}) - \mathbb{C}_{\text{mix}}(\mathcal{A}_{\text{test}}) \\ &\geq 1 - V_{\text{test}}. \end{aligned}$$

Since $0 \leq h_{\text{test}}^2 \leq u_0 < 2$,

$$u_0(1 - u_0/4) - h_{\text{test}}^2(1 - h_{\text{test}}^2/4) = \frac{(u_0 - h_{\text{test}}^2)(4 - u_0 - h_{\text{test}}^2)}{4} \geq 0.$$

Monotonicity of the square root consequently yields

$$1 - \sqrt{u_0(1 - u_0/4)} \leq \mathbb{B}_{\text{mix}}(\mathcal{A}_{\text{test}}) + \mathbb{C}_{\text{mix}}(\mathcal{A}_{\text{test}}^c).$$

5. The threshold separation gives the pointwise loss bounds, for every index z ,

$$\begin{aligned} \frac{s_0}{2} \mathbf{1}_{\mathcal{A}_{\text{test}}}(w) &\leq |\hat{\Psi}(w) - \Psi(\mathbb{B}_z)|, \\ \frac{s_0}{2} \mathbf{1}_{\mathcal{A}_{\text{test}}^c}(w) &\leq |\hat{\Psi}(w) - \Psi(\mathbb{C}_z)|. \end{aligned}$$

Indeed, on $\mathcal{A}_{\text{test}}$ the decision lies below a_{test} , while $\Psi(\mathbb{B}_z)$ lies at least $s_0/2$ above it. On its complement the decision lies at or above the threshold, while $\Psi(\mathbb{C}_z)$ lies at least $s_0/2$ below it. Outside the respective events, the displayed left-hand sides vanish.

Integrating these inequalities under the component product laws gives

$$\begin{aligned}\frac{s_0}{2} \mathbb{B}_z^{\otimes v_0}(\mathcal{A}_{\text{test}}) &\leq \mathcal{R}_{\text{test}}(\mathbb{B}_z) \leq \mathcal{S}_{\text{test}}, \\ \frac{s_0}{2} \mathbb{C}_z^{\otimes v_0}(\mathcal{A}_{\text{test}}^c) &\leq \mathcal{R}_{\text{test}}(\mathbb{C}_z) \leq \mathcal{S}_{\text{test}}.\end{aligned}$$

Kernel event probabilities are measurable functions of the index. Integrating their bounds against the probability prior therefore gives

$$\begin{aligned}\frac{s_0}{2} \mathbb{B}_{\text{mix}}(\mathcal{A}_{\text{test}}) &= \int \frac{s_0}{2} \mathbb{B}_z^{\otimes v_0}(\mathcal{A}_{\text{test}}) d\omega(z) \leq \int \mathcal{S}_{\text{test}} d\omega = \mathcal{S}_{\text{test}}, \\ \frac{s_0}{2} \mathbb{C}_{\text{mix}}(\mathcal{A}_{\text{test}}^c) &= \int \frac{s_0}{2} \mathbb{C}_z^{\otimes v_0}(\mathcal{A}_{\text{test}}^c) d\omega(z) \leq \int \mathcal{S}_{\text{test}} d\omega = \mathcal{S}_{\text{test}}.\end{aligned}$$

Adding yields the extended nonnegative inequality

$$\frac{s_0}{2} \{ \mathbb{B}_{\text{mix}}(\mathcal{A}_{\text{test}}) + \mathbb{C}_{\text{mix}}(\mathcal{A}_{\text{test}}^c) \} \leq 2\mathcal{S}_{\text{test}}.$$

6. Combining the testing and risk inequalities gives

$$\frac{s_0}{2} \left\{ 1 - \sqrt{u_0(1 - u_0/4)} \right\} \leq 2\mathcal{S}_{\text{test}}.$$

The bracket is nonnegative under $0 \leq u_0 < 2$. Using

$$\frac{s_0}{2} \left\{ 1 - \sqrt{u_0(1 - u_0/4)} \right\} = 2 \left[\frac{s_0}{4} \left\{ 1 - \sqrt{u_0(1 - u_0/4)} \right\} \right]$$

and cancelling the finite positive factor 2 in $[0, \infty]$, we obtain

$$\frac{s_0}{4} \left\{ 1 - \sqrt{u_0(1 - u_0/4)} \right\} \leq \mathcal{S}_{\text{test}}.$$

Finally, every member of $\mathcal{M}_{\text{test}}$ equals $\mathbb{B}_z$ or $\mathbb{C}_z$ for some index z ; either membership hypothesis places that measure in $\mathcal{M}$. Thus

$$\mathcal{M}_{\text{test}} \subseteq \mathcal{M}, \quad \mathcal{S}_{\text{test}} \leq \sup_{P \in \mathcal{M}} \int |\hat{\Psi}(w) - \Psi(P)| dP^{\otimes v_0}(w).$$

Transitivity proves the asserted bound, including when the supremum is infinite.

□

The marginal in [Definition 16](#) is the sampling law of each outcome-free record. The next result establishes model membership, target separation, and the information bound for the prescribed smooth-frame construction. Its occupancy condition restricts the expected number of treatment records at the fine spatial scale.

Theorem 4. `[thm:marked-component-certificate]` (Marked component mixture certificate)
Fix the primitive class $\mathcal{M}$ of [Definition 3](#), with public parameters satisfying

$$d \in \mathbb{N}, \quad d \geq 1, \quad \alpha, \beta \in (0, 1], \quad \gamma \in [1, 3], \quad L > \frac{1}{2}, \quad \varepsilon \in (0, \frac{1}{2}).$$

There exist constants $c \in (0, 1]$, $c_0 > 0$, and $C > 0$, depending only on these public parameters, such that the following holds for every choice satisfying:

- **(Record counts.)** $n, m \in \mathbb{N}$, $n \geq 2$, with total record count $N = n + m$.
- **(Scales.)** $0 < \delta \leq h \leq \frac{1}{2}$.
- **(Propensity amplitude.)** $0 < a \leq c\delta^\alpha$.
- **(Response amplitude.)** $0 < b \leq c\delta^\beta$.
- **(Effect calibration.)** $ab \leq ch^\gamma$.
- **(Occupancy.)** $N\delta^d \leq c_0$.

The prescribed construction $\mathfrak{C}$ of [Algorithm 3](#), evaluated at d, h, δ, a, b , has the following properties for both hypothesis indices $\theta \in \{-1, +1\}$.

Its weights on the finite sign arrays are nonnegative and normalized:

$$w_\theta(\lambda, \eta) \geq 0 \quad \text{for every } \lambda, \eta \in \{-1, +1\}^{\mathcal{I}}, \quad \sum_{\lambda, \eta \in \{-1, +1\}^{\mathcal{I}}} w_\theta(\lambda, \eta) = 1.$$

Every constructed law $P_{\theta, \lambda, \eta}$ belongs to $\mathcal{M}$, and its canonical contrast agrees with the deterministic constructed contrast everywhere on the unit cube:

$$\tau_{P_{\theta, \lambda, \eta}}(x) = \tau_\theta(x) \quad (x \in [0, 1]^d).$$

Thus the finite prior ϖ_θ is supported on $\mathcal{M}$. The designated propensity functions, restricted to the unit cube, have the same distribution under ϖ_{+1} and ϖ_{-1} .

At the interior center $x_0 = (1/2, \dots, 1/2)$, the contrasts satisfy

$$\tau_\theta(x_0) = -2\theta ab, \quad \tau_{-1}(x_0) - \tau_{+1}(x_0) = 4ab, \quad cab \leq 4ab.$$

For the fixed-size original-record mixture laws M_{+1} and M_{-1} , the unhalved squared Hellinger distance of [Definition 18](#) satisfies

$$H^2(M_{+1}, M_{-1}) \leq CnNh^d\delta^d a^2 b^2.$$

For every integer $k_{\text{aux}} \geq 0$, the auxiliary-record mixtures agree:

$$\int P_{XA}^{\otimes k_{\text{aux}}} d\varpi_{+1}(P) = \int P_{XA}^{\otimes k_{\text{aux}}} d\varpi_{-1}(P).$$

Their explicit conditional auxiliary kernels also agree at every collection of k_{aux} covariates in $[0, 1]^d$. For either hypothesis and every $x \in [0, 1]^d$, the singleton labeled mixture conditional on $X = x$ is uniform on the four values of $(A, Y) \in \{0, 1\}^2$.

Conditional on every collection of all N covariates in the unit cube, form the graph whose vertices are all N records. Join two distinct vertices when both covariates lie in the localization cube C_h and their maximum-norm distance is at most 2δ ; records exterior to C_h are isolated vertices. The conditional mixed density equals the product of the component densities over all connected components of this graph, including the exterior singleton components. Every auxiliary-only component and every component consisting of a single labeled record has the same conditional density, and hence the same conditional law, under the two hypotheses.

The information bound in [Theorem 4](#) reflects where the two hypotheses can be distinguished. Their common propensity distribution gives identical auxiliary mixtures for every auxiliary sample size, including equality conditional on the auxiliary covariates. Compensation in the response means also makes a singleton labeled mixture uniform under either hypothesis. Conditional component factorization therefore locates distinguishing information in components containing a labeled record and at least one additional nearby record.

The factor nN in the distance bound corresponds to record pairs involving a labeled member, while $h^d \delta^d$ records the localization and proximity scales. The squared perturbation factor is $a^2 b^2$, and the target separation is $4ab$. Under the stated sparse-occupancy gate $N\delta^d \leq c_0$, with occupancy constant c_0 , the certificate bounds the aggregate information by this pair-scale expression. These clauses give the component structure needed by [Lemma 4](#) and the separation needed by the testing bound of [Lemma 5](#).

The next result packages the smoothness, contrast, and occupancy calibrations into a minimax lower bound. Its sample-size interval is the range in which the unequal-count rate is at least the supplied-propensity order.

Lemma 6. [`lem:nuisance-frontier-converse`] (Nuisance frontier lower bound) *Fix $d \in \mathbb{N}$ and $\alpha, \beta, \gamma, L, \varepsilon \in \mathbb{R}$, where ε is the overlap level. Define the nuisance smoothness sum S , the smoothness threshold S_{crit} , the record-growth exponent q_* , and the rate denominator Δ by*

$$S = \alpha + \beta, \quad S_{\text{crit}} = \frac{\gamma d}{2\gamma + d}, \quad q_* = \frac{\gamma d}{(\alpha + \beta)(2\gamma + d)}, \quad \Delta = 2\gamma + d + \frac{\gamma d}{\alpha + \beta}.$$

Suppose that:

- (*Dimension.*) $d \geq 1$.
- (*Nuisance smoothness.*) $0 < \alpha \leq 1$ and $0 < \beta \leq 1$.
- (*Effect smoothness.*) $1 \leq \gamma \leq 3$.
- (*Radius.*) $L > 1/2$.
- (*Overlap level.*) $0 < \varepsilon < 1/2$.
- (*Frontier regime.*) $S < S_{\text{crit}}$.

There exists a constant $c > 0$, depending on the fixed public parameters $d, \alpha, \beta, \gamma, L, \varepsilon$, such that for every $n, m \in \mathbb{N}$ with $n \geq 2$, the total treatment-record count

$$N = n + m$$

satisfying $N \leq n^{q_*}$ obeys the following bound for the minimax absolute-error risk $R(n, m)$ of [Definition 5](#), evaluated at these fixed parameters:

$$R(n, m) \geq c(nN)^{-\gamma/\Delta}.$$

Proof of Lemma 6. 1. *Public constants.* Take the constants supplied by [Theorem 4](#), writing its amplitude constant as c_{cert} :

$$0 < c_{\text{cert}} \leq 1, \quad c_0 > 0, \quad C > 0.$$

They depend only on the fixed public parameters. The parameter conditions give

$$\gamma > 0, \quad S > 0, \quad \Delta > 0.$$

Define

$$c_h = \frac{1}{4}, \quad c_\delta = \min\{c_h, c_0\}, \quad \rho_0 = \min\left\{c_{\text{cert}}, 1, c_{\text{cert}}c_h^\gamma, \frac{1}{16C}\right\}.$$

Thus

$$0 < c_\delta \leq c_h \leq \frac{1}{2}, \quad c_\delta \leq c_0, \quad 0 < \rho_0 \leq 1,$$

and

$$\rho_0 \leq c_{\text{cert}}, \quad \rho_0 \leq c_{\text{cert}}c_h^\gamma, \quad C\rho_0 \leq \frac{1}{16}.$$

Since $d \geq 1$, $S > 0$, and $c_\delta, c_h \leq 1$,

$$c_\delta^d \leq c_\delta \leq c_0, \quad c_\delta^S \leq 1, \quad c_h^d \leq 1, \quad c_\delta^{d+2S} \leq 1.$$

Also,

$$\rho_0^2 \leq \rho_0, \quad \rho_0^4 \leq \rho_0,$$

so the information constant satisfies

$$C\rho_0^4 c_h^d c_\delta^{d+2S} \leq C\rho_0 \leq \frac{1}{16}.$$

Finally, define the separation constant and the claimed lower-bound constant by

$$c_{\text{sep}} = \rho_0^2 c_\delta^S > 0, \quad c = \frac{3}{4}c_{\text{sep}} > 0.$$

2. *Admissible scales.* Fix arbitrary $n, m \in \mathbb{N}$ in the stated range, and write

$$N = n + m, \quad n \geq 2, \quad n \leq N \leq n^{q*}.$$

In particular, $n, N > 0$ and $nN \geq 1$. Choose

$$h = c_h(nN)^{-1/\Delta}, \quad \delta = c_\delta(nN)^{-\gamma/(S\Delta)}, \quad a = \rho_0\delta^\alpha, \quad b = \rho_0\delta^\beta.$$

The frontier condition implies

$$S < S_{\text{crit}} = \frac{\gamma d}{2\gamma + d} < \gamma,$$

because $2\gamma + d > d > 0$. Consequently,

$$-\frac{\gamma}{S\Delta} \leq -\frac{1}{\Delta} \leq 0.$$

Monotonicity of positive-base powers for $nN \geq 1$ now gives

$$0 < \delta \leq c_\delta(nN)^{-1/\Delta} \leq c_h(nN)^{-1/\Delta} = h \leq c_h \leq \frac{1}{2}.$$

The amplitude definitions then give $a, b > 0$.

3. *Separation and contrast calibration.* Using $\delta > 0$ and $S = \alpha + \beta$, multiplication of positive-base powers yields

$$\begin{aligned} ab &= \rho_0^2 \delta^{\alpha+\beta} \\ &= \rho_0^2 c_\delta^S (nN)^{-\gamma S/(S\Delta)} \\ &= c_{\text{sep}} (nN)^{-\gamma/\Delta}. \end{aligned}$$

Likewise,

$$h^\gamma = c_h^\gamma (nN)^{-\gamma/\Delta}.$$

The bounds $\rho_0^2 \leq \rho_0$, $c_\delta^S \leq 1$, and $\rho_0 \leq c_{\text{cert}} c_h^\gamma$ therefore imply

$$\begin{aligned} ab &= \rho_0^2 c_\delta^S (nN)^{-\gamma/\Delta} \\ &\leq \rho_0 (nN)^{-\gamma/\Delta} \\ &\leq c_{\text{cert}} c_h^\gamma (nN)^{-\gamma/\Delta} = c_{\text{cert}} h^\gamma. \end{aligned}$$

4. *Occupancy, class membership, and mixture distance.* The definitions of the rate parameters give

$$q_* = \frac{S_{\text{crit}}}{S} \geq 0, \quad \Delta = (2\gamma + d)(1 + q_*), \quad \frac{\gamma d}{S\Delta} = \frac{q_*}{1 + q_*}.$$

Substituting the scale δ and collecting powers therefore gives

$$\begin{aligned} N\delta^d &= c_\delta^d n^{-q_*/(1+q_*)} N^{1/(1+q_*)} \\ &= c_\delta^d \left(\frac{N}{n^{q_*}} \right)^{1/(1+q_*)} \leq c_\delta^d \leq c_0. \end{aligned}$$

Here the boundary condition makes the positive ratio N/n^{q_*} at most one, and $1/(1+q_*) > 0$. The remaining amplitude hypotheses follow from $\rho_0 \leq c_{\text{cert}}$:

$$0 < a \leq c_{\text{cert}} \delta^\alpha, \quad 0 < b \leq c_{\text{cert}} \delta^\beta.$$

All hypotheses of [Theorem 4](#) are thus satisfied for the construction of [Algorithm 3](#). For its finite lattice index set $\mathcal{I}$, write

$$\mathcal{S}_{\text{sign}} = \{-1, +1\}^{\mathcal{I}} \times \{-1, +1\}^{\mathcal{I}}.$$

The certificate supplies probability priors ϖ_θ and, for every $\theta \in \{-1, +1\}$, $(\lambda, \eta) \in \mathcal{S}_{\text{sign}}$, and $x \in [0, 1]^d$,

$$P_{\theta, \lambda, \eta} \in \mathcal{M}, \quad \tau_{P_{\theta, \lambda, \eta}}(x) = \tau_\theta(x).$$

At the evaluation profile this gives

$$\tau_{P_{-1, \lambda, \eta}}(x_0) = 2ab, \quad \tau_{P_{+1, \lambda, \eta}}(x_0) = -2ab.$$

Define the mixtures of the original data blocks by

$$M_\theta = \int \mathcal{L}_P(\mathcal{D}_{n,m}) d\varpi_\theta(P), \quad \theta \in \{-1, +1\}.$$

The certificate also supplies

$$H^2(M_{+1}, M_{-1}) \leq CnNh^d\delta^d a^2 b^2.$$

To evaluate its right-hand side, first note that

$$\delta^d a^2 b^2 = \rho_0^4 \delta^{d+2S}, \quad d + \frac{\gamma(d+2S)}{S} = d + \frac{\gamma d}{S} + 2\gamma = \Delta.$$

Hence

$$\begin{aligned} nNh^d\delta^d a^2 b^2 &= \rho_0^4 c_h^d c_\delta^{d+2S} (nN)^{1-d/\Delta-\gamma(d+2S)/(S\Delta)} \\ &= \rho_0^4 c_h^d c_\delta^{d+2S}. \end{aligned}$$

The inequality $C\rho_0^4 c_h^d c_\delta^{d+2S} \leq 1/16$ proves

$$H^2(M_{+1}, M_{-1}) \leq C\rho_0^4 c_h^d c_\delta^{d+2S} \leq \frac{1}{16}.$$

5. *The target as a functional of the experiment law.* The randomized observation space and the marked experiment model are

$$\Omega_{\text{obs}} = ([0, 1]^d \times \{0, 1\}^2)^n \times ([0, 1]^d \times \{0, 1\})^m \times [0, 1],$$

$$\mathcal{P}_{\text{exp}} = \{\mathbb{E}_{n,m}(P_{\theta,\lambda,\eta}) : \theta \in \{-1, +1\}, (\lambda, \eta) \in \mathcal{S}_{\text{sign}}\}.$$

When $m = 0$, the auxiliary-record factor is the one-point empty product.

We show that equal marked experiment laws have equal point targets. Take two constructed laws

$$P = P_{\theta,\lambda,\eta}, \quad P' = P_{\theta',\lambda',\eta'},$$

where both hypothesis indices belong to $\{-1, +1\}$ and both sign pairs belong to $\mathcal{S}_{\text{sign}}$, and suppose

$$\mathbb{E}_{n,m}(P) = \mathbb{E}_{n,m}(P').$$

Since $n > 0$, projection onto the first labeled record gives

$$\mathcal{L}_P(X, A, Y) = \mathcal{L}_{P'}(X, A, Y).$$

Both covariate laws are uniform. Uniqueness of conditional probability kernels therefore gives

$$\mathcal{L}_P(A, Y \mid X = x) = \mathcal{L}_{P'}(A, Y \mid X = x) \quad \text{for Lebesgue-almost every } x \in [0, 1]^d.$$

For the independent Bernoulli construction, with the response selection of [Definition 2](#), the four conditional cells are

$$\begin{aligned} \Pr_P(A = 1, Y = 1 \mid X = x) &= e_P(x)\mu_{1,P}(x), \\ \Pr_P(A = 1, Y = 0 \mid X = x) &= e_P(x)(1 - \mu_{1,P}(x)), \\ \Pr_P(A = 0, Y = 1 \mid X = x) &= (1 - e_P(x))\mu_{0,P}(x), \\ \Pr_P(A = 0, Y = 0 \mid X = x) &= (1 - e_P(x))(1 - \mu_{0,P}(x)). \end{aligned}$$

The same formulas hold for P' . Summing the first two equal cells identifies the propensity. The equal cells with $Y = 1$, together with overlap, then identify the arm means:

$$e_P(x) = e_{P'}(x), \quad \mu_{0,P}(x) = \mu_{0,P'}(x), \quad \mu_{1,P}(x) = \mu_{1,P'}(x) \quad \text{almost everywhere.}$$

Indeed, the factors used for division satisfy

$$e_P(x) \geq \varepsilon > 0, \quad 1 - e_P(x) \geq \varepsilon > 0.$$

Conditional independence in the prescribed construction now gives

$$\begin{aligned} \mathcal{L}_P(A, Y(0), Y(1) \mid X = x) &= \text{Bernoulli}(e_P(x)) \otimes \text{Bernoulli}(\mu_{0,P}(x)) \otimes \text{Bernoulli}(\mu_{1,P}(x)) \\ &= \mathcal{L}_{P'}(A, Y(0), Y(1) \mid X = x) \quad \text{almost everywhere.} \end{aligned}$$

Integration against their common uniform covariate law proves equality of the full primitive laws.

Their common primitive law thus gives identical Bernoulli margin laws for $A, Y(0), Y(1)$, so admissible conditional-margin versions agree almost everywhere. The positive Hölder orders give continuity of $e_P, \mu_{0,P}, \tau_P$, and

$$\mu_{1,P} = \mu_{0,P} + \tau_P \quad \text{on } [0, 1]^d$$

gives continuity of the treated mean; the same holds for P' . Since the uniform law has full support on the cube, these continuous versions agree everywhere. In particular,

$$\tau_P(x_0) = \tau_{P'}(x_0).$$

We may consequently define a functional on all nonnegative Borel measures on Ω_{obs} by

$$\Psi(\mathbb{P}) = \begin{cases} \tau_{P_{\theta, \lambda, \eta}}(x_0), & \mathbb{P} = \mathbb{E}_{n,m}(P_{\theta, \lambda, \eta}) \text{ for some } \theta \in \{-1, +1\}, (\lambda, \eta) \in \mathcal{S}_{\text{sign}}, \\ 0, & \mathbb{P} \notin \mathcal{P}_{\text{exp}}. \end{cases}$$

The equality just proved makes the first case independent of the representative. Thus

$$\Psi(\mathbb{E}_{n,m}(P_{\theta, \lambda, \eta})) = \tau_{P_{\theta, \lambda, \eta}}(x_0)$$

throughout the marked family.

6. *A common prior for testing.* Use the finite index space

$$\mathcal{Z}_{\text{test}} = \mathcal{S}_{\text{sign}} \times \mathcal{S}_{\text{sign}}, \quad z_{\text{test}} = ((\lambda_-, \eta_-), (\lambda_+, \eta_+)).$$

Define the common prior by its point masses:

$$\omega(\{z_{\text{test}}\}) = w_{-1}(\lambda_-, \eta_-) w_{+1}(\lambda_+, \eta_+).$$

The nonnegative weights and their unit sums from [Theorem 4](#) make ω a probability measure. Define the two probability kernels by

$$\mathbb{B}_{z_{\text{test}}} = \mathbb{E}_{n,m}(P_{-1, \lambda_-, \eta_-}), \quad \mathbb{C}_{z_{\text{test}}} = \mathbb{E}_{n,m}(P_{+1, \lambda_+, \eta_+}).$$

Their values belong to $\mathcal{P}_{\text{exp}}$. For every $z_{\text{test}}, z'_{\text{test}} \in \mathcal{Z}_{\text{test}}$, the target identities give

$$\Psi(\mathbb{B}_{z_{\text{test}}}) - \Psi(\mathbb{C}_{z'_{\text{test}}}) = 2ab - (-2ab) = 4ab > 0.$$

Summing out the unused sign pair in each kernel gives the mixture identities

$$\begin{aligned} \int \mathbb{B}_{z_{\text{test}}} d\omega(z_{\text{test}}) &= M_{-1} \otimes \text{Unif}([0, 1]), \\ \int \mathbb{C}_{z_{\text{test}}} d\omega(z_{\text{test}}) &= M_{+1} \otimes \text{Unif}([0, 1]). \end{aligned}$$

The uniform factor is the independent randomizer. By the common-factor identity in [Lemma 4](#) and symmetry of the squared difference in [Definition 18](#),

$$\begin{aligned} \mathfrak{h}^2 \left(\int \mathbb{B}_{z_{\text{test}}} d\omega, \int \mathbb{C}_{z_{\text{test}}} d\omega \right) &= H^2(M_{-1}, M_{+1}) \\ &= H^2(M_{+1}, M_{-1}) \leq \frac{1}{16}. \end{aligned}$$

7. *Testing implies a minimax bound.* Apply [Lemma 5](#) to the model $\mathcal{P}_{\text{exp}}$, the functional Ψ , and the preceding kernels and prior, with

$$v_0 = 1, \quad s_0 = 4ab > 0, \quad u_0 = \frac{1}{16} \in [0, 2).$$

Here one observation is the entire randomized data block; the one-fold product space is identified with that block by its sole coordinate. For every Borel real-valued decision T on Ω_{obs} , the testing bound gives

$$\sup_{\mathbb{P} \in \mathcal{P}_{\text{exp}}} \int_{\Omega_{\text{obs}}} |T(w_{\text{obs}}) - \Psi(\mathbb{P})| d\mathbb{P}(w_{\text{obs}}) \geq ab \left(1 - \frac{\sqrt{63}}{32} \right).$$

The numerical overlap factor satisfies

$$63 \leq 64, \quad \frac{\sqrt{63}}{32} \leq \frac{1}{4}, \quad 1 - \frac{\sqrt{63}}{32} \geq \frac{3}{4}.$$

Moreover, the target representation and [Definition 4](#) give

$$\begin{aligned} &\sup_{\mathbb{P} \in \mathcal{P}_{\text{exp}}} \int_{\Omega_{\text{obs}}} |T(w_{\text{obs}}) - \Psi(\mathbb{P})| d\mathbb{P}(w_{\text{obs}}) \\ &= \sup_{\substack{\theta \in \{-1, +1\} \\ (\lambda, \eta) \in \mathcal{S}_{\text{sign}}}} \mathcal{R}_{n,m}(T, P_{\theta, \lambda, \eta}) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P), \end{aligned}$$

where the last inequality uses the certified class membership. Thus every such decision satisfies

$$\sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P) \geq \frac{3}{4}ab.$$

Taking the infimum over the decisions in [Definition 5](#) proves

$$R(n, m) \geq \frac{3}{4}ab.$$

All loss integrals and suprema here retain their extended nonnegative interpretation.

8. *Substitution of the separation scale.* The identity $ab = c_{\text{sep}}(nN)^{-\gamma/\Delta}$ now yields

$$R(n, m) \geq \frac{3}{4}ab = \frac{3}{4}c_{\text{sep}}(nN)^{-\gamma/\Delta} = c(nN)^{-\gamma/\Delta}.$$

The positive constant c depends only on the fixed public parameters, and the sample sizes were arbitrary in the stated range. □

Under the low-nuisance-smoothness condition in [Lemma 6](#), the exponent q_* exceeds one. The stated interval consequently covers total record counts from the fully labeled experiment through the crossover to supplied-propensity precision. Within this interval, the lower bound captures the joint contribution of labeled and outcome-free records through their count product.

The complete minimax characterization in [Theorem 2](#) organizes the lower bounds into complementary ranges. When $S < S_{\text{crit}}$ and $n \leq N \leq n^{q_*}$, [Lemma 6](#) supplies the unequal-count term. Supplying the propensity enlarges the information available to a decision, so the supplied-propensity risk in [Definition 6](#) is a lower bound for the original-record minimax risk in [Definition 5](#). For $N \geq n^{q_*}$, the supplied-propensity lower bound in [Theorem 3](#) supplies the order $n^{-\gamma/(2\gamma+d)}$, which is the larger term in the sharp rate. When $S \geq S_{\text{crit}}$, that same order is the larger term for every $N \geq n$. Thus the supplied-propensity floor resolves the ranges complementary to the sparse-occupancy construction.

Together with [Theorem 1](#), these lower bounds yield the matching characterization in [Theorem 2](#). For the formal crosswalk, the rate and attaining decision are collected in the ordered pair of [Definition 20](#) in [Section D](#).

The statistical interpretation and count consequences of this pair are stated once in [Theorem 2](#) and [Proposition 1](#).

C Acquisition consequences and supervised comparison

The sample requirements in [Proposition 1](#) follow from the two terms of the sharp rate. Throughout the acquisition analysis, fix $d \geq 1$, $\alpha, \beta \in (0, 1]$, $\gamma \in [1, 3]$, $L > 1/2$, and an overlap level $\varepsilon \in (0, 1/2)$, as in [Theorems 2](#) and [3](#). Retain the total treatment-record count $N = n + m$, summed nuisance smoothness $S = \alpha + \beta$, rate denominator Δ , smoothness threshold S_{crit} , and total-record exponent q_* defined there. The identities below express how labeled responses and treatment records jointly determine precision.

C.1 Complete acquisition statement

Proposition 1. `[prop:annotation-threshold]` (Annotation thresholds and acquisition requirements) *Fix public parameters satisfying*

- (*Dimension.*) $d \in \mathbb{N}$ and $d \geq 1$.
- (*Nuisance smoothness.*) $0 < \alpha \leq 1$ and $0 < \beta \leq 1$.
- (*Effect smoothness.*) $1 \leq \gamma \leq 3$.
- (*Radius.*) $L > 1/2$.
- (*Overlap.*) $0 < \varepsilon < 1/2$.

Let $R(n, m)$ and $R^e(n, m)$ be the minimax and supplied-propensity risks of [Definitions 5](#) and [6](#), and let $r_*(n, m)$ be the sharp rate of [Definition 14](#), with the fixed parameter arguments suppressed. Define the total treatment-record count N , the summed nuisance smoothness S , the tuning denominator Δ , the smoothness cutoff S_{crit} , the total-record exponent q_* , and the oracle rate r_o by

$$N = n + m, \quad S = \alpha + \beta, \quad \Delta = 2\gamma + d + \frac{\gamma d}{S}, \quad S_{\text{crit}} = \frac{\gamma d}{2\gamma + d}, \quad q_* = \frac{\gamma d}{S(2\gamma + d)}, \quad r_o(n) = n^{-\gamma/(2\gamma + d)}.$$

All constants below may depend on the fixed public parameters.

For every sequence $m_n \in \mathbb{N}$, the following equivalences hold:

$$\begin{aligned} \frac{R(n, m_n)}{R^e(n, m_n)} = O(1) &\iff r_*(n, m_n) = O(r_o(n)) \\ &\iff \exists \zeta \in [1, \infty) \\ \forall n \text{ sufficiently large,} \\ (n, m_n) \in \mathcal{B}_*(\zeta) &\iff \liminf_{n \rightarrow \infty} \frac{n + m_n}{n^{q_*}} > 0, \end{aligned}$$

where $\mathcal{B}_*(\zeta)$ is the region of [Definition 15](#); the lower limit is understood in the extended nonnegative reals. Moreover,

$$\begin{aligned} \frac{R(n, m_n)}{R(n, 0)} \rightarrow 0 &\iff \frac{r_*(n, m_n)}{r_*(n, 0)} \rightarrow 0 \\ &\iff S < S_{\text{crit}} \text{ and} \\ \frac{m_n}{n} &\rightarrow \infty. \end{aligned}$$

All sequence limits and order relations are as $n \rightarrow \infty$. If $S < S_{\text{crit}}$, then $q_* > 1$. For every sequence with $m_n/n = O(1)$,

$$R(n, m_n) = O(R(n, 0)), \quad R(n, 0) = O(R(n, m_n)).$$

If $S \geq S_{\text{crit}}$, then every auxiliary-size sequence satisfies

$$\frac{R(n, m_n)}{R^e(n, m_n)} = O(1).$$

There exist real constants c, C with $0 < c \leq C$ such that, for every $n, m \in \mathbb{N}$ with $n \geq 2$,

$$c r_*(n, m) \leq R(n, m) \leq C r_*(n, m).$$

For these sample sizes, the exact rate branches are

$$r_*(n, m) = \begin{cases} r_o(n), & S \geq S_{\text{crit}}, \\ (nN)^{-\gamma/\Delta}, & S < S_{\text{crit}}, \ N \leq n^{q_*}, \\ r_o(n), & S < S_{\text{crit}}, \ N \geq n^{q_*}. \end{cases}$$

At every boundary pair with $N = n^{q_*}$,

$$(nN)^{-\gamma/\Delta} = r_o(n).$$

In particular, for every $n \geq 2$,

$$r_*(n, 0) = \begin{cases} n^{-2\gamma/\Delta}, & S < S_{\text{crit}}, \\ r_o(n), & S \geq S_{\text{crit}}. \end{cases}$$

Writing $s_{\text{pub}} = S/2$ for the average nuisance smoothness, the numerical supervised benchmark coincides with $r_*(n, 0)$: for every $n \in \mathbb{N}$,

$$\begin{cases} n^{-1/(1+d/(2\gamma)+d/(4s_{\text{pub}}))}, & s_{\text{pub}} < \frac{d/4}{1+d/(2\gamma)}, \\ n^{-1/(2+d/\gamma)}, & s_{\text{pub}} \geq \frac{d/4}{1+d/(2\gamma)} \end{cases} = r_*(n, 0).$$

In this identity at $n = 0$, negative powers of zero are assigned the value zero.

Finally, there exist real constants $c, C > 0$, separate from the preceding pair, such that the following acquisition guarantees hold for every target absolute-error scale $0 < \eta < 1$ and every $n, m \in \mathbb{N}$ with $n \geq 2$. Let $\mathcal{M}$ be the primitive class at the fixed parameters from [Definition 3](#), and use the risk of [Definition 4](#).

If there exists a Borel decision T satisfying

$$\mathcal{R}_{n,m}(T, P) \leq \eta \quad \text{for every } P \in \mathcal{M},$$

then

$$c\eta^{-(2+d/\gamma)} \leq n, \quad c\eta^{-(2+d/\gamma+d/S)} \leq n(n+m).$$

Conversely, if

$$C\eta^{-(2+d/\gamma)} \leq n, \quad C\eta^{-(2+d/\gamma+d/S)} \leq n(n+m),$$

then the sharp decision T_* of [Definition 13](#) is Borel and satisfies

$$\mathcal{R}_{n,m}(T_*, P) \leq \eta \quad \text{for every } P \in \mathcal{M}.$$

The exact rate-level acquisition condition is

$$r_*(n, m) \leq \eta \iff \eta^{-(2+d/\gamma)} \leq n \text{ and } \eta^{-(2+d/\gamma+d/S)} \leq n(n+m).$$

The total treatment-record count also satisfies $n + m \geq n$.

C.2 Oracle attainment and strict benefit

Since $\Delta = (2\gamma + d)(1 + q_*)$, the sharp rate in [Definition 14](#) satisfies

$$\frac{r_*(n, m)}{r_o(n)} = \max \left\{ 1, \left(\frac{n^{q_*}}{N} \right)^{\gamma/\Delta} \right\}.$$

Thus, for any finite tolerance $\zeta \geq 1$, the annotation region in [Definition 15](#) has the exact characterization

$$(n, m) \in \mathcal{B}_*(\zeta) \iff N \geq \zeta^{-\Delta/\gamma} n^{q_*}, \quad n \geq 2, \quad m \geq 0.$$

The tolerance measures the remaining nuisance contribution relative to the supplied-propensity order. At $\zeta = 1$, the count threshold is the point at which the two terms coincide.

For an auxiliary-size sequence (m_n) , write $N_n = n + m_n$. The uniform risk comparisons in [Theorems 2](#) and [3](#) give

$$\frac{R(n, m_n)}{R^e(n, m_n)} \asymp \frac{r_*(n, m_n)}{r_o(n)}.$$

Consequently, bounded relative risk is equivalent to an eventual positive lower bound on N_n/n^{q_*} , or, equivalently,

$$\frac{R(n, m_n)}{R^e(n, m_n)} = O(1) \iff \liminf_{n \rightarrow \infty} \frac{N_n}{n^{q_*}} > 0.$$

This is the oracle-attainment equivalence in [Proposition 1](#). When $S \geq S_{\text{crit}}$, the inequality $q_* \leq 1$, together with $N_n \geq n$, places every auxiliary-size sequence in this regime.

Strict benefit compares precision with labeled records alone. In the regime $S < S_{\text{crit}}$, one has $q_* > 1$ and $r_*(n, 0) = n^{-2\gamma/\Delta}$. Hence

$$\frac{r_*(n, m_n)}{r_*(n, 0)} = \max \left\{ n^{-\gamma(q_*-1)/\Delta}, \left(\frac{n}{N_n} \right)^{\gamma/\Delta} \right\}.$$

The first term tends to zero, and the second tends to zero exactly when $m_n/n \rightarrow +\infty$. For $S \geq S_{\text{crit}}$, the rate ratio equals one. The uniform minimax comparisons therefore yield the criterion in [Proposition 1](#):

$$\frac{R(n, m_n)}{R(n, 0)} \rightarrow 0 \iff S < S_{\text{crit}} \quad \text{and} \quad \frac{m_n}{n} \rightarrow +\infty.$$

Strict benefit and oracle attainment thus describe distinct acquisition requirements. In the low-smoothness regime, a diverging auxiliary-to-labeled ratio produces a strict improvement, while attainment of the supplied-propensity order requires total treatment information of order at least n^{q_*} . A bounded auxiliary-to-labeled ratio preserves the supervised minimax order.

C.3 Target-error acquisition

For a target absolute-error scale $0 < \eta < 1$, inversion of the two rate terms gives the exact identity

$$r_*(n, m) \leq \eta \iff \begin{cases} n \geq \eta^{-(2+d/\gamma)}, \\ nN \geq \eta^{-(2+d/\gamma+d/S)}. \end{cases}$$

The first inequality specifies the response-label requirement. The second specifies the joint requirement on labeled and treatment-record counts.

To connect these numerical conditions to risk, let $c_R, C_R > 0$ be comparison constants from [Theorem 2](#), so that

$$c_R r_*(n, m) \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P) \leq C_R r_*(n, m).$$

A Borel decision whose risk is at most η uniformly over $\mathcal{M}$ necessarily satisfies

$$n \geq (c_R/\eta)^{2+d/\gamma}, \quad nN \geq (c_R/\eta)^{2+d/\gamma+d/S}.$$

Conversely, the sufficient conditions

$$n \geq (C_R/\eta)^{2+d/\gamma}, \quad nN \geq (C_R/\eta)^{2+d/\gamma+d/S}$$

ensure that the decision T_* in [Definition 13](#) has uniform risk at most η . Absorbing the fixed powers of these comparison constants gives the necessity and sufficiency constants in [Proposition 1](#).

At the rate level, a fixed labeled count satisfying the first inequality requires

$$N \geq \max \left\{ n, \frac{\eta^{-(2+d/\gamma+d/S)}}{n} \right\}.$$

In particular, choosing $n \asymp \eta^{-(2+d/\gamma)}$ gives the total-record requirement

$$N \gtrsim \max \left\{ \eta^{-(2+d/\gamma)}, \eta^{-d/S} \right\}.$$

The smoothness threshold compares these two powers. Below S_{crit} , the treatment-record requirement grows faster than the labeled requirement; at and above the threshold, the labeled count already supplies the required treatment information.

C.4 The supervised numerical benchmark

At $m = 0$, the sharp rate becomes

$$r_*(n, 0) = \max \left\{ n^{-\gamma/(2\gamma+d)}, n^{-2\gamma/\Delta} \right\}.$$

Its exponents and smoothness cutoff coincide with the numerical benchmark in [Kennedy et al. \[2024, Section 3, Theorem 1\]](#).

The cited theorem concerns a supervised model with a covariate density bounded above by a constant, an α -smooth propensity, a β -smooth control regression, and a γ -smooth contrast. It gives a minimax mean-absolute-error lower bound for sample sizes larger than a constant depending on $(\alpha, \beta, \gamma, d)$. Its implicit multiplicative constant is positive and independent of sample size, with the density bound and Hölder constants fixed; these are source-model constants. The density restriction accommodates the construction-specific designs used in the source testing argument [[Kennedy et al., 2024, Section 3.1](#)]. The following identities express the numerical comparison in the notation of this paper.

Lemma 7. [`lem:published-cate-benchmark`] (Published CATE benchmark identities) *Let the dimension $d \in \mathbb{N}$ and smoothness orders $\alpha, \beta, \gamma \in \mathbb{R}$ satisfy:*

- **(Dimension.)** $1 \leq d$.
- **(First nuisance smoothness.)** $0 < \alpha$.
- **(Second nuisance smoothness.)** $0 < \beta$.
- **(Effect smoothness.)** $1 \leq \gamma$.

Put $s_{\text{pub}} = (\alpha + \beta)/2$, the average nuisance smoothness. The numerical exponents and smoothness cutoff displayed in [Kennedy et al. \[2024, Section 3, Theorem 1\]](#) satisfy

$$\frac{1}{1 + d/(2\gamma) + d/(4s_{\text{pub}})} = \frac{2\gamma}{2\gamma + d + \gamma d/(\alpha + \beta)}, \quad \frac{1}{2 + d/\gamma} = \frac{\gamma}{2\gamma + d},$$

and

$$s_{\text{pub}} < \frac{d/4}{1 + d/(2\gamma)} \iff \alpha + \beta < \frac{\gamma d}{2\gamma + d}.$$

For every $n \in \mathbb{N}$ with $n \geq 2$, the numerical piecewise benchmark satisfies

$$\begin{cases} n^{-1/(1+d/(2\gamma)+d/(4s_{\text{pub}}))}, & s_{\text{pub}} < (d/4)/(1 + d/(2\gamma)), \\ n^{-1/(2+d/\gamma)}, & s_{\text{pub}} \geq (d/4)/(1 + d/(2\gamma)) \end{cases} = \max \left\{ n^{-\gamma/(2\gamma+d)}, n^{-2\gamma/(2\gamma+d+\gamma d/(\alpha+\beta))} \right\}.$$

Proof of Lemma 7. 1. The hypotheses give

$$d > 0, \quad \gamma > 0, \quad \alpha + \beta > 0.$$

Define the real denominators

$$D_{\text{or}} := 2\gamma + d, \quad D_{\text{pair}} := 2\gamma + d + \frac{\gamma d}{\alpha + \beta}.$$

Both are strictly positive:

$$D_{\text{or}} > 0, \quad D_{\text{pair}} > 0.$$

2. Clearing the positive denominators and simplifying gives

$$\frac{1}{1 + d/(2\gamma) + d/(4s_{\text{pub}})} = \frac{2\gamma}{D_{\text{pair}}}, \quad \frac{1}{2 + d/\gamma} = \frac{\gamma}{D_{\text{or}}}.$$

Likewise,

$$\frac{d/4}{1 + d/(2\gamma)} = \frac{1}{2} \frac{\gamma d}{D_{\text{or}}}.$$

Since $s_{\text{pub}} = (\alpha + \beta)/2$, multiplication by $2 > 0$ therefore yields

$$s_{\text{pub}} < \frac{d/4}{1 + d/(2\gamma)} \iff \alpha + \beta < \frac{\gamma d}{D_{\text{or}}}.$$

These are the asserted exponent identities and cutoff equivalence.

3. Fix $n \in \mathbb{N}$ with $n \geq 2$, so that

$$n \geq 1.$$

After substituting the exponent identities, the numerical piecewise benchmark becomes

$$\begin{cases} n^{-2\gamma/D_{\text{pair}}}, & s_{\text{pub}} < (d/4)/(1 + d/(2\gamma)), \\ n^{-\gamma/D_{\text{or}}}, & s_{\text{pub}} \geq (d/4)/(1 + d/(2\gamma)). \end{cases}$$

We compare these two powers by splitting at the equivalent cutoff in the preceding step.

4. Suppose first that

$$\alpha + \beta < \frac{\gamma d}{D_{\text{or}}}.$$

Multiplying by $D_{\text{or}} > 0$, then dividing by $\alpha + \beta > 0$, gives

$$(\alpha + \beta)D_{\text{or}} < \gamma d, \quad D_{\text{or}} < \frac{\gamma d}{\alpha + \beta}.$$

Consequently,

$$2D_{\text{or}} \leq D_{\text{or}} + \frac{\gamma d}{\alpha + \beta} = D_{\text{pair}}.$$

Multiplying by $\gamma > 0$ and dividing by the positive product $D_{\text{pair}}D_{\text{or}}$ yields

$$\frac{2\gamma}{D_{\text{pair}}} \leq \frac{\gamma}{D_{\text{or}}}, \quad -\frac{\gamma}{D_{\text{or}}} \leq -\frac{2\gamma}{D_{\text{pair}}}.$$

For a base $n \geq 1$, real powers are nondecreasing in their exponent. Hence

$$n^{-\gamma/D_{\text{or}}} \leq n^{-2\gamma/D_{\text{pair}}},$$

and thus

$$\max \left\{ n^{-\gamma/D_{\text{or}}}, n^{-2\gamma/D_{\text{pair}}} \right\} = n^{-2\gamma/D_{\text{pair}}}.$$

The cutoff equivalence selects precisely this first branch of the piecewise benchmark.

5. In the complementary case,

$$\frac{\gamma d}{D_{\text{or}}} \leq \alpha + \beta.$$

Multiplication by $D_{\text{or}} > 0$, followed by division by $\alpha + \beta > 0$, gives

$$\gamma d \leq (\alpha + \beta)D_{\text{or}}, \quad \frac{\gamma d}{\alpha + \beta} \leq D_{\text{or}}.$$

It follows that

$$D_{\text{pair}} = D_{\text{or}} + \frac{\gamma d}{\alpha + \beta} \leq 2D_{\text{or}}.$$

Multiplying by $\gamma > 0$ and dividing by $D_{\text{or}}D_{\text{pair}} > 0$ now gives

$$\frac{\gamma}{D_{\text{or}}} \leq \frac{2\gamma}{D_{\text{pair}}}, \quad -\frac{2\gamma}{D_{\text{pair}}} \leq -\frac{\gamma}{D_{\text{or}}}.$$

Monotonicity of real powers for $n \geq 1$ therefore implies

$$n^{-2\gamma/D_{\text{pair}}} \leq n^{-\gamma/D_{\text{or}}},$$

so

$$\max \left\{ n^{-\gamma/D_{\text{or}}}, n^{-2\gamma/D_{\text{pair}}} \right\} = n^{-\gamma/D_{\text{or}}}.$$

The cutoff equivalence selects the second branch, including equality at the cutoff. Substituting the definitions of D_{or} and D_{pair} proves the asserted maximum identity in both cases. □

Lemma 7 identifies both the exponents and the boundary between their regimes. Equality at the cutoff places the supervised benchmark at the supplied-propensity order. Under the present public-parameter restrictions, its maximum is exactly $r_*(n, 0)$.

For the known-uniform primitive class in **Definition 3**, the minimax interpretation of this expression follows from the original-record lower bounds. The construction in **Theorem 4** and the converse in **Lemma 6** provide the nuisance contribution in its governing range. The same-class subfamily with $e_P = \mu_{0,P} = 1/2$ in **Theorem 3** provides the labeled-response floor. Together with **Theorem 1**, these results establish the sharp risk comparison used throughout the acquisition analysis, while **Lemma 7** supplies the exponent-and-cutoff identity with the published supervised benchmark.

D Verification note

The formal results checked in Lean 4 are the estimator guarantee in [Theorem 1](#), the supplied-propensity benchmark in [Theorem 3](#), the smooth-frame construction certificate in [Theorem 4](#), and the nuisance lower bound in [Lemma 6](#). Their combination yields the minimax sandwich in [Theorem 2](#) and the acquisition consequences in [Proposition 1](#). Auxiliary checked results cover local polynomial projection, population positivity, rectangular sampling error, and Hellinger calculus in [Lemmas 1 to 4](#).

The companion artifact for this paper version, whose working-paper number and version appear in the first-page stamp, is available from https://causal-smith.org/papers/stat-twosample-pointcate-annotation-frontier_v1. Its interactive formal links provide the theorem-to-file crosswalk and exact source excerpts. In the source archive, the checked modules are rooted at `CausalSmith/Stat/STAT-TwosamplePointcateAnnotationFrontier_Research`; the generated file `presentation_crosswalk.json` records the mapping from every displayed result to its Lean module and declaration. The crosswalk is regenerated with each paper version, so it describes the results as printed in this version.

The primitive-model and sampling restrictions are assumed inputs. These comprise the known uniform covariate design, conditional exchangeability, a fixed public overlap level $\varepsilon \in (0, 1/2)$, binary potential responses, Hölder restrictions, and probability-valued arm-mean bounds defining [Definition 3](#), together with the independent product experiment and randomizer in [Assumption 9](#). Each guarantee retains its stated public-parameter, sample-size, scale, amplitude, and occupancy conditions.

The absolute-loss testing result in [Lemma 5](#) is proved in the formal artifact without external formal assumptions; [Kennedy et al. \[2024, Section 3, Lemma 1\]](#) remains its bibliographic source and statistical motivation. For the supervised comparison, [Lemma 7](#) checks the numerical exponent identities, smoothness cutoff, and piecewise-rate equality associated with [Kennedy et al. \[2024, Section 3, Theorem 1\]](#). The published statistical lower bound belongs to that source’s model and assumptions. The minimax interpretation for the present known-uniform primitive class is established by [Theorem 2](#), through this paper’s estimator guarantee and original-record lower bounds. The formal numerical identity also assigns totalized negative powers the value zero at $n = 0$; every statistical risk and acquisition guarantee in the paper uses $n \geq 2$, so this representation value has no precision interpretation.

For the machine-readable crosswalk, the verified rate and decision are also packaged in the following representation-level handle; no statistical argument depends on this packaging.

Definition 20. `[def:frontier-handle]` (Rate and decision handle) *The frontier handle is the ordered pair (r_*, T_*) , parameterized by $d, n, m \in \mathbb{N}$ and $\alpha, \beta, \gamma, \varepsilon \in \mathbb{R}$, where ε is the overlap parameter. Its components are the sharp rate and decision of [Definitions 13](#) and [14](#):*

$$r_* = \max \left\{ n^{-\gamma/(2\gamma+d)}, (n(n+m))^{-\gamma/(2\gamma+d+\gamma d/(\alpha+\beta))} \right\}, \quad T_*(\mathcal{D}_{n,m}, U) = \widehat{T}_{\text{up}}(\mathcal{D}_{n,m}).$$

Here $\mathcal{D}_{n,m}$ is the original dataset and $U \in \mathbb{R}$ is the randomizer coordinate. The decision component uses the publicly tuned estimator of [Definition 12](#) with parameters $d, \alpha, \beta, \gamma, \varepsilon, n, m$.

E Proofs of the main results

Proof of [Theorem 1](#). Fix the public parameters in the statement. Every constant constructed below depends only on these parameters.

1. Fix admissible n, m, h, J and $P \in \mathcal{M}$. Define

$$N = n + m, \quad S = \alpha + \beta, \quad \delta = \frac{h}{J}, \quad q = \binom{d+2}{2}, \quad k = qJ^d.$$

In particular, $N \geq n \geq 2$ and $\delta > 0$. By [Lemma 1](#), there is a constant $0 < C_{\text{loc}} < \infty$, uniform over these choices, and a coefficient vector $\vartheta \in \mathbb{R}^q$ such that

$$p_\vartheta(x) = r(x)^\top \vartheta, \quad \sup_{x \in C_h} |\tau_P(x) - p_\vartheta(x)| \leq C_{\text{loc}} h^\gamma, \quad \|\vartheta\|_2 \leq C_{\text{loc}}, \quad r_0^\top \vartheta = \tau_P(x_0).$$

The same result supplies

$$\|r_0\|_2 \leq C_{\text{loc}}, \quad \|e_P r_u - \Pi_J(e_P r_u)\|_{L_2(\nu_h)} \leq C_{\text{loc}} \delta^\alpha, \quad \|\mu_{0,P} - \Pi_J \mu_{0,P}\|_{L_2(\nu_h)} \leq C_{\text{loc}} \delta^\beta.$$

Here $x_0 \in C_h$, so the bound on r_0 follows from the stated uniform coarse-basis bound.

The finite Borel basis functions and the finite moment sums in [Algorithm 2](#) make both empirical moments finite Borel functions of the dataset. The eigenvalue cutoff is Borel: its matrix superlevel set is the intersection of the closed unit-vector quadratic-form inequalities. Matrix inversion on the invertible branch and clipping are Borel. Thus the decision in [Definition 11](#), with value zero on the failed-cutoff branch, is finite and Borel on every dataset. Composing with the dataset coordinate gives a Borel decision on the original dataset and randomizer.

2. Define the population moments and the localized Taylor remainder by

$$\bar{R} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{R}_{h,J}, \quad Q_{P,h,J} = \mathbb{E}_{\mathbb{E}_{n,m}(P)} \hat{Q}_{h,J}, \quad g_{\text{rem}} : C_h \longrightarrow \mathbb{R}, \quad g_{\text{rem}}(x) = \tau_P(x) - p_\vartheta(x).$$

All localized integrals below are over C_h .

The observed-response identity and the exchangeability condition in [Definition 3](#) give, almost surely,

$$\mathbb{E}_P[AY \mid X] = e_P(X) \mu_{1,P}(X), \quad \mathbb{E}_P[Y \mid X] = \mu_{0,P}(X) + e_P(X) \tau_P(X).$$

Indeed, $AY = AY(1)$, and conditional exchangeability factors its conditional mean. Applying the same argument to $(1 - A)Y(0)$ and adding gives the second identity.

The role counts in [Algorithm 1](#) satisfy

$$\ell = \lfloor n/2 \rfloor \geq 1, \quad t = N - \ell \geq 1.$$

Their records are independent under the sampling law in [Assumption 9](#). Consequently, taking expectations in [Algorithm 2](#), using uniform design and factoring each rectangular product, yields

$$\begin{aligned} \bar{R}_u &= \int r_u e_P \mu_{1,P} d\nu_h - \int e_P r_u \Pi_J(\mu_{0,P} + e_P \tau_P) d\nu_h, \\ (Q_{P,h,J})_{uv} &= \int r_u e_P r_v d\nu_h - \int e_P r_u \Pi_J(e_P r_v) d\nu_h. \end{aligned}$$

The response in the rectangular correction is Y , so its conditional mean is the second conditional mean displayed above.

For completeness, writing the orthonormal fine basis from [Definition 10](#) as

$$\mathbf{b}_J = (b_{J,\iota})_{\iota=1}^k,$$

its finite expansion gives, for arbitrary $f_1, f_2 \in L_2(\nu_h)$,

$$\int f_1 \Pi_J f_2 d\nu_h = \sum_{\iota=1}^k \left(\int b_{J,\iota} f_1 d\nu_h \right) \left(\int b_{J,\iota} f_2 d\nu_h \right) = \int (\Pi_J f_1) f_2 d\nu_h.$$

Orthonormality also gives

$$\int f_1 f_2 d\nu_h - \int f_1 \Pi_J f_2 d\nu_h = \int (f_1 - \Pi_J f_1)(f_2 - \Pi_J f_2) d\nu_h,$$

and

$$\|\Pi_J f_1\|_{L_2(\nu_h)}^2 + \|f_1 - \Pi_J f_1\|_{L_2(\nu_h)}^2 = \|f_1\|_{L_2(\nu_h)}^2.$$

These identities show that the population matrix displayed above is symmetric, so it agrees with the expectation of the symmetrized empirical matrix.

Multiplying its entries by ϑ , substituting $\mu_{1,P} = \mu_{0,P} + \tau_P$, and using the residual identity gives

$$\begin{aligned} \bar{R}_u - (Q_{P,h,J}\vartheta)_u &= \int (e_P r_u - \Pi_J(e_P r_u))(\mu_{0,P} - \Pi_J \mu_{0,P}) d\nu_h \\ &\quad + \int r_u e_P g_{\text{rem}} d\nu_h - \int \Pi_J(e_P r_u) e_P g_{\text{rem}} d\nu_h. \end{aligned}$$

Cauchy–Schwarz and the two residual bounds from the preceding step imply

$$\left| \int (e_P r_u - \Pi_J(e_P r_u))(\mu_{0,P} - \Pi_J \mu_{0,P}) d\nu_h \right| \leq C_{\text{loc}}^2 \delta^S.$$

The coarse coordinates are orthonormal, overlap bounds $|e_P|$ by one, and projection contraction follows from the displayed Pythagorean identity. Hence

$$\|r_u\|_{L_2(\nu_h)} = 1, \quad \|e_P r_u\|_{L_2(\nu_h)} \leq 1, \quad \|\Pi_J(e_P r_u)\|_{L_2(\nu_h)} \leq 1, \quad \|e_P g_{\text{rem}}\|_{L_2(\nu_h)} \leq C_{\text{loc}} h^\gamma.$$

Applying Cauchy–Schwarz to the remaining two integrals therefore gives

$$|\bar{R}_u - (Q_{P,h,J}\vartheta)_u| \leq C_{\text{loc}}^2 \delta^S + 2C_{\text{loc}} h^\gamma \leq (C_{\text{loc}}^2 + 2C_{\text{loc}})(h^\gamma + \delta^S).$$

Define

$$C_{\text{bias}} = \sqrt{q} (C_{\text{loc}}^2 + 2C_{\text{loc}}).$$

Squaring the coordinate bounds, summing over the q coordinates, and taking square roots proves

$$\|\bar{R} - Q_{P,h,J}\vartheta\|_2 \leq C_{\text{bias}}(h^\gamma + \delta^S).$$

3. Define the squared deviations, the bias scale, and the good event by

$$\begin{aligned} Z_R(\mathcal{D}_{n,m}, U) &= \|\hat{R}_{h,J} - \bar{R}\|_2^2, \\ Z_Q(\mathcal{D}_{n,m}, U) &= \|\hat{Q}_{h,J} - Q_{P,h,J}\|_F^2, \\ b_{\text{bias}} &= h^\gamma + \delta^S, \\ g_\varepsilon &= \frac{\varepsilon(1-\varepsilon)}{2}, \\ G_{\text{good}} &= \{(\mathcal{D}_{n,m}, U) : \lambda_{\min}(\hat{Q}_{h,J}) \geq g_\varepsilon\}. \end{aligned}$$

The first two functions ignore U .

On G_{good} , the Rayleigh bound is

$$\xi_{\text{inv}}^\top \widehat{Q}_{h,J} \xi_{\text{inv}} \geq g_\varepsilon \|\xi_{\text{inv}}\|_2^2 \quad (\xi_{\text{inv}} \in \mathbb{R}^q).$$

It excludes a nonzero kernel and thus makes the matrix invertible. For arbitrary $\xi_{\text{rhs}} \in \mathbb{R}^q$, set

$$\xi_{\text{inv}} = \widehat{Q}_{h,J}^{-1} \xi_{\text{rhs}}.$$

Cauchy–Schwarz then gives

$$g_\varepsilon \|\xi_{\text{inv}}\|_2^2 \leq \xi_{\text{inv}}^\top \xi_{\text{rhs}} \leq \|\xi_{\text{inv}}\|_2 \|\xi_{\text{rhs}}\|_2,$$

and consequently

$$\|\widehat{Q}_{h,J}^{-1} \xi_{\text{rhs}}\|_2 \leq g_\varepsilon^{-1} \|\xi_{\text{rhs}}\|_2.$$

For $\xi_{\text{inv}} = 0$ this is immediate; otherwise division by its positive norm proves it.

The inverse identity and the residual decomposition are

$$\widehat{Q}_{h,J}^{-1} \widehat{R}_{h,J} - \vartheta = \widehat{Q}_{h,J}^{-1} (\widehat{R}_{h,J} - \widehat{Q}_{h,J} \vartheta),$$

$$\widehat{R}_{h,J} - \widehat{Q}_{h,J} \vartheta = (\widehat{R}_{h,J} - \bar{R}) + (\bar{R} - Q_{P,h,J} \vartheta) + (Q_{P,h,J} - \widehat{Q}_{h,J}) \vartheta.$$

The triangle inequality and rowwise Cauchy–Schwarz imply

$$\|\widehat{R}_{h,J} - \widehat{Q}_{h,J} \vartheta\|_2 \leq \sqrt{Z_R} + C_{\text{bias}} b_{\text{bias}} + C_{\text{loc}} \sqrt{Z_Q}.$$

Because both arm means lie in $[0, 1]$,

$$\tau_P(x_0) \in [-1, 1].$$

Clipping toward this interval decreases absolute error: below its lower endpoint the clipped endpoint lies between the value and the target, above its upper endpoint the same holds, and inside the interval clipping preserves the value. Thus, on G_{good} ,

$$\begin{aligned} |\widehat{T}_{h,J} - \tau_P(x_0)| &\leq g_\varepsilon^{-1} \|r_0\|_2 \|\widehat{R}_{h,J} - \widehat{Q}_{h,J} \vartheta\|_2 \\ &\leq g_\varepsilon^{-1} C_{\text{loc}} \left(\sqrt{Z_R} + C_{\text{bias}} b_{\text{bias}} + C_{\text{loc}} \sqrt{Z_Q} \right). \end{aligned}$$

Define the positive finite constant

$$C_{\text{pt}} = g_\varepsilon^{-1} C_{\text{loc}} (1 + C_{\text{bias}} + C_{\text{loc}}).$$

All three coefficients in the preceding parentheses are at most $1 + C_{\text{bias}} + C_{\text{loc}}$. On the complementary event the decision is zero and its error is at most 1. Combining these branches gives, on every sample,

$$|\widehat{T}_{h,J} - \tau_P(x_0)| \leq C_{\text{pt}} (b_{\text{bias}} + \sqrt{Z_R} + \sqrt{Z_Q}) + \mathbf{1}_{G_{\text{good}}^c}.$$

4. By [Lemma 2](#),

$$v_{\text{unit}}^\top Q_{P,h,J} v_{\text{unit}} \geq \varepsilon(1 - \varepsilon) = 2g_\varepsilon \quad (v_{\text{unit}} \in \mathbb{R}^q, \|v_{\text{unit}}\|_2 = 1).$$

For each such vector, finite Cauchy–Schwarz gives

$$\left| v_{\text{unit}}^\top (\hat{Q}_{h,J} - Q_{P,h,J}) v_{\text{unit}} \right|^2 \leq Z_Q.$$

If $Z_Q < g_\varepsilon^2$, every unit-vector quadratic form of $\hat{Q}_{h,J}$ is at least g_ε , by subtracting this deviation from $2g_\varepsilon$. Its infimum is therefore at least g_ε . Consequently,

$$G_{\text{good}}^c \implies Z_Q \geq g_\varepsilon^2, \quad \mathbf{1}_{G_{\text{good}}^c} \leq g_\varepsilon^{-1} \sqrt{Z_Q}.$$

Define

$$\kappa_{\text{mean}} = C_{\text{pt}} + g_\varepsilon^{-1}.$$

The preceding pointwise bound now becomes

$$|\hat{T}_{h,J} - \tau_P(x_0)| \leq \kappa_{\text{mean}}(b_{\text{bias}} + \sqrt{Z_R} + \sqrt{Z_Q}).$$

Define the variance scale by

$$V_{\text{var}} = \frac{1}{nh^d} + \frac{qJ^d}{nNh^{2d}}.$$

By [Lemma 3](#), there is a uniform positive finite constant C_{samp} such that

$$\mathbb{E}_{\mathbb{E}_{n,m}(P)} Z_R + \mathbb{E}_{\mathbb{E}_{n,m}(P)} Z_Q \leq C_{\text{samp}} V_{\text{var}}.$$

Both deviations are nonnegative. Cauchy–Schwarz under the probability sampling law therefore gives

$$\begin{aligned} \mathbb{E}_{\mathbb{E}_{n,m}(P)} \sqrt{Z_R} &\leq \sqrt{\mathbb{E}_{\mathbb{E}_{n,m}(P)} Z_R} \leq \sqrt{C_{\text{samp}}} \sqrt{V_{\text{var}}}, \\ \mathbb{E}_{\mathbb{E}_{n,m}(P)} \sqrt{Z_Q} &\leq \sqrt{\mathbb{E}_{\mathbb{E}_{n,m}(P)} Z_Q} \leq \sqrt{C_{\text{samp}}} \sqrt{V_{\text{var}}}. \end{aligned}$$

The decision and target both belong to $[-1, 1]$, so their absolute difference is at most 2 and is integrable. Integrating the pointwise inequality and defining

$$C_{\text{mean}} = \kappa_{\text{mean}}(1 + 2\sqrt{C_{\text{samp}}})$$

proves

$$\begin{aligned} \mathcal{R}_{n,m}(\hat{T}_{h,J}, P) &\leq \kappa_{\text{mean}} \left(b_{\text{bias}} + 2\sqrt{C_{\text{samp}}} \sqrt{V_{\text{var}}} \right) \\ &\leq C_{\text{mean}}(b_{\text{bias}} + \sqrt{V_{\text{var}}}). \end{aligned}$$

5. Define

$$\rho_{\text{lab}} = (nh^d)^{-1/2}, \quad \rho_{\text{rect}} = (nNh^d \delta^d)^{-1/2}, \quad a_{\text{noise}} = \rho_{\text{lab}} + \rho_{\text{rect}}, \quad \kappa_{\text{rank}} = 1 + \sqrt{q}.$$

All these quantities are nonnegative, and $\kappa_{\text{rank}} \geq 1$. Since $\delta = h/J$,

$$\rho_{\text{lab}}^2 = \frac{1}{nh^d}, \quad \rho_{\text{rect}}^2 = \frac{J^d}{nNh^{2d}}, \quad V_{\text{var}} = \rho_{\text{lab}}^2 + q\rho_{\text{rect}}^2.$$

It follows that

$$V_{\text{var}} \leq (\rho_{\text{lab}} + \sqrt{q} \rho_{\text{rect}})^2, \quad \rho_{\text{lab}} + \sqrt{q} \rho_{\text{rect}} \leq \kappa_{\text{rank}} a_{\text{noise}},$$

and hence

$$\sqrt{V_{\text{var}}} \leq \kappa_{\text{rank}} a_{\text{noise}}.$$

Define

$$C_{\text{fixed}} = C_{\text{mean}} \kappa_{\text{rank}} + 2.$$

Using also $b_{\text{bias}} \leq \kappa_{\text{rank}} b_{\text{bias}}$, the preceding risk bound gives

$$\mathcal{R}_{n,m}(\hat{T}_{h,J}, P) \leq C_{\text{fixed}}(b_{\text{bias}} + a_{\text{noise}}).$$

If $b_{\text{bias}} + a_{\text{noise}} \leq 1$, this is the desired bound with the minimum. If $b_{\text{bias}} + a_{\text{noise}} > 1$, the pointwise diameter bound gives

$$\mathcal{R}_{n,m}(\hat{T}_{h,J}, P) \leq 2 \leq C_{\text{fixed}}.$$

Thus, uniformly over $P \in \mathcal{M}$,

$$\mathcal{R}_{n,m}(\hat{T}_{h,J}, P) \leq C_{\text{fixed}} \min \left\{ 1, h^\gamma + \delta^S + (nh^d)^{-1/2} + (nNh^d\delta^d)^{-1/2} \right\}.$$

6. We next establish the public tuning bound. Define

$$\Delta_{\text{o}} = 2\gamma + d, \quad \Delta = 2\gamma + d + \frac{\gamma d}{S}, \quad c_{\text{lab}} = 2^{\Delta_{\text{o}}/2}, \quad c_{\text{rect}} = 2^{d/2} 2^{\Delta/2} + 2^{\Delta_{\text{o}}}, \quad C_{\text{bal}} = 2 + c_{\text{lab}} + c_{\text{rect}}.$$

The parameter domain gives $S > 0$, $\gamma > 0$, $\Delta_{\text{o}} > 0$, and $\Delta > 0$, so these constants are positive and finite.

For fixed $n \geq 2$ and $m \geq 0$, define the two bandwidth orders

$$u_{\text{o}} = n^{-1/\Delta_{\text{o}}}, \quad u_{\text{rect}} = (nN)^{-1/\Delta}.$$

Because $n \geq 1$ and $nN \geq 1$,

$$0 < u_{\text{o}} \leq 1, \quad 0 < u_{\text{rect}} \leq 1.$$

The definition in [Definition 12](#) consequently gives

$$h_* = \frac{1}{2} \max\{u_{\text{o}}, u_{\text{rect}}\}, \quad 0 < h_* \leq \frac{1}{2}, \quad u_{\text{o}} \leq 2h_*, \quad u_{\text{rect}} \leq 2h_*.$$

Moreover,

$$\begin{aligned} h_*^\gamma &\leq \max\{u_{\text{o}}^\gamma, u_{\text{rect}}^\gamma\} \\ &= \max \left\{ n^{-\gamma/\Delta_{\text{o}}}, (nN)^{-\gamma/\Delta} \right\} = r_{\text{up}}(n, m), \end{aligned}$$

using [Definition 14](#).

Taking logarithms of $u_{\text{o}} \leq 2h_*$ gives

$$-\log n \leq \Delta_{\text{o}}(\log 2 + \log h_*).$$

Therefore

$$\begin{aligned}\log((nh_*^d)^{-1/2}) &= -\frac{1}{2}(\log n + d \log h_*) \\ &\leq \frac{\Delta_o}{2} \log 2 + \gamma \log h_*,\end{aligned}$$

where $\Delta_o - d = 2\gamma$. Exponentiating yields

$$(nh_*^d)^{-1/2} \leq c_{\text{lab}} h_*^\gamma.$$

7. Suppose first that $S < \gamma$. Define

$$z_{\text{grid}} = h_*^{-(\gamma/S-1)}.$$

Then $z_{\text{grid}} \geq 1$, and [Definition 12](#) gives

$$J_* = \lceil z_{\text{grid}} \rceil, \quad 1 \leq J_*, \quad z_{\text{grid}} \leq J_* < z_{\text{grid}} + 1 \leq 2z_{\text{grid}}.$$

Since $h_*/z_{\text{grid}} = h_*^{\gamma/S}$, these inequalities imply

$$0 < \frac{h_*}{J_*}, \quad \frac{1}{2}h_*^{\gamma/S} \leq \frac{h_*}{J_*} \leq h_*^{\gamma/S}, \quad \left(\frac{h_*}{J_*}\right)^S \leq h_*^\gamma.$$

The lower cell-side bound gives

$$\log(h_*/J_*) \geq \frac{\gamma}{S} \log h_* - \log 2.$$

Substituting this inequality into the logarithm of the rectangular sampling term and exponentiating proves

$$\left(nNh_*^d(h_*/J_*)^d\right)^{-1/2} \leq 2^{d/2} \left(nNh_*^{d+\gamma d/S}\right)^{-1/2}.$$

Similarly, $u_{\text{rect}} \leq 2h_*$ gives

$$-\log(nN) \leq \Delta(\log 2 + \log h_*).$$

As $\Delta - (d + \gamma d/S) = 2\gamma$,

$$\begin{aligned}\log \left(\left(nNh_*^{d+\gamma d/S}\right)^{-1/2} \right) &= -\frac{1}{2}(\log(nN) + (d + \gamma d/S) \log h_*) \\ &\leq \frac{\Delta}{2} \log 2 + \gamma \log h_*.\end{aligned}$$

Consequently,

$$\left(nNh_*^d(h_*/J_*)^d\right)^{-1/2} \leq 2^{d/2} 2^{\Delta/2} h_*^\gamma \leq c_{\text{rect}} h_*^\gamma.$$

8. Suppose instead that $S \geq \gamma$, including equality. The exponent in [Definition 12](#) is then zero, so

$$J_* = 1, \quad 0 < \frac{h_*}{J_*} = h_*, \quad \left(\frac{h_*}{J_*}\right)^S = h_*^S \leq h_*^\gamma,$$

because $0 < h_* \leq 1$.

The oracle bandwidth inequality also reads

$$(n^2)^{-1/(2\Delta_o)} = u_o \leq 2h_*.$$

Taking logarithms gives

$$-\log(n^2) \leq 2\Delta_o(\log 2 + \log h_*).$$

Using $2\Delta_o - 2d = 4\gamma$, we obtain

$$\begin{aligned} \log \left((n^2 h_*^{2d})^{-1/2} \right) &= -\frac{1}{2} (\log(n^2) + 2d \log h_*) \\ &\leq \Delta_o \log 2 + 2\gamma \log h_*. \end{aligned}$$

Since $N \geq n$ and $h_*^{2\gamma} \leq h_*^\gamma$,

$$\begin{aligned} \left(n N h_*^d (h_*/J_*)^d \right)^{-1/2} &= (n N h_*^{2d})^{-1/2} \\ &\leq (n^2 h_*^{2d})^{-1/2} \\ &\leq 2^{\Delta_o} h_*^{2\gamma} \\ &\leq c_{\text{rect}} h_*^\gamma. \end{aligned}$$

Thus both cases give admissible tuning and the four-term balance

$$\begin{aligned} 0 < h_* \leq \frac{1}{2}, \quad 1 \leq J_*, \\ h_*^\gamma + (h_*/J_*)^S + (n h_*^d)^{-1/2} + \left(n N h_*^d (h_*/J_*)^d \right)^{-1/2} \\ \leq (2 + c_{\text{lab}} + c_{\text{rect}}) h_*^\gamma \leq C_{\text{bal}} r_{\text{up}}(n, m). \end{aligned}$$

9. Finally, define one constant for both assertions:

$$C = C_{\text{fixed}}(C_{\text{bal}} + 1).$$

It is positive and finite, and

$$C_{\text{fixed}} \leq C, \quad C_{\text{fixed}} C_{\text{bal}} \leq C.$$

The fixed-tuning envelope has a nonnegative minimum, so enlarging its constant from C_{fixed} to C and taking the supremum over $P \in \mathcal{M}$ proves the first risk assertion.

For the second assertion, the tuning bounds make (h_*, J_*) admissible for the fixed-tuning result. The identity in [Definition 12](#) supplies

$$\hat{T}_{\text{up}} = \hat{T}_{h_*, J_*},$$

and hence also its finiteness and Borel measurability. Using the minimum's upper bound by its four-term argument gives, for every $P \in \mathcal{M}$,

$$\begin{aligned} \mathcal{R}_{n,m}(\hat{T}_{\text{up}}, P) &\leq C_{\text{fixed}} \left\{ h_*^\gamma + (h_*/J_*)^S + (n h_*^d)^{-1/2} + \left(n N h_*^d (h_*/J_*)^d \right)^{-1/2} \right\} \\ &\leq C_{\text{fixed}} C_{\text{bal}} r_{\text{up}}(n, m) \\ &\leq C r_{\text{up}}(n, m). \end{aligned}$$

The rate is nonnegative as the maximum of two positive powers of positive sample counts. All constants are uniform over the primitive class, so taking the supremum proves the tuned assertion. $\square$

Proof of Theorem 2. Fix the public parameters as in the statement. Throughout the proof, use

$$N = n + m, \quad S = \alpha + \beta, \quad \Delta = 2\gamma + d + \frac{\gamma d}{S}, \quad S_{\text{crit}} = \frac{\gamma d}{2\gamma + d}, \quad q_* = \frac{\gamma d}{S(2\gamma + d)},$$

and

$$r_o(n) = n^{-\gamma/(2\gamma+d)}, \quad r_* = \max\{r_o(n), (nN)^{-\gamma/\Delta}\},$$

where the last equality is Definition 14. All sample sizes below satisfy $n, m \in \mathbb{N}$ and $n \geq 2$.

1. *Supplied-propensity lower bound.* By Theorem 3, there is a constant $c_O > 0$, depending on the fixed public parameters, such that

$$c_O r_o(n) \leq R^e(n, m) \quad \text{for every } n, m.$$

Every admissible decision T induces a supplied-propensity decision by

$$T^e(\mathcal{D}_{n,m}, U, e) = T(\mathcal{D}_{n,m}, U)$$

for every function $e : [0, 1]^d \rightarrow \mathbb{R}$. For each supplied function this decision is Borel, and its loss under each $P \in \mathcal{M}$ is

$$\int |T^e(\mathcal{D}_{n,m}, U, e_P) - \tau_P(x_0)| d\mathbb{E}_{n,m}(P) = \mathcal{R}_{n,m}(T, P).$$

Taking the supremum over the same primitive class and then the infimum over decisions, using Definitions 5 and 6, gives

$$R^e(n, m) \leq R(n, m).$$

Consequently,

$$c_O r_o(n) \leq R(n, m) \quad \text{for every } n, m.$$

2. *A uniform lower constant.* First suppose $S < S_{\text{crit}}$. By Lemma 6, there is a constant $c_B > 0$, depending on the fixed public parameters, such that

$$c_B (nN)^{-\gamma/\Delta} \leq R(n, m) \quad \text{whenever } n \leq N \leq n^{q_*}.$$

Choose

$$c = \min\{c_O, c_B\} > 0.$$

We establish the rate comparisons used in this case and in the remaining smoothness case. The parameter conditions imply

$$S > 0, \quad \gamma > 0, \quad 2\gamma + d > 0, \quad \Delta > 0, \quad q_* > 0, \quad N \geq n \geq 2.$$

Direct substitution gives

$$q_* = \frac{\gamma d}{S(2\gamma + d)} = \frac{S_{\text{crit}}}{S},$$

and

$$\Delta = (2\gamma + d) + \frac{\gamma d}{S} = (2\gamma + d) \left(1 + \frac{\gamma d}{S(2\gamma + d)} \right) = (2\gamma + d)(1 + q_*).$$

Since $n > 0$, the laws of real powers therefore give the boundary identity

$$\begin{aligned} (n n^{q_*})^{-\gamma/\Delta} &= (n^{1+q_*})^{-\gamma/\Delta} \\ &= n^{-(1+q_*)\gamma/\Delta} \\ &= n^{-\gamma/(2\gamma+d)} = r_o(n). \end{aligned}$$

The exponent $-\gamma/\Delta$ is negative. Thus, when $N \geq n^{q_*}$,

$$0 < n n^{q_*} \leq nN \implies (nN)^{-\gamma/\Delta} \leq (n n^{q_*})^{-\gamma/\Delta} = r_o(n).$$

When $N \leq n^{q_*}$, instead,

$$0 < nN \leq n n^{q_*} \implies r_o(n) = (n n^{q_*})^{-\gamma/\Delta} \leq (nN)^{-\gamma/\Delta}.$$

For completeness, if $S \geq S_{\text{crit}}$, then

$$q_* = \frac{S_{\text{crit}}}{S} \leq 1, \quad n^{q_*} \leq n^1 = n \leq N,$$

where the power comparison uses $n \geq 1$. Taking the maximum in the definition of r_* consequently gives

$$\begin{aligned} S \geq S_{\text{crit}} &\implies r_* = r_o(n), \\ S < S_{\text{crit}}, N \leq n^{q_*} &\implies r_* = (nN)^{-\gamma/\Delta}, \\ S < S_{\text{crit}}, N \geq n^{q_*} &\implies r_* = r_o(n). \end{aligned}$$

The boundary identity also gives

$$N = n^{q_*} \implies (nN)^{-\gamma/\Delta} = r_o(n).$$

Return to the case $S < S_{\text{crit}}$. If $N \leq n^{q_*}$, then $N \geq n$, so the nuisance lower bound applies and yields

$$cr_* = c(nN)^{-\gamma/\Delta} \leq c_B(nN)^{-\gamma/\Delta} \leq R(n, m).$$

If $N > n^{q_*}$, the supplied-propensity lower bound yields

$$cr_* = cr_o(n) \leq c_O r_o(n) \leq R(n, m).$$

Here the comparisons of constants preserve the inequalities because both rate terms are nonnegative.

In the remaining case $S \geq S_{\text{crit}}$, choose

$$c = c_O > 0.$$

Then $r_* = r_o(n)$, and hence

$$cr_* = c_O r_o(n) \leq R(n, m).$$

This constructs a positive constant c , depending on the fixed public parameters and valid simultaneously for every admissible sample-size pair, such that

$$cr_* \leq R(n, m).$$

3. *Attainment and the upper constant.* By [Theorem 1](#), there is a real constant $C_U > 0$, depending on the fixed public parameters, such that the publicly tuned decision is Borel measurable and its risk under every $P \in \mathcal{M}$ satisfies

$$\mathcal{R}_{n,m}(\hat{T}_{\text{up}}, P) \leq C_U r_{\text{up}}(n, m).$$

By [Definitions 13](#) and [14](#),

$$T_* = \hat{T}_{\text{up}}, \quad r_{\text{up}}(n, m) = r_*.$$

Set

$$C = \max\{c, C_U\}.$$

Then $0 < c \leq C < \infty$. For every admissible sample-size pair, T_* is Borel measurable, and taking the supremum of the pointwise risk bound gives

$$\sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P) \leq C_U r_* \leq Cr_*.$$

The last inequality follows from $C_U \leq C$ and $r_* \geq 0$.

4. *Minimax comparison and arbitrary decisions.* Since T_* belongs to the decision class in [Definition 5](#),

$$R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P).$$

Together with the bounds already established, this proves

$$cr_* \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P) \leq Cr_*.$$

For any Borel real-valued decision T using the original dataset and the independent uniform randomizer, the same infimum comparison gives

$$cr_* \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P).$$

The rate comparisons and boundary identity established above supply all three stated branches and their equality at $N = n^{q*}$.

□

Proof of [Theorem 3](#). Fix the public parameters in the stated domain. Throughout, inverse-propensity quotients are assigned value zero when their denominator is zero:

$$\frac{t_{\text{num}}}{0} := 0 \quad (t_{\text{num}} \in \mathbb{R}).$$

The designated propensity is extended by zero outside $[0, 1]^d$.

1. Define the constant-nuisance subfamily and its supplied-propensity risk by

$$\mathcal{M}_{1/2} = \left\{ P \in \mathcal{M} : e_P(x) = \mu_{0,P}(x) = \frac{1}{2} \text{ for every } x \in [0, 1]^d \right\},$$

$$\mathcal{R}_{n,m}^e(T^e, P) = \int |T^e(\mathcal{D}_{n,m}, U, e_P) - \tau_P(x_0)| d\mathbf{E}_{n,m}(P), \quad R_{1/2}^e(n, m) = \inf_{T^e} \sup_{P \in \mathcal{M}_{1/2}} \mathcal{R}_{n,m}^e(T^e, P).$$

These loss integrals may take the value $+\infty$. Since the subfamily uses the same decision class as [Definition 6](#), restriction of the supremum gives

$$R_{1/2}^e(n, m) \leq R^e(n, m).$$

It therefore suffices to establish a lower bound for $R_{1/2}^e(n, m)$, a uniform upper bound for the displayed smoother, and its admissibility. We first construct the lower-bound family.

2. Use the smooth profile from [Algorithm 3](#), with the explicit formulas

$$\chi(v) = \begin{cases} \exp(-1/v), & v > 0, \\ 0, & v \leq 0, \end{cases} \quad v \in \mathbb{R}, \quad B(w) = \prod_{\iota_{\text{cov}}=1}^d \frac{\chi(1 - 4w_{\iota_{\text{cov}}}^2)}{\chi(1)}, \quad w \in \mathbb{R}^d.$$

For this argument, extend the scaled-bump formula to every positive localization width and every covariate vector:

$$B_h(x) = B((x - x_0)/h), \quad C_h = \prod_{j=1}^d \left[\frac{1}{2} - \frac{h}{2}, \frac{1}{2} + \frac{h}{2} \right], \quad h > 0, \quad x \in \mathbb{R}^d.$$

Smoothness of the exponential glue function and finite products give

$$B \in C^\infty(\mathbb{R}^d), \quad \text{supp}(B) \subseteq [-1/2, 1/2]^d, \quad 0 \leq B \leq 1, \quad B(0) = 1.$$

In particular,

$$0 \leq B_h \leq 1, \quad B_h(x) = 0 \quad (x \notin C_h), \quad B_h(x_0) = 1.$$

We need a uniform Hölder bound after scaling. Define

$$p_* = \lceil \gamma \rceil - 1, \quad s_o = \gamma - p_*, \quad 0 \leq p_* \leq 2, \quad 0 < s_o \leq 1.$$

Write the iterated derivatives as

$$D^0 B = B, \quad D^{\iota_{\text{der}}+1} B = D(D^{\iota_{\text{der}}} B), \quad \iota_{\text{der}} \in \mathbb{N},$$

and define their finite operator-norm bounds by

$$C_{B, \iota_{\text{der}}} = \sup_{w \in \mathbb{R}^d} \|D^{\iota_{\text{der}}} B(w)\|, \quad \iota_{\text{der}} \in \mathbb{N}.$$

Their finiteness follows from continuity and compact support. Set

$$A_B = \max\{C_{B, p_*}, C_{B, p_*+1}\}, \quad M_B = 2A_B.$$

The derivative bound and the mean-value inequality give, for every $w, w' \in \mathbb{R}^d$,

$$\|D^{p*}B(w) - D^{p*}B(w')\| \leq A_B \|w - w'\|_2, \quad \|D^{p*}B(w) - D^{p*}B(w')\| \leq 2A_B.$$

If $\|w - w'\|_2 \leq 1$, use $\|w - w'\|_2 \leq \|w - w'\|_2^{s_o}$; if $\|w - w'\|_2 > 1$, use $1 \leq \|w - w'\|_2^{s_o}$. Thus both cases yield

$$\|D^{p*}B(w) - D^{p*}B(w')\| \leq M_B \|w - w'\|_2^{s_o}.$$

Translation and dilation satisfy

$$D^{\iota_{\text{der}}} B_h(x) = h^{-\iota_{\text{der}}} D^{\iota_{\text{der}}} B((x - x_0)/h).$$

Evaluating these multilinear maps in coordinate directions gives the bounds

$$|\partial^\kappa B_h(x)| \leq d^{\iota_{\text{der}}} C_{B, \iota_{\text{der}}} h^{-\iota_{\text{der}}} \quad (|\kappa| = \iota_{\text{der}} \leq p_*),$$

and

$$|\partial^\kappa B_h(x) - \partial^\kappa B_h(x')| \leq d^{p*} M_B h^{-\gamma} \|x - x'\|_2^{s_o} \quad (|\kappa| = p_*).$$

Here the dimension factors are valid because every coordinate direction has norm at most d . Define

$$S_B = \sum_{\iota_{\text{der}}=0}^{p_*} d^{\iota_{\text{der}}} C_{B, \iota_{\text{der}}}, \quad D_B = S_B + d^{p*} M_B + 1.$$

For $0 < h \leq 1$ and $\iota_{\text{der}} \leq p_* < \gamma$, one has $h^{-\iota_{\text{der}}} \leq h^{-\gamma}$. Applying the norm definition in [Definition 1](#) to the preceding derivative and modulus bounds proves

$$D_B > 0, \quad \|B_h\|_{H^\gamma} \leq D_B h^{-\gamma} \quad (0 < h \leq 1).$$

This also covers the integer smoothness endpoints, where $s_o = 1$.

3. Choose the public amplitude constant

$$\eta_o = \frac{1}{128(D_B + 1)}.$$

It satisfies

$$\eta_o > 0, \quad \eta_o(D_B + 1) = \frac{1}{128}, \quad \eta_o \leq \frac{1}{8}, \quad \eta_o \leq \frac{1}{16}, \quad \eta_o D_B \leq L.$$

For each $n \geq 2$, define

$$h_n = n^{-1/(2\gamma+d)}, \quad b_n = \eta_o r_o(n).$$

Since $2\gamma + d > 0$ and $n \geq 2$, the power identities give

$$0 < h_n \leq 1, \quad h_n^\gamma = r_o(n), \quad n r_o(n)^2 h_n^d = 1, \quad 0 < b_n \leq \frac{1}{8}.$$

Consequently,

$$\|b_n B_{h_n}\|_{H^\gamma} \leq b_n D_B h_n^{-\gamma} = \eta_o D_B \leq L.$$

For $\theta \in \{-1, +1\}$, define the primitive law $P_{o, \theta}$ by

$$X \sim \text{Unif}([0, 1]^d),$$

$$\mathcal{L}_{P_{o,\theta}}(A, Y(0), Y(1) \mid X = x) = \text{Bernoulli}(1/2) \otimes \text{Bernoulli}(1/2) \otimes \text{Bernoulli}(1/2 + \theta b_n B_{h_n}(x)), \quad x \in [0, 1]^d$$

These conditional margins are Borel. Conditional independence gives exchangeability, and their bounds give

$$e(x) = \mu_0(x) = \frac{1}{2}, \quad \frac{1}{8} \leq \frac{1}{2} + \theta b_n B_{h_n}(x) \leq \frac{7}{8}.$$

The constant propensity and control functions have Hölder norms at most $1/2$, while multiplication by either sign preserves the preceding contrast-norm bound. Thus the construction satisfies every condition of [Definition 3](#).

The designated versions agree with these admissible margins on the cube. Indeed, conditional-margin uniqueness gives almost-sure equality, and continuity together with the full support of the uniform design promotes that equality to pointwise equality. Hence

$$P_{o,\theta} \in \mathcal{M}_{1/2}, \quad e_{P_{o,\theta}}(x) = \mu_{0,P_{o,\theta}}(x) = \frac{1}{2}, \quad \tau_{P_{o,\theta}}(x) = \theta b_n B_{h_n}(x) \quad (x \in [0, 1]^d),$$

so that

$$\tau_{P_{o,\theta}}(x_0) = \theta b_n.$$

4. Define the observed-record and full-experiment laws by

$$P_{o,\theta}^{\text{obs}} = \mathcal{L}_{P_{o,\theta}}(X, A, Y), \quad M_{o,\theta} = \mathbf{E}_{n,m}(P_{o,\theta}).$$

Use the common probability reference measure

$$\nu_o = \text{Unif}([0, 1]^d) \otimes \text{Bernoulli}(1/2) \otimes \text{Bernoulli}(1/2)$$

on the covariate, treatment, and observed-response coordinates. Relative to this reference, the observed-record density is

$$\ell_{o,\theta}(x, a_{\text{obs}}, y_{\text{obs}}) = \begin{cases} 1, & a_{\text{obs}} = 0, \\ 1 + 2(2y_{\text{obs}} - 1)\theta b_n B_{h_n}(x), & a_{\text{obs}} = 1, \end{cases}$$

for $x \in [0, 1]^d$ and $a_{\text{obs}}, y_{\text{obs}} \in \{0, 1\}$. The amplitude bound implies

$$\frac{1}{2} \leq \ell_{o,\theta} \leq 2.$$

For any two density values at least $1/2$, the squared sum of their square roots is at least one. The difference-of-squares identity therefore gives, pointwise,

$$(\sqrt{\ell_{o,-1}} - \sqrt{\ell_{o,+1}})^2 \leq (\ell_{o,-1} - \ell_{o,+1})^2 \leq 16b_n^2 B_{h_n}(x)^2.$$

The Hellinger normalization in [Definition 18](#) can be evaluated under ν_o : the densities relative to the sum of the two laws are $\ell_{o,\theta}/(\ell_{o,-1} + \ell_{o,+1})$, and the common denominator cancels. Thus

$$h^2(P_{o,-1}^{\text{obs}}, P_{o,+1}^{\text{obs}}) = \int (\sqrt{\ell_{o,-1}} - \sqrt{\ell_{o,+1}})^2 d\nu_o.$$

Since $B_{h_n}^2 \leq \mathbf{1}_{C_{h_n}}$, the containment $C_{h_n} \subseteq [0, 1]^d$ and the product volume of this cube give

$$\int_{[0,1]^d} B_{h_n}(x)^2 dx \leq \text{vol}(C_{h_n}) = h_n^d.$$

Combining the preceding displays proves

$$\mathbf{h}^2(P_{o,-1}^{\text{obs}}, P_{o,+1}^{\text{obs}}) \leq 16b_n^2 h_n^d.$$

The auxiliary covariate-treatment marginal is identical under the two laws:

$$P_{o,-1,XA} = P_{o,+1,XA} = \text{Unif}([0, 1]^d) \otimes \text{Bernoulli}(1/2).$$

The product bound and common-factor identity in [Lemma 4](#) therefore yield

$$\mathbf{h}^2(M_{o,-1}, M_{o,+1}) \leq n \mathbf{h}^2(P_{o,-1}^{\text{obs}}, P_{o,+1}^{\text{obs}}) \leq 16nb_n^2 h_n^d = 16\eta_o^2 \leq \frac{1}{16}.$$

This applies to every $m \geq 0$; at $m = 0$, the auxiliary product is the probability law on the empty record block. The independent uniform randomizer is another common probability factor. Finally, symmetry of the squared square-root difference gives the orientation used below:

$$\mathbf{h}^2(M_{o,+1}, M_{o,-1}) = \mathbf{h}^2(M_{o,-1}, M_{o,+1}) \leq \frac{1}{16}.$$

5. The target is a well-defined functional of these full-experiment laws. To see this, equality of two such laws implies equality of their first labeled-record marginals, since $n > 0$. Under either constructed law, the observed conditional probabilities identify the arm means through

$$\mu_{1,P_{o,\theta}}(x) = 2P_{o,\theta}(A = 1, Y = 1 \mid X = x), \quad \mu_{0,P_{o,\theta}}(x) = 2P_{o,\theta}(A = 0, Y = 1 \mid X = x)$$

almost everywhere under the uniform design. Thus equality of the observed laws identifies all three conditional margins. Their conditional product construction then identifies the primitive laws, and uniqueness of the canonical continuous contrast gives

$$M_{o,\theta} = M_{o,\theta'} \implies \tau_{P_{o,\theta}}(x_0) = \tau_{P_{o,\theta'}}(x_0).$$

In particular, $M_{o,+1}$ and $M_{o,-1}$ are distinct because $b_n > 0$. Define a functional on all measures on the full sample space by

$$\Psi_o(\mathbb{Q}_o) = \begin{cases} b_n, & \mathbb{Q}_o = M_{o,+1}, \\ -b_n, & \mathbb{Q}_o = M_{o,-1}, \\ 0, & \mathbb{Q}_o \notin \{M_{o,+1}, M_{o,-1}\}. \end{cases}$$

It satisfies

$$\Psi_o(M_{o,\theta}) = \tau_{P_{o,\theta}}(x_0).$$

The supplied propensity is the same function for both hypotheses, including its extension outside the cube:

$$e_o(x) = \begin{cases} 1/2, & x \in [0, 1]^d, \\ 0, & x \notin [0, 1]^d. \end{cases}$$

For any admissible supplied-propensity decision, its section at e_o is measurable. Apply [Lemma 5](#), the absolute-loss bound of [[Kennedy et al., 2024](#), Section 3, Lemma 1], with singleton priors, one observation consisting of the entire record block and randomizer, and the choices

$$\begin{aligned}\omega_o &= \text{Dirac}(*), & \mathbb{B}_*^o &= M_{o,+1}, & \mathbb{C}_*^o &= M_{o,-1}, & \mathcal{P}_o^{\text{block}} &= \{M_{o,+1}, M_{o,-1}\}, \\ v_{\text{test}} &= 1, & s_{\text{test}} &= 2b_n, & u_{\text{test}} &= \frac{1}{16}.\end{aligned}$$

The separation is $2b_n > 0$, and the preceding Hellinger bound supplies the distance hypothesis. Consequently,

$$\sup_{\theta \in \{-1, +1\}} \mathcal{R}_{n,m}^e(T^e, P_{o,\theta}) \geq \frac{b_n}{2} \left(1 - \frac{\sqrt{63}}{32}\right).$$

Since $63 \leq 64$,

$$1 - \frac{\sqrt{63}}{32} \geq \frac{3}{4}, \quad \frac{b_n}{2} \left(1 - \frac{\sqrt{63}}{32}\right) \geq \frac{3}{8}b_n.$$

The two primitive laws belong to $\mathcal{M}_{1/2}$, so taking the infimum over decisions proves

$$c = \frac{3}{8}\eta_o > 0, \quad c r_o(n) \leq R_{1/2}^e(n, m) \quad (n \geq 2, m \geq 0).$$

6. We next establish the upper bound. For every $h > 0$, define the localization measure and weight by

$$d\nu_h(x) = h^{-d} \mathbf{1}_{C_h}(x) dx, \quad w_h(x) = h^{-d} \mathbf{1}_{C_h}(x).$$

The measure ν_h is a probability measure because $\text{vol}(C_h) = h^d$. For $0 < h \leq 1$, its support lies in the design cube.

To make the coarse basis explicit, define

$$\mathcal{I}_2 = \{\kappa \in \mathbb{N}^d : |\kappa| \leq 2\}, \quad q = |\mathcal{I}_2|,$$

and the one-dimensional polynomials

$$P_0(v) = 1, \quad P_1(v) = \sqrt{12}v, \quad P_2(v) = \sqrt{5}(6v^2 - 1/2), \quad v \in \mathbb{R}.$$

At the current localization width h , the basis and evaluation vector are

$$r_\kappa(x) = \prod_{\iota_{\text{cov}}=1}^d P_{\kappa_{\iota_{\text{cov}}}}((x_{\iota_{\text{cov}}} - 1/2)/h), \quad r(x) = (r_\kappa(x))_{\kappa \in \mathcal{I}_2}, \quad r_0 = r(x_0),$$

for every $x \in \mathbb{R}^d$. Define the evaluation kernel and its dimension constant by

$$\mathcal{K}_h(x) = r_0^\top r(x), \quad K_{\text{ev}} = q(4^d)^2.$$

The constant coordinate belongs to $\mathcal{I}_2$, so $q \geq 1$ and $K_{\text{ev}} > 0$.

Direct integration of the three one-dimensional formulas gives

$$\int_{-1/2}^{1/2} \mathbf{P}_{j_{\text{pol}}}(v) \mathbf{P}_{j'_{\text{pol}}}(v) dv = \mathbf{1}_{\{j_{\text{pol}}=j'_{\text{pol}}\}} \quad (j_{\text{pol}}, j'_{\text{pol}} \in \{0, 1, 2\}).$$

Scaling each coordinate and multiplying these identities yields

$$\int r_{\kappa}(x) r_{\kappa'}(x) d\nu_h(x) = \mathbf{1}_{\{\kappa=\kappa'\}}.$$

Expanding the finite sums consequently gives polynomial reproduction and kernel energy:

$$\int \mathcal{K}_h(x) r(x)^{\top} v d\nu_h(x) = r_0^{\top} v \quad (v \in \mathbb{R}^q),$$

$$\int \mathcal{K}_h(x)^2 d\nu_h(x) = \|r_0\|_2^2.$$

The displayed polynomial formulas also give $|\mathbf{P}_{j_{\text{pol}}}(v)| \leq 4$ on $[-1/2, 1/2]$. Thus

$$|r_{\kappa}(x)| \leq 4^d \quad (x \in C_h), \quad |\mathcal{K}_h(x)| \leq K_{\text{ev}} \quad (x \in C_h), \quad \|r_0\|_2^2 \leq K_{\text{ev}}.$$

All these identities hold for every $h > 0$.

7. Fix $P \in \mathcal{M}$. Define the supplied-propensity response, its localized record contribution, the raw average, and its mean by

$$Z_P(x, a_{\text{obs}}, y_{\text{obs}}) = \frac{a_{\text{obs}} y_{\text{obs}}}{e_P(x)} - \frac{(1 - a_{\text{obs}}) y_{\text{obs}}}{1 - e_P(x)},$$

$$F_{P,h}(x, a_{\text{obs}}, y_{\text{obs}}) = w_h(x) \mathcal{K}_h(x) Z_P(x, a_{\text{obs}}, y_{\text{obs}}),$$

$$\bar{T}_{P,h}(\mathcal{D}_{n,m}, U) = \frac{1}{n} \sum_{i=1}^n F_{P,h}(X_i, A_i, Y_i), \quad \mu_{P,h} = \int \bar{T}_{P,h} d\mathbf{E}_{n,m}(P).$$

These definitions apply to all $x \in \mathbb{R}^d$, binary record coordinates, and every dataset and randomizer, using the quotient convention stated above.

On the design cube, overlap and the binary record coordinates give

$$|Z_P(x, a_{\text{obs}}, y_{\text{obs}})| \leq \varepsilon^{-1}.$$

Outside the cube, the zero extension of the propensity gives

$$Z_P(x, a_{\text{obs}}, y_{\text{obs}}) = -(1 - a_{\text{obs}}) y_{\text{obs}}, \quad |Z_P(x, a_{\text{obs}}, y_{\text{obs}})| \leq 1 \leq \varepsilon^{-1}.$$

Hence the response bound is global, and

$$|F_{P,h}(x, a_{\text{obs}}, y_{\text{obs}})| \leq \varepsilon^{-1} h^{-d} K_{\text{ev}}.$$

Borel measurability and this bound supply all integrability and finite second moments used below.

For $0 < h \leq 1$, exchangeability and the conditional-margin identities in [Definition 3](#), followed by cancellation of the overlap denominators, give

$$\begin{aligned}\mathbb{E}_P \left[w_h(X) \mathcal{K}_h(X) \frac{AY}{e_P(X)} \right] &= \int_{[0,1]^d} w_h(x) \mathcal{K}_h(x) \mu_{1,P}(x) dx, \\ \mathbb{E}_P \left[w_h(X) \mathcal{K}_h(X) \frac{(1-A)Y}{1-e_P(X)} \right] &= \int_{[0,1]^d} w_h(x) \mathcal{K}_h(x) \mu_{0,P}(x) dx.\end{aligned}$$

Subtracting and using the canonical contrast identity gives

$$\mathbb{E}_P[F_{P,h}(X, A, Y)] = \int \mathcal{K}_h(x) \tau_P(x) d\nu_h(x).$$

Every labeled coordinate has this expectation, while the auxiliary block and randomizer integrate out as probability factors. Since $n > 0$,

$$\mu_{P,h} = \int \mathcal{K}_h(x) \tau_P(x) d\nu_h(x).$$

8. We now bound this mean's bias over the full range $0 < h \leq 1$. Define the Taylor polynomial

$$p(x) = \sum_{|\kappa| \leq p_*} \frac{\partial^\kappa \tau_P(x_0)}{\kappa!} (x - x_0)^\kappa, \quad \kappa! = \prod_{\iota_{\text{cov}}=1}^d \kappa_{\iota_{\text{cov}}}!.$$

Because $p_* \leq 2$, the coarse polynomial basis spans p . Thus there exists a coefficient vector $\vartheta_{\text{Tay}} \in \mathbb{R}^q$ with

$$p_{\vartheta_{\text{Tay}}}(x) = r(x)^\top \vartheta_{\text{Tay}} = p(x) \quad (x \in \mathbb{R}^d), \quad p_{\vartheta_{\text{Tay}}}(x_0) = \tau_P(x_0).$$

If $\gamma = 1$, then $p_* = 0$ and $p(x) = \tau_P(x_0)$. The Hölder norm condition gives the Lipschitz bound, and the geometry of C_h gives

$$|\tau_P(x) - p(x)| \leq L \|x - x_0\|_2 \leq L \sqrt{d} h \leq L(\sqrt{d} + 1)h \quad (x \in C_h).$$

If $1 < \gamma \leq 3$, then $1 \leq p_* \leq 2$. For each $x \in C_h$, define the segment function

$$g_{P,x}(\sigma_{\text{seg}}) = \tau_P(x_0 + \sigma_{\text{seg}}(x - x_0)), \quad 0 \leq \sigma_{\text{seg}} \leq 1.$$

The segment lies in the design cube. Expanding its p_* -th derivative in coordinate partials and using their s_o -Hölder modulus gives

$$\left| g_{P,x}^{(p_*)}(\sigma_{\text{seg}}) - g_{P,x}^{(p_*)}(0) \right| \leq (d+1)^{p_*} L \|x - x_0\|_2^{p_* + s_o} \sigma_{\text{seg}}^{s_o}.$$

Indeed, the multinomial expansion contributes at most $(d+1)^{p_*} \|x - x_0\|_2^{p_*}$, and each top-partial difference is bounded by $L \sigma_{\text{seg}}^{s_o} \|x - x_0\|_2^{s_o}$. The one-dimensional Lagrange remainder at degree $p_* - 1$, followed by subtraction of the degree- p_* Taylor term, supplies some $\xi_{P,x} \in (0, 1)$ such that

$$\tau_P(x) - p(x) = \frac{g_{P,x}^{(p_*)}(\xi_{P,x}) - g_{P,x}^{(p_*)}(0)}{p_*!}.$$

Since $\xi_{P,x}^{s_o} \leq 1$ and $p_*! \geq 1$,

$$|\tau_P(x) - p(x)| \leq (d+1)^{p_*} L \|x - x_0\|_2^\gamma.$$

Using $\|x - x_0\|_2 \leq (d+1)h$ and $\gamma = p_* + s_o \leq p_* + 1$ yields

$$|\tau_P(x) - p(x)| \leq L(d+1)^{2p_*+1} h^\gamma \quad (x \in C_h).$$

Define the positive public constants

$$C_{\text{Tay}} = \begin{cases} L(\sqrt{d} + 1), & \gamma = 1, \\ L(d+1)^{2p_*+1}, & 1 < \gamma \leq 3, \end{cases} \quad C_{\text{bias}} = K_{\text{ev}} C_{\text{Tay}}.$$

Both cases establish

$$\sup_{x \in C_h} |\tau_P(x) - p_{\vartheta_{\text{Tay}}}(x)| \leq C_{\text{Tay}} h^\gamma.$$

Polynomial reproduction and the mean identity therefore imply

$$\begin{aligned} |\mu_{P,h} - \tau_P(x_0)| &= \left| \int \mathcal{K}_h(x) (\tau_P(x) - p_{\vartheta_{\text{Tay}}}(x)) d\nu_h(x) \right| \\ &\leq \int |\mathcal{K}_h(x)| |\tau_P(x) - p_{\vartheta_{\text{Tay}}}(x)| d\nu_h(x) \\ &\leq C_{\text{bias}} h^\gamma. \end{aligned}$$

At $h = h_n$, this becomes

$$|\mu_{P,h_n} - \tau_P(x_0)| \leq C_{\text{bias}} r_o(n).$$

9. The global response bound gives the recordwise squared bound

$$F_{P,h}(X, A, Y)^2 \leq \varepsilon^{-2} (w_h(X) \mathcal{K}_h(X))^2.$$

Under the uniform design, localization and the kernel-energy identity give

$$\begin{aligned} \mathbb{E}_P[F_{P,h}(X, A, Y)^2] &\leq \varepsilon^{-2} \int_{[0,1]^d} (w_h(x) \mathcal{K}_h(x))^2 dx \\ &= \varepsilon^{-2} h^{-d} \int \mathcal{K}_h(x)^2 d\nu_h(x) \\ &= \varepsilon^{-2} h^{-d} \|r_0\|_2^2. \end{aligned}$$

Independence of the n labeled records yields

$$\begin{aligned} \int (\bar{T}_{P,h} - \mu_{P,h})^2 d\mathbf{E}_{n,m}(P) &= \frac{1}{n} \text{Var}_P(F_{P,h}(X, A, Y)) \\ &\leq \frac{1}{n} \mathbb{E}_P[F_{P,h}(X, A, Y)^2] \\ &\leq \frac{\varepsilon^{-2} \|r_0\|_2^2}{nh^d}. \end{aligned}$$

The first inequality uses

$$\text{Var}_P(F_{P,h}) = \mathbb{E}_P[F_{P,h}^2] - (\mathbb{E}_P[F_{P,h}])^2 \leq \mathbb{E}_P[F_{P,h}^2].$$

Cauchy–Schwarz on the probability experiment then gives

$$\int |\bar{T}_{P,h} - \mu_{P,h}| d\mathbf{E}_{n,m}(P) \leq \left(\int (\bar{T}_{P,h} - \mu_{P,h})^2 d\mathbf{E}_{n,m}(P) \right)^{1/2} \leq \sqrt{\frac{\varepsilon^{-2} \|r_0\|_2^2}{nh^d}}.$$

At $h = h_n$, use the target-coordinate bound and

$$(nh_n^d)^{-1/2} = r_o(n)$$

to obtain

$$\int |\bar{T}_{P,h_n} - \mu_{P,h_n}| d\mathbf{E}_{n,m}(P) \leq \sqrt{\varepsilon^{-2} K_{\text{ev}}} r_o(n) \leq C_{\text{dev}} r_o(n), \quad C_{\text{dev}} = \sqrt{\varepsilon^{-2} K_{\text{ev}}} + 1 > 0.$$

The estimate is uniform in P , $n \geq 2$, and $m \geq 0$.

10. The probability-valued arm-mean conditions in [Definition 3](#) imply

$$-1 \leq \mu_{1,P}(x_0) - \mu_{0,P}(x_0) = \tau_P(x_0) \leq 1.$$

The displayed smoother equals

$$T^e(\mathcal{D}_{n,m}, U, e_P) = \text{clip}_{[-1,1]}(\bar{T}_{P,h_n}(\mathcal{D}_{n,m}, U)).$$

Projection onto this interval contracts distance to any target inside it. Thus, for every dataset and randomizer,

$$\begin{aligned} |T^e(\mathcal{D}_{n,m}, U, e_P) - \tau_P(x_0)| &\leq |\bar{T}_{P,h_n}(\mathcal{D}_{n,m}, U) - \tau_P(x_0)| \\ &\leq |\bar{T}_{P,h_n}(\mathcal{D}_{n,m}, U) - \mu_{P,h_n}| + |\mu_{P,h_n} - \tau_P(x_0)|. \end{aligned}$$

Integrating and applying the preceding two bounds gives

$$\mathcal{R}_{n,m}^e(T^e, P) \leq (C_{\text{dev}} + C_{\text{bias}}) r_o(n).$$

Define

$$C_{\text{up}} = C_{\text{dev}} + C_{\text{bias}} > 0.$$

The bounded measurable raw average and continuous clipping map give measurability of the supplied-propensity section. The clipping formula itself gives the pointwise guarantee

$$|T^e(\mathcal{D}_{n,m}, U, e_P)| \leq 1$$

for every dataset and randomizer.

11. For completeness, the smoother has a total admissible extension to arbitrary supplied functions. Define

$$\tilde{T}^e(\mathcal{D}_{n,m}, U, e) = \begin{cases} T^e(\mathcal{D}_{n,m}, U, e), & e \text{ is Borel,} \\ 0, & e \text{ is not Borel.} \end{cases}$$

For every fixed Borel e , the totalized reciprocal, record-coordinate maps, finite sum, and clipping are Borel; the other section is constant. Since each designated e_P is Borel, this

extension agrees with the displayed smoother at every admitted law. It is therefore a candidate in [Definition 6](#), and the uniform bound proves

$$R^e(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n, m}^e(\tilde{T}^e, P) \leq C_{\text{up}} r_o(n).$$

Finally, retain the lower constant already constructed and set

$$C = \max\{c, C_{\text{up}}\}.$$

These are finite real constants depending on the fixed public parameters, with

$$0 < c \leq C.$$

Because $r_o(n) \geq 0$, enlarging the upper constant preserves both the minimax and individual-decision bounds. Combining the lower bound with subfamily monotonicity gives, for every $n \geq 2$ and $m \geq 0$,

$$c r_o(n) \leq R_{1/2}^e(n, m) \leq R^e(n, m) \leq C r_o(n),$$

and, for every $P \in \mathcal{M}$,

$$\mathcal{R}_{n, m}^e(T^e, P) \leq C r_o(n).$$

Together with the measurability and pointwise clipping bound above, these are all the asserted conclusions. □

Proof of [Theorem 4](#). Fix the public parameters. All constants constructed below depend only on these parameters. We use the functions and sign arrays prescribed by [Algorithm 3](#), extending the displayed function formulas to $\mathbb{R}^d$ when taking analytic bounds.

1. *Regularity and membership.* Set

$$D_d := 2^d.$$

The function χ is smooth and flat at zero: each of its derivatives on the positive half-line is a polynomial in $1/u$ multiplied by $\exp(-1/u)$, and this expression tends to zero as $u \downarrow 0$. The denominator defining the scalar angular transition in [Algorithm 3](#) is positive everywhere. Consequently the prescribed sine and cosine pieces join smoothly, including at their support endpoints. The bump B is also smooth and compactly supported, and

$$0 \leq B_h(x) \leq 1, \quad B_h(x_0) = 1.$$

Define the finite derivative bounds

$$K_\omega := 1 + \sup_{u \in \mathbb{R}} |\omega'_0(u)|, \quad K_B := 1 + \sup_{w \in \mathbb{R}^d} \|\nabla B(w)\|_2,$$

and put

$$K_\phi := K_B + dK_\omega, \quad K_F := 2D_d K_\phi.$$

Translation invariance gives the same derivative bound for every ω_z . Telescoping a product of factors bounded by one, and then using $\delta \leq h$, yields

$$|\phi_{\mathbf{z}}(x) - \phi_{\mathbf{z}}(x')| \leq \left(\frac{K_B}{h} + \frac{dK_\omega}{\delta} \right) \|x - x'\|_2 \leq \frac{K_\phi}{\delta} \|x - x'\|_2.$$

At each point, a nonzero frame entry has, in each coordinate, one of the two lattice indices adjacent to the scaled coordinate. Thus at most D_d entries are nonzero. Each entry has absolute value at most one, so

$$|F_\lambda(x)| \leq D_d, \quad |G_\eta(x)| \leq D_d.$$

For a difference between two points, entries outside the union of their active index sets contribute zero. That union has at most $2D_d$ elements. Therefore

$$|F_\lambda(x) - F_\lambda(x')| \leq \frac{K_F}{\delta} \|x - x'\|_2, \quad |G_\eta(x) - G_\eta(x')| \leq \frac{K_F}{\delta} \|x - x'\|_2.$$

For a Hölder order $s_{\text{ord}} \in (0, 1]$, define

$$C_{F, s_{\text{ord}}} := 2D_d + K_F + 1.$$

When $\|x - x'\|_2 \leq \delta$, the Lipschitz bound and

$$\frac{\|x - x'\|_2}{\delta} \leq \left(\frac{\|x - x'\|_2}{\delta} \right)^{s_{\text{ord}}}$$

give the required Hölder difference bound. When $\|x - x'\|_2 > \delta$, use the bound $2D_d$ instead. Since $\delta \leq 1$, the supremum term is covered as well. With the norm of [Definition 1](#), this proves

$$\|F_\lambda\|_{H^{s_{\text{ord}}}} \leq C_{F, s_{\text{ord}}} \delta^{-s_{\text{ord}}}, \quad \|G_\eta\|_{H^{s_{\text{ord}}}} \leq C_{F, s_{\text{ord}}} \delta^{-s_{\text{ord}}}.$$

For the macro bump, use its fixed compact profile B^2 . For every integer $j \geq 0$, define

$$D_{B, j} := 1 + \sup_{w \in \mathbb{R}^d} \|D^j(B^2)(w)\|_{\text{op}},$$

where D^j is the total derivative of order j , with the order-zero norm interpreted as absolute value. These constants are finite by smoothness and compact support. For $s_{\text{ord}} > 0$, define

$$j_{\text{der}} := \lceil s_{\text{ord}} \rceil - 1, \quad \xi_{\text{der}} := s_{\text{ord}} - j_{\text{der}} \in (0, 1],$$

$$M_{B, s_{\text{ord}}} := 2 \max\{D_{B, j_{\text{der}}}, D_{B, j_{\text{der}}+1}\},$$

$$C_{B, s_{\text{ord}}} := \sum_{j=0}^{j_{\text{der}}} D_{B, j} d^j + M_{B, s_{\text{ord}}} d^{j_{\text{der}}} + 1.$$

The derivative of order j_{der} has both a bounded supremum and a Lipschitz bound supplied by the next derivative. Splitting at distance one gives

$$\|D^{j_{\text{der}}}(B^2)(w) - D^{j_{\text{der}}}(B^2)(w')\|_{\text{op}} \leq M_{B, s_{\text{ord}}} \|w - w'\|_2^{\xi_{\text{der}}}.$$

Dilation by h multiplies a derivative of order j by h^{-j} . Evaluating a derivative on coordinate directions bounds a partial derivative by its operator norm times d^j . Hence, for $0 < h \leq 1$,

$$|\partial^\kappa(B_h^2)(x)| \leq D_{B,|\kappa|} d^{|\kappa|} h^{-|\kappa|} \leq C_{B,s_{\text{ord}}} h^{-s_{\text{ord}}} \quad (|\kappa| \leq j_{\text{der}}),$$

and

$$|\partial^\kappa(B_h^2)(x) - \partial^\kappa(B_h^2)(x')| \leq M_{B,s_{\text{ord}}} d^{j_{\text{der}}} h^{-s_{\text{ord}}} \|x - x'\|_2^{\xi_{\text{der}}} \quad (|\kappa| = j_{\text{der}}).$$

Thus, in particular for the orders β and γ ,

$$\|B_h^2\|_{H^{s_{\text{ord}}}} \leq C_{B,s_{\text{ord}}} h^{-s_{\text{ord}}}.$$

Define

$$K_M := D_d + C_{F,\alpha} + C_{F,\beta} + C_{B,\gamma} + C_{B,\beta} + 1, \quad \rho := \min \left\{ 1, L - 1/2, \frac{1}{2} - \varepsilon \right\}, \quad c_M := \frac{\rho}{32K_M}.$$

The public domain gives $0 < \rho \leq 1$. Hence $0 < c_M \leq \rho/32 \leq 1/32$, and multiplying any of the five summands preceding the final 1 in K_M by c_M gives at most $\rho/32$.

Suppose the amplitude and contrast inequalities hold with c_M . Since $0 < \delta \leq h \leq 1/2$,

$$\begin{aligned} a &\leq c_M, & b &\leq c_M, & ab &\leq c_M, \\ a\delta^{-\alpha} &\leq c_M, & b\delta^{-\beta} &\leq c_M, & abh^{-\gamma} &\leq c_M. \end{aligned}$$

Also $\beta \leq \gamma$ and $h \leq 1$, so

$$ab \leq c_M h^\gamma \leq c_M h^\beta, \quad abh^{-\beta} \leq c_M.$$

The prescribed probabilities therefore satisfy

$$\begin{aligned} |e_\lambda(x) - 1/2| &\leq aD_d \leq \frac{\rho}{32} \leq \frac{1}{4}, & aD_d &\leq \frac{\rho}{32} \leq \frac{1}{2} - \varepsilon, \\ |\mu_{0,\theta,\eta}(x) - 1/2|, |\mu_{1,\theta,\eta}(x) - 1/2| &\leq bD_d + ab \leq \frac{\rho}{32} + c_M \leq \frac{1}{16} \leq \frac{3}{8}. \end{aligned}$$

These bounds imply the overlap and both probability-valued arm-mean conditions.

Subadditivity and homogeneity of the Hölder norm give

$$\begin{aligned} \|aF_\lambda\|_{H^\alpha} &\leq C_{F,\alpha} c_M, & \|bG_\eta\|_{H^\beta} &\leq C_{F,\beta} c_M, \\ \|\tau_\theta\|_{H^\gamma} &\leq 2C_{B,\gamma} c_M, & \|\theta abB_h^2\|_{H^\beta} &\leq C_{B,\beta} c_M. \end{aligned}$$

Consequently

$$\begin{aligned} \|e_\lambda\|_{H^\alpha} &\leq \frac{1}{2} + \frac{\rho}{32} \leq L, \\ \|\mu_{0,\theta,\eta}\|_{H^\beta} &\leq \frac{1}{2} + \frac{\rho}{16} \leq L, & \|\tau_\theta\|_{H^\gamma} &\leq \frac{\rho}{16} \leq L. \end{aligned}$$

All the prescribed functions are Borel. The independent conditional Bernoulli construction gives uniform covariates and conditional exchangeability, and its conditional margins are

exactly the prescribed functions. Together with the preceding bounds, these facts establish every condition in [Definition 3](#).

To identify the canonical contrast, the prescribed control mean is continuous, and the prescribed treated mean is continuous because it is the sum of the control mean and τ_θ . Their admissible designated versions agree with them almost surely by uniqueness of conditional margins. Two continuous versions that agree almost everywhere under the uniform design agree everywhere on the cube: a nonzero continuous difference would persist on a relative neighborhood of positive uniform measure, including at a boundary point. Thus

$$P_{\theta,\lambda,\eta} \in \mathcal{M}, \quad \tau_{P_{\theta,\lambda,\eta}}(x) = \mu_{1,\theta,\eta}(x) - \mu_{0,\theta,\eta}(x) = \tau_\theta(x) \quad (x \in [0, 1]^d).$$

2. *Normalization and singleton cancellation.* Define expectation over the finite sign arrays by

$$\mathbb{E}_\theta Q := \sum_{\lambda,\eta} w_\theta(\lambda,\eta) Q(\lambda,\eta).$$

At each index, direct summation over the four sign pairs gives

$$\frac{1 + \theta l s_\eta / 2}{4} \geq 0 \quad (l, s_\eta \in \{-1, +1\}), \quad \sum_{l, s_\eta \in \{-1, +1\}} \frac{1 + \theta l s_\eta / 2}{4} = 1.$$

Multiplying these identities over the finite index set proves

$$w_\theta(\lambda, \eta) \geq 0, \quad \sum_{\lambda, \eta} w_\theta(\lambda, \eta) = 1.$$

The same finite-product summation gives

$$\mathbb{E}_\theta \lambda_{\mathbf{z}} = \mathbb{E}_\theta \eta_{\mathbf{z}} = 0, \quad \mathbb{E}_\theta [\lambda_{\mathbf{z}} \eta_{\mathbf{w}}] = \begin{cases} \theta/2, & \mathbf{z} = \mathbf{w}, \\ 0, & \mathbf{z} \neq \mathbf{w}. \end{cases}$$

Indeed, matching indices use the scalar sign-pair correlation, while distinct indices include a zero-mean scalar factor.

The sine-cosine identity on each lattice interval gives a sum of squares equal to one. Tensoring it and multiplying by B_h^2 yields

$$\sum_{\mathbf{z} \in \mathcal{I}} \phi_{\mathbf{z}}(x)^2 = B_h(x)^2.$$

For $x \in C_h$, all nonzero tensor entries lie in the prescribed finite index set; for $x \notin C_h$, both sides vanish. Expanding the finite signed sums therefore gives

$$\mathbb{E}_\theta F_\lambda(x) = \mathbb{E}_\theta G_\eta(x) = 0, \quad \mathbb{E}_\theta [F_\lambda(x) G_\eta(x)] = \frac{\theta}{2} B_h(x)^2.$$

Set

$$c_S := \frac{1}{64D_d}.$$

If the amplitude inequalities hold with c_S , then

$$a, b \leq c_S, \quad aD_d, bD_d \leq \frac{1}{64}, \quad ab \leq \frac{1}{4096}.$$

Thus the raw propensity and both raw arm probabilities lie in $[0, 1]$, at every covariate, so their Bernoulli atom formulas apply directly.

For a labeled record, define the treatment and observed-response spins by

$$s^A := 2A - 1, \quad Y := AY(1) + (1 - A)Y(0), \quad s^Y := 2Y - 1.$$

Relative to the uniform probability measure on the four spin pairs, its likelihood is

$$L_{\theta, \lambda, \eta}(x; s^A, s^Y) := (1 + 2s^A a F_\lambda(x))(1 + 2s^Y b G_\eta(x) + s^A s^Y \tau_\theta(x)).$$

This follows by multiplying the two Bernoulli atom probabilities, each divided by its fair atom probability. Expanding the product gives

$$\begin{aligned} \mathbb{E}_\theta L_{\theta, \lambda, \eta}(x; s^A, s^Y) &= 1 + s^A s^Y \tau_\theta(x) \\ &\quad + (2s^A a + 2(s^A)^2 s^Y a \tau_\theta(x)) \mathbb{E}_\theta F_\lambda(x) \\ &\quad + 2s^Y b \mathbb{E}_\theta G_\eta(x) + 4s^A s^Y ab \mathbb{E}_\theta [F_\lambda(x) G_\eta(x)] \\ &= 1 + s^A s^Y (-2\theta ab B_h(x)^2 + 2\theta ab B_h(x)^2) = 1. \end{aligned}$$

Hence every atom of the singleton labeled mixture has probability $1/4$. This proves its uniform conditional law for every hypothesis and every cube covariate.

3. *Equality on uninformative components.* For integers $n \geq 2$ and $m \geq 0$, write

$$N := n + m, \quad \mathbf{x} := (x_1, \dots, x_N) \in ([0, 1]^d)^N.$$

The first n indices are labeled. A spin assignment has domain

$$\mathbf{s} \in (\{-1, +1\}^2)^n \times \{-1, +1\}^m.$$

Define the record likelihoods by

$$R_{\theta, \lambda, \eta, i}(\mathbf{x}, \mathbf{s}) := \begin{cases} L_{\theta, \lambda, \eta}(x_i; s_i^A, s_i^Y), & i \leq n, \\ 1 + 2s_i^A a F_\lambda(x_i), & i > n, \end{cases}$$

and, for every subset $\mathcal{V} \subseteq \{1, \dots, N\}$, define

$$g_{\theta, \mathcal{V}}(\mathbf{x}, \mathbf{s}) := \mathbb{E}_\theta \prod_{i \in \mathcal{V}} R_{\theta, \lambda, \eta, i}(\mathbf{x}, \mathbf{s}).$$

Empty products equal one, so $g_{\theta, \emptyset} = 1$.

The sign flip

$$\mathcal{F}_\eta(\lambda, \eta) := (\lambda, -\eta)$$

is a bijection, and its weights satisfy

$$w_{+1}(\lambda, \eta) = w_{-1}(\lambda, -\eta).$$

Every auxiliary likelihood is unchanged by this flip, because it depends on λ alone. Reindexing the finite sum consequently gives

$$g_{+1,\mathcal{V}}(\mathbf{x}, \mathbf{s}) = g_{-1,\mathcal{V}}(\mathbf{x}, \mathbf{s}) \quad \text{if } \mathcal{V} \subseteq \{n+1, \dots, N\}.$$

If $\mathcal{V} = \{i\}$ with $i \leq n$, singleton cancellation instead gives

$$g_{+1,\{i\}}(\mathbf{x}, \mathbf{s}) = g_{-1,\{i\}}(\mathbf{x}, \mathbf{s}) = 1.$$

Thus we may take

$$c_U := c_S.$$

Under its amplitude conditions, every auxiliary-only component and every singleton labeled component has equal densities under the two hypotheses. Equality for all spin assignments gives equality of their full conditional laws.

4. *The component density difference.* Define

$$c_\Delta := \frac{1}{64D_d}, \quad C_\Delta := 8D_d^2 + 6.$$

Assume the amplitude inequalities with c_Δ . Fix $\mathbf{x}, \mathbf{s}$ and a subset $\mathcal{V}$, and define its size by

$$p_{\text{cmp}} := |\mathcal{V}|.$$

For every record, introduce the contrast-deleted factors

$$f_{\lambda,i} := 1 + 2s_i^A a F_\lambda(x_i), \quad g_{\eta,i} := \begin{cases} 1 + 2s_i^Y b G_\eta(x_i), & i \leq n, \\ 1, & i > n. \end{cases}$$

The bounds from the previous steps give

$$|f_{\lambda,i} - 1| \leq 2aD_d, \quad |g_{\eta,i} - 1| \leq 2bD_d, \quad |f_{\lambda,i}|, |g_{\eta,i}| \leq \frac{9}{8}.$$

For labeled records,

$$R_{\theta,\lambda,\eta,i} = f_{\lambda,i}(g_{\eta,i} + s_i^A s_i^Y \tau_\theta(x_i)), \quad |\tau_\theta(x_i)| \leq 2ab.$$

For auxiliary records, $R_{\theta,\lambda,\eta,i} = f_{\lambda,i}$. Since $ab \leq 1/4096$, both cases satisfy

$$|R_{\theta,\lambda,\eta,i}| \leq \frac{3}{2}, \quad |R_{\theta,\lambda,\eta,i} - f_{\lambda,i} g_{\eta,i}| \leq 3ab, \quad |f_{\lambda,i} g_{\eta,i}| \leq \frac{3}{2}.$$

Define

$$\begin{aligned} \mathcal{A}_\mathcal{V}(\lambda) &:= \prod_{i \in \mathcal{V}} f_{\lambda,i}, & \mathcal{G}_\mathcal{V}(\eta) &:= \prod_{i \in \mathcal{V}} g_{\eta,i}, \\ U_\mathcal{V} &:= p_{\text{cmp}}(2aD_d)(9/8)^{p_{\text{cmp}}}, & W_\mathcal{V} &:= p_{\text{cmp}}(2bD_d)(9/8)^{p_{\text{cmp}}}, \\ D_\mathcal{V} &:= p_{\text{cmp}}(3ab)(3/2)^{p_{\text{cmp}}}. \end{aligned}$$

Telescoping a product, using an upper bound $D_{\text{prod}} \geq 1$ on both sets of factors, gives

$$\left| \prod_{i \in \mathcal{V}} f_{\text{tel},i} - \prod_{i \in \mathcal{V}} g_{\text{tel},i} \right| \leq D_{\text{prod}}^{|\mathcal{V}|} \sum_{i \in \mathcal{V}} |f_{\text{tel},i} - g_{\text{tel},i}|.$$

Here the telescoping sum replaces one factor at a time; each remaining product is bounded by $D_{\text{prod}}^{|\mathcal{V}|-1}$, which is at most the displayed power. For the empty set both sides are zero. Applying this inequality gives

$$|\mathcal{A}_{\mathcal{V}} - 1| \leq U_{\mathcal{V}}, \quad |\mathcal{G}_{\mathcal{V}} - 1| \leq W_{\mathcal{V}},$$

$$\left| \prod_{i \in \mathcal{V}} R_{\theta, \lambda, \eta, i} - \mathcal{A}_{\mathcal{V}} \mathcal{G}_{\mathcal{V}} \right| \leq D_{\mathcal{V}}.$$

Flipping η preserves every function of λ and changes the positive weights to the negative weights. Flipping λ similarly preserves every function of η . Consequently

$$\mathbb{E}_{+1} \mathcal{A}_{\mathcal{V}} = \mathbb{E}_{-1} \mathcal{A}_{\mathcal{V}}, \quad \mathbb{E}_{+1} \mathcal{G}_{\mathcal{V}} = \mathbb{E}_{-1} \mathcal{G}_{\mathcal{V}}.$$

Expanding the centered product and using normalization therefore yields

$$\begin{aligned} & \mathbb{E}_{+1}[\mathcal{A}_{\mathcal{V}} \mathcal{G}_{\mathcal{V}}] - \mathbb{E}_{-1}[\mathcal{A}_{\mathcal{V}} \mathcal{G}_{\mathcal{V}}] \\ &= \mathbb{E}_{+1}[(\mathcal{A}_{\mathcal{V}} - 1)(\mathcal{G}_{\mathcal{V}} - 1)] - \mathbb{E}_{-1}[(\mathcal{A}_{\mathcal{V}} - 1)(\mathcal{G}_{\mathcal{V}} - 1)]. \end{aligned}$$

Each expectation on the right has absolute value at most $U_{\mathcal{V}} W_{\mathcal{V}}$. Adding the two contrast-deletion errors gives

$$|g_{+1, \mathcal{V}} - g_{-1, \mathcal{V}}| \leq D_{\mathcal{V}} + 2U_{\mathcal{V}} W_{\mathcal{V}} + D_{\mathcal{V}}.$$

Finally,

$$(9/8)^{p_{\text{cmp}}} (9/8)^{p_{\text{cmp}}} \leq (3/2)^{p_{\text{cmp}}}, \quad p_{\text{cmp}} \leq p_{\text{cmp}}^2 \quad (p_{\text{cmp}} \in \mathbb{N}).$$

Substitution proves

$$|g_{+1, \mathcal{V}}(\mathbf{x}, \mathbf{s}) - g_{-1, \mathcal{V}}(\mathbf{x}, \mathbf{s})| \leq C_{\Delta} (3/2)^{p_{\text{cmp}}} p_{\text{cmp}}^2 ab.$$

For $p_{\text{cmp}} = 0$, both densities equal one and the bound is zero.

5. *A positive floor and component Hellinger distance.* Define

$$c_H := \min \left\{ c_{\Delta}, \frac{1}{64D_d} \right\}, \quad K_{\text{info}} := C_{\Delta}^2.$$

Under the amplitude inequalities with c_H ,

$$aD_d \leq \frac{1}{16}, \quad bD_d + ab \leq \frac{1}{16}.$$

Thus the treatment factor of a labeled likelihood is at least $7/8$, and its outcome factor is also at least $7/8$. Their product is at least $1/2$. An auxiliary likelihood is at least $7/8$, hence at least $1/2$. Averaging the resulting product floor with nonnegative normalized weights gives

$$g_{\theta, \mathcal{V}}(\mathbf{x}, \mathbf{s}) \geq (1/2)^{p_{\text{cmp}}}.$$

Define the common probability reference on the record spins by

$$\mu_{\text{spin}} := \bigotimes_{i=1}^n \text{Unif}(\{-1, +1\}^2) \otimes \bigotimes_{i=n+1}^N \text{Unif}(\{-1, +1\}),$$

and define

$$H_{\mathcal{V}}^2(\mathbf{x}) := \int \left(\sqrt{g_{+1,\mathcal{V}}(\mathbf{x}, \mathbf{s})} - \sqrt{g_{-1,\mathcal{V}}(\mathbf{x}, \mathbf{s})} \right)^2 d\mu_{\text{spin}}(\mathbf{s}).$$

For nonnegative numbers bounded below by $t_{\text{floor}} > 0$, the identity

$$(f_{\text{den}} - g_{\text{den}})^2 = (\sqrt{f_{\text{den}}} - \sqrt{g_{\text{den}}})^2 (\sqrt{f_{\text{den}}} + \sqrt{g_{\text{den}}})^2$$

and the denominator bound

$$(\sqrt{f_{\text{den}}} + \sqrt{g_{\text{den}}})^2 \geq 4t_{\text{floor}}$$

give

$$(\sqrt{f_{\text{den}}} - \sqrt{g_{\text{den}}})^2 \leq \frac{(f_{\text{den}} - g_{\text{den}})^2}{4t_{\text{floor}}}.$$

Apply this with the preceding component floor and density difference, and integrate against the probability reference. We obtain

$$\begin{aligned} H_{\mathcal{V}}^2(\mathbf{x}) &\leq \frac{(C_{\Delta}(3/2)^{p_{\text{cmp}}} p_{\text{cmp}}^2 ab)^2}{4(1/2)^{p_{\text{cmp}}}} \\ &= \frac{C_{\Delta}^2}{4} (9/2)^{p_{\text{cmp}}} p_{\text{cmp}}^4 a^2 b^2 \\ &\leq K_{\text{info}} (9/2)^{p_{\text{cmp}}} p_{\text{cmp}}^4 a^2 b^2. \end{aligned}$$

6. *Conditional factorization and the count bound.* For fixed covariates, define the augmented graph

$$\mathcal{G}_{\mathbf{x}} := (\{1, \dots, N\}, \{\{i, j\} : i \neq j, x_i, x_j \in C_h, \|x_i - x_j\|_{\infty} \leq 2\delta\}),$$

and let

$$\mathcal{K}_{\mathbf{x}} := \{\mathcal{V} : \mathcal{V} \text{ is the vertex set of a connected component of } \mathcal{G}_{\mathbf{x}}\}.$$

The cube C_h is that of [Definition 7](#). Records outside C_h are isolated in this augmented graph and have likelihood one.

If $\phi_{\mathbf{z}}(x_i) \neq 0$, then

$$x_i \in C_h, \quad \left| \frac{x_{i,j_{\text{coord}}} - x_{0,j_{\text{coord}}}}{\delta} - z_{j_{\text{coord}}} \right| < 1 \quad (j_{\text{coord}} \in \{1, \dots, d\}).$$

Two records sharing a nonzero frame entry therefore satisfy

$$\|x_i - x_j\|_{\infty} \leq 2\delta,$$

and belong to the same component. Assign each frame index active at any record to this unique component; inactive indices contribute a factor one. The product prior is independent across indices, and each component likelihood depends solely on its assigned indices. Regrouping the finite sign sums by these disjoint groups consequently proves

$$g_{\theta}(\mathbf{x}, \mathbf{s}) := \mathbb{E}_{\theta} \prod_{i=1}^N R_{\theta,\lambda,\eta,i}(\mathbf{x}, \mathbf{s}) = \prod_{\mathcal{V} \in \mathcal{K}_{\mathbf{x}}} g_{\theta,\mathcal{V}}(\mathbf{x}, \mathbf{s}).$$

Each component density depends solely on its component spins. Integrating the fixed-sign record product over those fair spins gives one, because every record likelihood is normalized. Averaging signs therefore gives

$$g_{\theta, \nu} \geq 0, \quad \int g_{\theta, \nu} d\mu_{\text{spin}} = 1.$$

Define the component affinities by

$$\mathcal{A}_{\text{aff}, \nu}(\mathbf{x}) := \int \sqrt{g_{+1, \nu} g_{-1, \nu}} d\mu_{\text{spin}}.$$

Normalization and $2\sqrt{f_{\text{den}} g_{\text{den}}} \leq f_{\text{den}} + g_{\text{den}}$ imply

$$0 \leq \mathcal{A}_{\text{aff}, \nu} \leq 1, \quad H_{\mathcal{V}}^2 = 2(1 - \mathcal{A}_{\text{aff}, \nu}).$$

Integration over disjoint spin blocks multiplies the affinities. The elementary finite-product inequality

$$1 - \prod_{\nu \in \mathcal{K}_{\mathbf{x}}} \mathcal{A}_{\text{aff}, \nu} \leq \sum_{\nu \in \mathcal{K}_{\mathbf{x}}} (1 - \mathcal{A}_{\text{aff}, \nu})$$

follows by induction from the fact that all the affinities lie in $[0, 1]$. Hence

$$\int (\sqrt{g_{+1}} - \sqrt{g_{-1}})^2 d\mu_{\text{spin}} \leq \sum_{\nu \in \mathcal{K}_{\mathbf{x}}} H_{\mathcal{V}}^2(\mathbf{x}).$$

Set

$$c_I := \min\{c_H, c_U\}.$$

Its amplitude conditions permit both the component bound and the uninformative-component equalities. A component containing a labeled record and having size less than two has size one, so its distance is zero by singleton cancellation. A component containing no labeled record is auxiliary-only, so its distance is also zero. Define the informative component counts, for every integer $p_{\text{cmp}} \geq 0$, by

$$C_{p_{\text{cmp}}}^{\text{L}}(\mathbf{x}) := \# \{ \mathcal{V} \in \mathcal{K}_{\mathbf{x}} : |\mathcal{V}| = p_{\text{cmp}}, \mathcal{V} \cap \{1, \dots, n\} \neq \emptyset \}.$$

Regrouping the remaining component sum by size gives

$$\int (\sqrt{g_{+1}} - \sqrt{g_{-1}})^2 d\mu_{\text{spin}} \leq K_{\text{info}} a^2 b^2 \sum_{p_{\text{cmp}}=2}^N (9/2)^{p_{\text{cmp}}} p_{\text{cmp}}^4 C_{p_{\text{cmp}}}^{\text{L}}(\mathbf{x}).$$

The graph $\mathcal{G}_{\mathbf{x}}$ is exactly the graph in the statement: it retains all N vertices, including the exterior singleton components, whose density factors equal one.

7. *The labeled-root tree series.* Define the common covariate probability measure by

$$\nu_{\text{cov}} := \text{Unif}([0, 1]^d)^{\otimes N}.$$

For $p_{\text{cmp}} \geq 2$, a component counted by $C_{p_{\text{cmp}}}^{\text{L}}$ has a labeled root and a spanning tree on its vertex set. For a fixed root, there are $\binom{N-1}{p_{\text{cmp}}-1}$ choices of the other vertices. Orient

a spanning tree toward the root. Each nonroot vertex chooses a parent among the p_{cmp} selected vertices, so the number of valid rooted trees is at most $p_{\text{cmp}}^{p_{\text{cmp}}-1}$.

For a fixed valid tree, integrate its leaf covariates successively. Each child is constrained to a maximum-norm box of side 4δ around its parent; intersection with the unit cube has uniform mass at most $(4\delta)^d$. The root lies in C_h , whose uniform mass is h^d . Thus the probability of this tree event is at most

$$h^d ((4\delta)^d)^{p_{\text{cmp}}-1}.$$

Dropping the remaining component conditions enlarges the event. Summing these tree witnesses over the n possible labeled roots gives

$$\int C_{p_{\text{cmp}}}^L(\mathbf{x}) d\nu_{\text{cov}}(\mathbf{x}) \leq n h^d \binom{N-1}{p_{\text{cmp}}-1} p_{\text{cmp}}^{p_{\text{cmp}}-1} ((4\delta)^d)^{p_{\text{cmp}}-1}.$$

For $p_{\text{cmp}} > N$, the count and the binomial coefficient are both zero.

The binomial bound and the exponential series imply

$$\binom{N-1}{p_{\text{cmp}}-1} \leq \frac{N^{p_{\text{cmp}}-1}}{(p_{\text{cmp}}-1)!}, \quad \frac{p_{\text{cmp}}^{p_{\text{cmp}}-1}}{(p_{\text{cmp}}-1)!} \leq \exp(p_{\text{cmp}}).$$

For the second inequality, its left side is one nonnegative summand of the exponential series at p_{cmp} . Consequently

$$\binom{N-1}{p_{\text{cmp}}-1} p_{\text{cmp}}^{p_{\text{cmp}}-1} \leq N^{p_{\text{cmp}}-1} \exp(p_{\text{cmp}}).$$

Define

$$\begin{aligned} \kappa_{\text{tree}} &:= \frac{9}{2}, & S_{\text{tree}} &:= \sum_{j=0}^{\infty} (j+2)^4 2^{-j}, & E_{\text{occ}} &:= \kappa_{\text{tree}} \exp(1) 4^d, \\ c_0 &:= \frac{1}{2E_{\text{occ}}}, & C_{\text{tree}} &:= (S_{\text{tree}} + 1) (\kappa_{\text{tree}} \exp(1))^2 4^d. \end{aligned}$$

The series defining S_{tree} converges because it is a shifted polynomial-geometric series. These definitions give $c_0 > 0$ and $C_{\text{tree}} > 0$.

Under $N\delta^d \leq c_0$, define

$$z_{\text{occ}} := E_{\text{occ}} N \delta^d, \quad 0 \leq z_{\text{occ}} \leq \frac{1}{2}.$$

Applying the binomial-tree bound with $p_{\text{cmp}} = j+2$ gives

$$\begin{aligned} \kappa_{\text{tree}}^{j+2} (j+2)^4 \binom{N-1}{j+1} (j+2)^{j+1} ((4\delta)^d)^{j+1} \\ \leq \kappa_{\text{tree}} \exp(1) (j+2)^4 z_{\text{occ}}^{j+1}. \end{aligned}$$

For every finite truncation,

$$\sum_{j=0}^{N-1} (j+2)^4 z_{\text{occ}}^{j+1} \leq z_{\text{occ}} \sum_{j=0}^{N-1} (j+2)^4 2^{-j} \leq z_{\text{occ}} S_{\text{tree}}.$$

Multiplying the tree-count inequality by its size weight and summing therefore yields

$$\begin{aligned} \sum_{j=0}^{N-1} \kappa_{\text{tree}}^{j+2} (j+2)^4 \int C_{j+2}^{\text{L}} d\nu_{\text{cov}} \\ \leq nh^d \kappa_{\text{tree}} \exp(1) z_{\text{occ}} S_{\text{tree}} \\ \leq C_{\text{tree}} n N h^d \delta^d. \end{aligned}$$

The final inequality replaces S_{tree} by $S_{\text{tree}} + 1$ and substitutes the definition of z_{occ} .

8. *Integration for the original experiment.* The covariate distribution is ν_{cov} for every sign realization and hypothesis. Conditional on these covariates, the original records have density g_θ against μ_{spin} . Thus, after grouping covariates and spins, the mixture laws have densities g_{+1} and g_{-1} against $\nu_{\text{cov}} \otimes \mu_{\text{spin}}$.

For two densities against a common dominating measure, substitution into [Definition 18](#) gives the integral of their squared square-root difference. Applying this identity and then integrating over covariates yields

$$H^2(M_{+1}, M_{-1}) = \int \left[\int (\sqrt{g_{+1}(\mathbf{x}, \mathbf{s})} - \sqrt{g_{-1}(\mathbf{x}, \mathbf{s})})^2 d\mu_{\text{spin}}(\mathbf{s}) \right] d\nu_{\text{cov}}(\mathbf{x}).$$

The finite graph counts are measurable, and all functions in the finite weighted count sum are integrable. Integrating the conditional bound therefore gives

$$H^2(M_{+1}, M_{-1}) \leq K_{\text{info}} a^2 b^2 \sum_{p_{\text{cmp}}=2}^N (9/2)^{p_{\text{cmp}}} p_{\text{cmp}}^4 \int C_{p_{\text{cmp}}}^{\text{L}} d\nu_{\text{cov}}.$$

Every term is nonnegative, so extending the size sum to the shifted range used above gives

$$\begin{aligned} H^2(M_{+1}, M_{-1}) &\leq K_{\text{info}} a^2 b^2 \sum_{j=0}^{N-1} (9/2)^{j+2} (j+2)^4 \int C_{j+2}^{\text{L}} d\nu_{\text{cov}} \\ &\leq K_{\text{info}} C_{\text{tree}} n N h^d \delta^d a^2 b^2. \end{aligned}$$

Define

$$C := K_{\text{info}} C_{\text{tree}} > 0.$$

This proves the required information estimate under the amplitude conditions with c_I and the occupancy condition with c_0 .

9. *Common calibration and the remaining equalities.* Choose

$$c := \min\{c_M, \min\{c_S, \min\{c_U, c_I\}\}\}.$$

All four constants are positive and at most one, so

$$0 < c \leq 1, \quad c \leq c_M, \quad c \leq c_S, \quad c \leq c_U, \quad c \leq c_I.$$

Fix a tuple satisfying the theorem's hypotheses with this c . Since $\delta > 0$ and $h > 0$, multiplication by their real powers preserves these comparisons. For each $c' \in \{c_M, c_S, c_U, c_I\}$,

$$a \leq c\delta^\alpha \leq c'\delta^\alpha, \quad b \leq c\delta^\beta \leq c'\delta^\beta, \quad ab \leq ch^\gamma \leq c'h^\gamma.$$

The membership, singleton, uninformative-component, and information estimates established above therefore apply simultaneously. Weight normalization makes the finite push-forward priors probability measures, and the pointwise membership conclusion supports them on $\mathcal{M}$.

For the propensity distribution, the sign flip preserves the covariate-treatment marginal:

$$(P_{+1,\lambda,\eta})_{XA} = (P_{-1,\lambda,-\eta})_{XA}.$$

The designated propensity witnesses of these admitted laws are continuous and are versions of the same conditional treatment probability. Uniqueness of conditional margins followed by continuity under the full-support uniform design gives

$$e_{P_{+1,\lambda,\eta}}(x) = e_{P_{-1,\lambda,-\eta}}(x) \quad (x \in [0, 1]^d).$$

Reindexing the finite pushforwards using $w_{+1}(\lambda, \eta) = w_{-1}(\lambda, -\eta)$ proves equality of the distributions of the entire propensity functions restricted to the cube.

At the target, $B_h(x_0) = 1$, and hence

$$\tau_\theta(x_0) = -2\theta ab, \quad \tau_{-1}(x_0) - \tau_{+1}(x_0) = 4ab.$$

Because $a, b > 0$ and $c \leq 1 \leq 4$,

$$cab = c(ab) \leq 4(ab) = 4ab.$$

Finally, for every integer $k_{\text{aux}} \geq 0$, reindexing by the same sign flip gives

$$\begin{aligned} \int P_{XA}^{\otimes k_{\text{aux}}} d\varpi_{+1}(P) &= \sum_{\lambda, \eta} w_{+1}(\lambda, \eta) (P_{+1,\lambda,\eta})_{XA}^{\otimes k_{\text{aux}}} \\ &= \sum_{\lambda, \eta} w_{-1}(\lambda, \eta) (P_{-1,\lambda,\eta})_{XA}^{\otimes k_{\text{aux}}} \\ &= \int P_{XA}^{\otimes k_{\text{aux}}} d\varpi_{-1}(P). \end{aligned}$$

For any fixed auxiliary covariates, their conditional mixture kernels are

$$\sum_{\lambda, \eta} w_\theta(\lambda, \eta) \bigotimes_{i=1}^{k_{\text{aux}}} \text{Bernoulli}(e_\lambda(x_i)).$$

The factors are unchanged by the sign flip, so these kernels also agree for every covariate collection. When $k_{\text{aux}} = 0$, the products are the unit probability measure on the empty tuple, and normalization gives the same equality.

The singleton calculation supplies the stated uniform labeled conditional laws. The conditional factorization and component equalities apply to every cube covariate collection and every spin assignment. Together with the calibrated information bound, these establish all assertions.

□

Proof of Proposition 1. Fix the public parameters in the stated domain. The frontier, oracle, and acquisition constants below are finite real numbers depending only on these parameters. In particular,

$$\gamma > 0, \quad S > 0, \quad 2\gamma + d > 0, \quad \Delta > 0.$$

For an arbitrary auxiliary-size sequence, write

$$N_n := n + m_n \quad (n \in \mathbb{N}).$$

All asymptotic arguments may be restricted to the tail $n \geq 2$.

1. By Theorems 2 and 3, choose constants satisfying

$$0 < c_F \leq C_F < \infty, \quad 0 < c_O \leq C_O < \infty$$

such that, uniformly for $n \geq 2$ and $m \geq 0$,

$$c_F r_*(n, m) \leq R(n, m) \leq C_F r_*(n, m), \quad c_O r_o(n) \leq R^e(n, m) \leq C_O r_o(n).$$

Since

$$r_*(n, m) \geq r_o(n) = n^{-\gamma/(2\gamma+d)} > 0,$$

these bounds make both risks positive and finite. Dividing the upper bound for one risk by the lower bound for the other gives

$$\begin{aligned} 0 &\leq \frac{R(n, m)}{R^e(n, m)} \leq \frac{C_F}{c_O} \frac{r_*(n, m)}{r_o(n)}, \\ 0 &\leq \frac{r_*(n, m)}{r_o(n)} \leq \frac{C_O}{c_F} \frac{R(n, m)}{R^e(n, m)}. \end{aligned}$$

The same calculation, using the frontier bounds at (n, m) and $(n, 0)$, gives

$$\begin{aligned} 0 &\leq \frac{R(n, m)}{R(n, 0)} \leq \frac{C_F}{c_F} \frac{r_*(n, m)}{r_*(n, 0)}, \\ 0 &\leq \frac{r_*(n, m)}{r_*(n, 0)} \leq \frac{C_F}{c_F} \frac{R(n, m)}{R(n, 0)}. \end{aligned}$$

The first pair transfers eventual boundedness in both directions. The second pair transfers convergence to zero by squeezing between zero and a fixed multiple of the other ratio. Moreover, positivity of $r_o(n)$ gives

$$\frac{r_*(n, m_n)}{r_o(n)} = O(1) \iff r_*(n, m_n) = O(r_o(n)).$$

Consequently,

$$\begin{aligned} \frac{R(n, m_n)}{R^e(n, m_n)} = O(1) &\iff r_*(n, m_n) = O(r_o(n)), \\ \frac{R(n, m_n)}{R(n, 0)} \longrightarrow 0 &\iff \frac{r_*(n, m_n)}{r_*(n, 0)} \longrightarrow 0. \end{aligned}$$

If the rate comparison holds, there is a real constant K_{rate} such that

$$r_*(n, m_n) \leq K_{\text{rate}} r_o(n) \quad \text{eventually.}$$

Set

$$\zeta := \max\{K_{\text{rate}}, 1\}.$$

Then $1 \leq \zeta < \infty$, and the defining inequality in [Definition 15](#) holds eventually. Conversely, eventual membership in that region supplies the rate comparison with constant ζ . Thus

$$r_*(n, m_n) = O(r_o(n)) \iff \exists \zeta \in [1, \infty) \text{ such that } (n, m_n) \in \mathcal{B}_*(\zeta) \text{ eventually.}$$

2. For the extended nonnegative lower limit, the tail characterization is

$$0 < \liminf_{n \rightarrow \infty} \frac{N_n}{n^{q_*}} \iff \exists c_{\text{count}} > 0 \text{ such that } c_{\text{count}} \leq \frac{N_n}{n^{q_*}} \text{ eventually.}$$

Indeed, a positive lower limit admits a smaller positive finite real number, which bounds every sufficiently late term from below; this also applies when the lower limit is $+\infty$. Conversely, an eventual positive lower bound bounds the lower limit from below.

We now translate annotation-region membership into precisely such a bound. Directly from the definitions,

$$q_* = \frac{S_{\text{crit}}}{S}, \quad \Delta = (2\gamma + d)(1 + q_*).$$

For $n \geq 2$, this yields

$$r_o(n)^{-\Delta/\gamma} = n^{\Delta/(2\gamma+d)} = n^{1+q_*} = n n^{q_*}.$$

Let $\zeta \geq 1$. Since $r_o(n) \leq \zeta r_o(n)$, the maximum defining r_* and [Definition 15](#) give

$$\begin{aligned} (n, m) \in \mathcal{B}_*(\zeta) &\iff (nN)^{-\gamma/\Delta} \leq \zeta r_o(n) \\ &\iff (\zeta r_o(n))^{-\Delta/\gamma} \leq nN \\ &\iff \zeta^{-\Delta/\gamma} \leq \frac{N}{n^{q_*}}. \end{aligned}$$

Here the second equivalence uses the strictly decreasing positive-power map with exponent $-\gamma/\Delta$, whose inverse has exponent $-\Delta/\gamma$; all bases are positive.

Eventual membership therefore supplies the positive lower bound

$$c_{\text{count}} := \zeta^{-\Delta/\gamma} > 0.$$

In the reverse direction, suppose an eventual lower bound $c_{\text{count}} > 0$ is given, and choose

$$\zeta := \max\left\{1, c_{\text{count}}^{-\gamma/\Delta}\right\}.$$

Then

$$1 \leq \zeta < \infty, \quad \zeta^{-\Delta/\gamma} \leq \left(c_{\text{count}}^{-\gamma/\Delta}\right)^{-\Delta/\gamma} = c_{\text{count}}.$$

The displayed count characterization gives eventual membership in $\mathcal{B}_*(\zeta)$. Combining this with [Item 1](#) proves all four oracle-order equivalences.

3. To characterize strict improvement, first suppose $S < S_{\text{crit}}$. The identities above imply

$$q_* = \frac{S_{\text{crit}}}{S} > 1, \quad \Delta = (2\gamma + d)(1 + q_*) > 2(2\gamma + d),$$

and hence

$$\frac{2\gamma}{\Delta} < \frac{\gamma}{2\gamma + d}.$$

Define the positive decay exponent

$$\eta_{\text{dec}} := \frac{\gamma}{2\gamma + d} - \frac{2\gamma}{\Delta} > 0.$$

For $n \geq 2$, the inequality $q_* > 1$ gives

$$n = n^1 \leq n^{q_*}.$$

The lower-count branch in [Theorem 2](#), applied with $m = 0$, therefore gives

$$r_*(n, 0) = (n^2)^{-\gamma/\Delta} = n^{-2\gamma/\Delta}.$$

Dividing the two terms in the maximum defining $r_*(n, m_n)$ by this positive rate yields

$$\frac{r_o(n)}{n^{-2\gamma/\Delta}} = n^{-\eta_{\text{dec}}}, \quad \frac{(nN_n)^{-\gamma/\Delta}}{n^{-2\gamma/\Delta}} = \left(\frac{N_n}{n}\right)^{-\gamma/\Delta}.$$

Thus

$$\frac{r_*(n, m_n)}{r_*(n, 0)} = \max \left\{ n^{-\eta_{\text{dec}}}, \left(\frac{N_n}{n}\right)^{-\gamma/\Delta} \right\}.$$

The second term tends to zero exactly when $N_n/n \rightarrow +\infty$. To see the reverse implication explicitly, let

$$b_{\text{thr}} \in \mathbb{R}, \quad x_{\text{thr}} := \max\{1, b_{\text{thr}}\} > 0.$$

If the second term tends to zero, then eventually

$$\left(\frac{N_n}{n}\right)^{-\gamma/\Delta} \leq x_{\text{thr}}^{-\gamma/\Delta}.$$

Inverting this decreasing power gives

$$\frac{N_n}{n} \geq \left(x_{\text{thr}}^{-\gamma/\Delta}\right)^{-\Delta/\gamma} = x_{\text{thr}} \geq b_{\text{thr}}.$$

This proves divergence to $+\infty$. The forward implication follows from the convergence of a strictly negative power to zero at $+\infty$. Since

$$\frac{N_n}{n} = 1 + \frac{m_n}{n} \quad (n \geq 2),$$

divergence of the total-to-labeled ratio is equivalent to divergence of the auxiliary-to-labeled ratio: an eventual lower bound at any real threshold for either ratio gives the corresponding bound for the other after shifting the threshold by one.

The first term $n^{-\eta_{\text{dec}}}$ tends to zero. Both terms are nonnegative, so convergence of their maximum to zero implies convergence of the second term by squeezing; conversely, convergence of both terms implies convergence of their maximum. We have proved, in this regime,

$$\frac{r_*(n, m_n)}{r_*(n, 0)} \rightarrow 0 \iff \frac{m_n}{n} \rightarrow +\infty.$$

For the complementary case $S \geq S_{\text{crit}}$, [Theorem 2](#) gives, for every $n \geq 2$,

$$r_*(n, m_n) = r_*(n, 0) = r_o(n) > 0, \quad \frac{r_*(n, m_n)}{r_*(n, 0)} = 1.$$

The ratio therefore has limit one, which excludes convergence to zero. Combining the two cases and the risk-to-rate equivalence proves

$$\frac{R(n, m_n)}{R(n, 0)} \rightarrow 0 \iff \frac{r_*(n, m_n)}{r_*(n, 0)} \rightarrow 0 \iff S < S_{\text{crit}} \text{ and } \frac{m_n}{n} \rightarrow +\infty.$$

4. The claimed exponent comparison follows from the same positive-denominator identity:

$$S < S_{\text{crit}} \implies q_* = \frac{S_{\text{crit}}}{S} > 1, \quad S > 0.$$

5. Suppose $m_n/n = O(1)$. Choose a real constant K_{aux} satisfying

$$\frac{m_n}{n} \leq K_{\text{aux}} \quad \text{eventually,}$$

and define

$$K_{\text{aux}}^+ := \max\{K_{\text{aux}}, 0\}, \quad M_{\text{aux}} := (1 + K_{\text{aux}}^+)^{\gamma/\Delta} \geq 1.$$

Eventually $m_n \leq K_{\text{aux}}^+ n$, and consequently

$$n^2 \leq nN_n \leq (1 + K_{\text{aux}}^+)n^2.$$

Taking the negative power $-\gamma/\Delta$ gives

$$(nN_n)^{-\gamma/\Delta} \leq (n^2)^{-\gamma/\Delta},$$

and

$$M_{\text{aux}}^{-1}(n^2)^{-\gamma/\Delta} \leq (nN_n)^{-\gamma/\Delta}, \quad (n^2)^{-\gamma/\Delta} \leq M_{\text{aux}}(nN_n)^{-\gamma/\Delta}.$$

The oracle term is common to both rates and satisfies

$$r_o(n) \leq M_{\text{aux}} r_o(n).$$

Taking maxima therefore gives

$$r_*(n, m_n) \leq r_*(n, 0) \leq M_{\text{aux}} r_*(n, m_n) \quad \text{eventually.}$$

Using the risk sandwiches,

$$\begin{aligned} R(n, m_n) &\leq C_F r_*(n, m_n) \leq C_F r_*(n, 0) \leq \frac{C_F}{c_F} R(n, 0), \\ R(n, 0) &\leq C_F r_*(n, 0) \leq C_F M_{\text{aux}} r_*(n, m_n) \leq \frac{C_F M_{\text{aux}}}{c_F} R(n, m_n). \end{aligned}$$

These eventual inequalities prove both stated big- O comparisons.

6. If $S \geq S_{\text{crit}}$, the frontier rate equals $r_o(n)$. For every $n \geq 2$ and $m \geq 0$, the frontier upper bound and the oracle lower bound consequently give

$$R(n, m) \leq C_F r_o(n) = \frac{C_F}{c_O} (c_O r_o(n)) \leq \frac{C_F}{c_O} R^e(n, m).$$

Dividing by the positive oracle risk yields

$$0 \leq \frac{R(n, m)}{R^e(n, m)} \leq \frac{C_F}{c_O}.$$

This uniform bound proves the oracle-order comparison for every auxiliary-size sequence.

7. For the finite-sample conclusions, [Theorem 2](#) supplies, simultaneously for all $n \geq 2$ and $m \geq 0$,

$$c_F r_*(n, m) \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P) \leq C_F r_*(n, m).$$

In particular, its upper bound follows by comparing the minimax value with the worst-case risk of T_* . The same cited statement supplies exactly

$$r_*(n, m) = \begin{cases} r_o(n), & S \geq S_{\text{crit}}, \\ (nN)^{-\gamma/\Delta}, & S < S_{\text{crit}}, \quad N \leq n^{q_*}, \\ r_o(n), & S < S_{\text{crit}}, \quad N \geq n^{q_*}, \end{cases}$$

together with

$$N = n^{q_*} \implies (nN)^{-\gamma/\Delta} = r_o(n).$$

Thus c_F, C_F furnish the asserted finite-sample constants, including the equality of the two count branches at their boundary.

8. For $n \geq 2$, specialize to $m = 0$. If $S < S_{\text{crit}}$, then $q_* > 1$, so

$$N = n \leq n^{q_*}, \quad r_*(n, 0) = (n^2)^{-\gamma/\Delta} = n^{-2\gamma/\Delta}.$$

If $S \geq S_{\text{crit}}$, the first branch of [Theorem 2](#) gives $r_*(n, 0) = r_o(n)$. Hence

$$r_*(n, 0) = \begin{cases} n^{-2\gamma/\Delta}, & S < S_{\text{crit}}, \\ r_o(n), & S \geq S_{\text{crit}}. \end{cases}$$

9. Define the average nuisance smoothness by

$$s_{\text{pub}} := \frac{S}{2} > 0.$$

By [Lemma 7](#), algebra with positive denominators gives

$$\frac{d/4}{1 + d/(2\gamma)} = \frac{S_{\text{crit}}}{2}, \quad \frac{1}{1 + d/(2\gamma) + d/(4s_{\text{pub}})} = \frac{2\gamma}{\Delta}, \quad \frac{1}{2 + d/\gamma} = \frac{\gamma}{2\gamma + d}.$$

In particular,

$$s_{\text{pub}} < \frac{d/4}{1 + d/(2\gamma)} \iff S < S_{\text{crit}}.$$

For $n \geq 2$, these identities identify the numerical benchmark's strict split and its two exponents with the supervised branches just established.

The remaining sample sizes are $n = 0$ and $n = 1$. At $n = 1$, all power bases equal one, so both the benchmark and $r_*(1, 0)$ equal one. At zero, the numerical formulas use the convention

$$0^{t_{\text{pow}}} = 0 \quad (t_{\text{pow}} \in \mathbb{R} \setminus \{0\}).$$

All exponents occurring here are strictly negative:

$$-\frac{\gamma}{2\gamma + d} < 0, \quad -\frac{\gamma}{\Delta} < 0, \quad -\frac{2\gamma}{\Delta} < 0.$$

Thus both branches of the benchmark equal zero at $n = 0$, and the two terms in the maximum defining $r_*(0, 0)$ also equal zero. This proves the benchmark equality for every $n \in \mathbb{N}$, with the stated formulas for $n \geq 1$.

10. For acquisition, define the positive exponents

$$p_0 := \frac{2\gamma + d}{\gamma} = 2 + \frac{d}{\gamma}, \quad p_1 := \frac{\Delta}{\gamma} = 2 + \frac{d}{\gamma} + \frac{d}{S}.$$

For positive η , the power-scaling identity is

$$\left(\frac{\eta}{c_{\text{scale}}} \right)^{-p_{\text{scale}}} = c_{\text{scale}}^{p_{\text{scale}}} \eta^{-p_{\text{scale}}} \quad (c_{\text{scale}} \in \{c_F, C_F\}, \quad p_{\text{scale}} \in \{p_0, p_1\}).$$

Choose

$$c_{\text{acq}} := \min\{c_F^{p_0}, c_F^{p_1}\} > 0, \quad C_{\text{acq}} := \max\{C_F^{p_0}, C_F^{p_1}\} > 0.$$

For any positive η , $n \geq 2$, and $m \geq 0$, the defining maximum in [Definition 14](#) gives

$$\begin{aligned} r_*(n, m) \leq \eta & \iff n^{-\gamma/(2\gamma+d)} \leq \eta \text{ and } (nN)^{-\gamma/\Delta} \leq \eta \\ & \iff \eta^{-p_0} \leq n \text{ and } \eta^{-p_1} \leq nN. \end{aligned}$$

The second equivalence again inverts strictly decreasing powers of positive bases.

Now let $0 < \eta < 1$, and suppose a Borel decision T satisfies

$$\mathcal{R}_{n,m}(T, P) \leq \eta \quad (P \in \mathcal{M}).$$

By [Definition 5](#) and the frontier lower bound,

$$c_F r_*(n, m) \leq R(n, m) \leq \sup_{P \in \mathcal{M}} \mathcal{R}_{n,m}(T, P) \leq \eta.$$

Therefore $r_*(n, m) \leq \eta/c_F$. Apply the exact rate equivalence at the positive scale η/c_F , and then the scaling identity, to obtain

$$c_F^{p_0} \eta^{-p_0} \leq n, \quad c_F^{p_1} \eta^{-p_1} \leq nN.$$

The choice of the minimum gives the common necessity constant:

$$c_{\text{acq}} \eta^{-p_0} \leq c_F^{p_0} \eta^{-p_0} \leq n, \quad c_{\text{acq}} \eta^{-p_1} \leq c_F^{p_1} \eta^{-p_1} \leq nN.$$

Conversely, suppose

$$C_{\text{acq}}\eta^{-p_0} \leq n, \quad C_{\text{acq}}\eta^{-p_1} \leq nN.$$

By the choice of the maximum and the scaling identity,

$$\left(\frac{\eta}{C_F}\right)^{-p_0} = C_F^{p_0}\eta^{-p_0} \leq C_{\text{acq}}\eta^{-p_0} \leq n,$$

and

$$\left(\frac{\eta}{C_F}\right)^{-p_1} = C_F^{p_1}\eta^{-p_1} \leq C_{\text{acq}}\eta^{-p_1} \leq nN.$$

The exact equivalence at scale η/C_F gives

$$r_*(n, m) \leq \frac{\eta}{C_F}.$$

By [Theorem 2](#), the decision T_* from [Definition 13](#) is Borel measurable, and, for every $P \in \mathcal{M}$,

$$\mathcal{R}_{n,m}(T_*, P) \leq \sup_{P' \in \mathcal{M}} \mathcal{R}_{n,m}(T_*, P') \leq C_F r_*(n, m) \leq \eta.$$

Thus $c_{\text{acq}}, C_{\text{acq}}$ supply the asserted acquisition constants. Finally,

$$n \leq n + m = N$$

because $m \geq 0$.

□

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